

ACTIVE DIRECTORY BLUE TEAM SECURITY LABORATORY

Active Directory Deployment, Attack Simulation, Detection Engineering
and SIEM Monitoring

Home-Lab Report

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Project Technologies:

Active Directory | Windows Server 2022 | Splunk Enterprise
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Phase 1 — Laboratory Infrastructure and Network Configuration

1.1 Introduction

The objective of Phase 1 was to establish the foundational infrastructure for the Active Directory Blue Team laboratory. The environment was designed to support controlled attack simulations against a Windows Active Directory environment and the subsequent collection, analysis, and detection of security telemetry using Splunk.

The laboratory was implemented using VMware virtualization and consists of four virtual machines:

Windows Server 2022 — future Active Directory Domain Controller

Windows 10 — Windows client endpoint

Kali Linux — attacker and penetration-testing system

Ubuntu Desktop — Splunk SIEM server

A dedicated VMware Host-only network was created to isolate communication between the laboratory systems. NAT networking was also configured on each virtual machine to provide Internet connectivity when required for software installation, updates, and administrative activities.

The infrastructure established in this phase serves as the foundation for all subsequent phases of the project.

1.2 Laboratory Architecture

The final laboratory consists of four virtual machines connected through VMware.

System	Operating System	Role	VMnet1 IP
DC01	Windows Server 2022	Active Directory Domain Controller	192.168.10.10
WIN10	Windows 10	Domain Client / Endpoint	192.168.10.20
KALI	Kali Linux	Attacker	192.168.10.30
SPLUNK01	Ubuntu Desktop	Splunk SIEM Server	192.168.10.40

Each virtual machine was configured with two network adapters:

NAT adapter: Used for Internet connectivity.

VMnet1 adapter: Used for communication within the isolated cybersecurity laboratory.

The resulting architecture separates normal Internet connectivity from internal laboratory traffic.

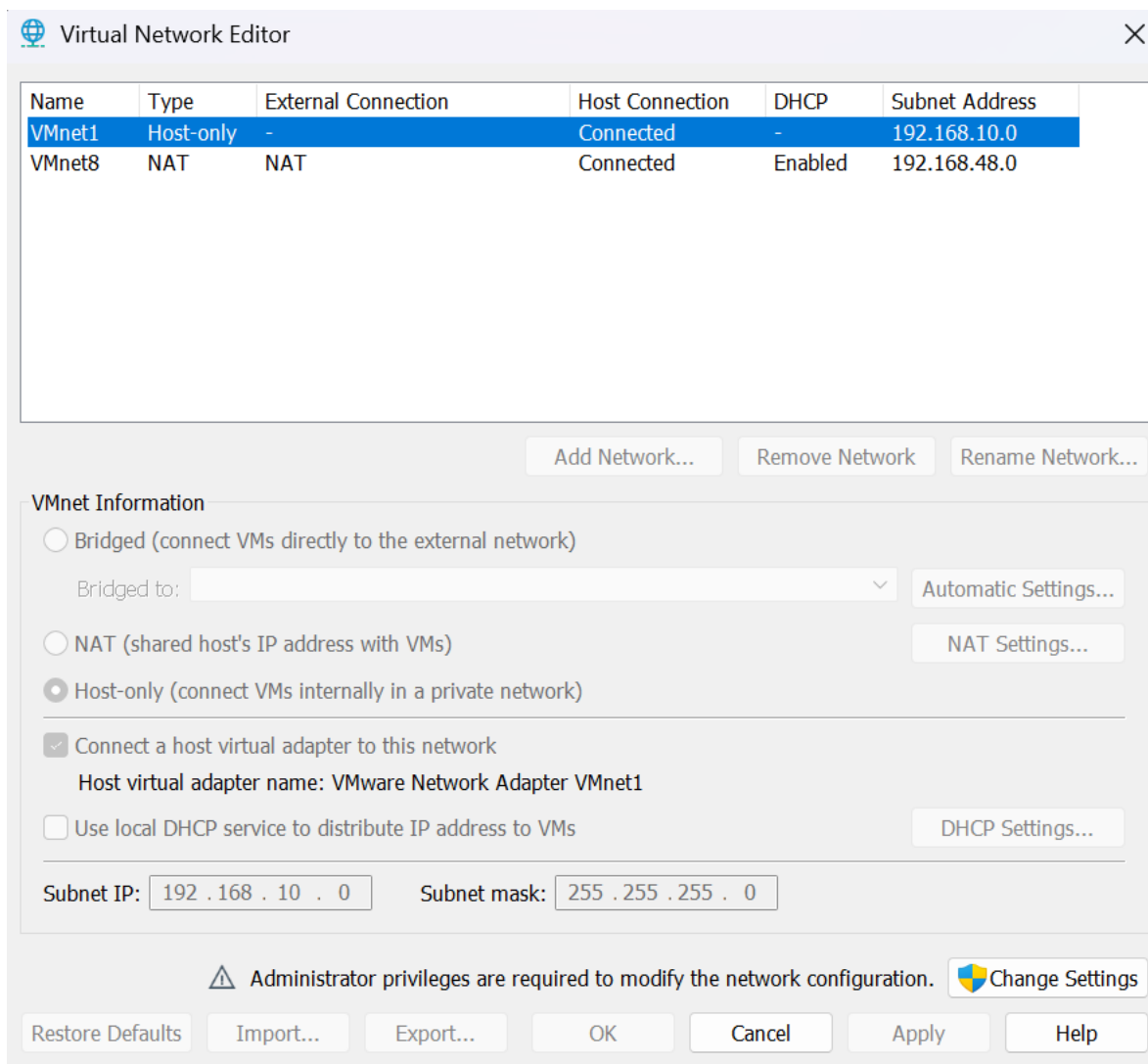


Figure 1: VMware VMnet1 Host-only network configuration.

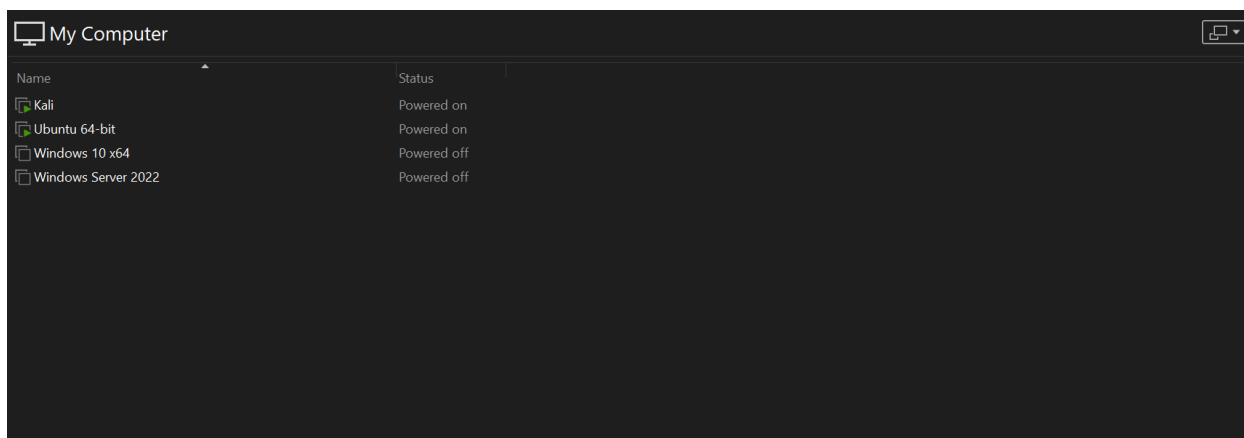


Figure 2: Virtual machines deployed for the cybersecurity laboratory.

1.3 VMware Host-only Network Configuration

A dedicated VMware Host-only network, VMnet1, was configured for the internal laboratory environment.

The network configuration was:

Network: 192.168.10.0/24

Subnet Mask: 255.255.255.0

DHCP: Disabled

DHCP was disabled intentionally because static addressing is required for predictable communication between the systems. Static addressing is particularly important for the future Active Directory Domain Controller because Active Directory and DNS depend heavily on stable network addressing.

The VMnet1 network is isolated from the physical network and provides the communication path between the attacker, Windows systems, and Splunk server.

1.4 Network Addressing Scheme

A structured static addressing scheme was implemented for the VMnet1 network.

192.168.10.0/24

The following addresses were assigned:

192.168.10.10 DC01

192.168.10.20 WIN10

192.168.10.30 KALI

192.168.10.40 SPLUNK01

The addressing scheme was selected to make the systems easy to identify during later attack simulations and security investigations.

The NAT interfaces remained configured separately using DHCP. Therefore, the VMnet1 interfaces provide predictable internal laboratory communication while the NAT interfaces provide Internet access.

1.5 Kali Linux Configuration

Kali Linux was configured as the attacker machine for the laboratory.

Two network interfaces were configured:

eth0 → NAT

- DHCP
- Internet connectivity

eth1 → VMnet1

- 192.168.10.30/24
- Static

The VMnet1 interface was configured without a default gateway. This ensures that internal laboratory traffic remains associated with the VMnet1 network while Internet-bound traffic continues to use the NAT interface.

The configuration was verified using the **ip addr** command.

```
root@kali:~# ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UNKNOWN group default qlen 1000
    link/ether 00:0c:29:99:6c:dd brd ff:ff:ff:ff:ff:ff
    inet 192.168.48.132/24 brd 192.168.48.255 scope global dynamic noprefixroute eth0
        valid_lft 1505sec preferred_lft 1505sec
    inet6 fe80::889f:d0fa:a1a1:e949/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
3: eth1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:99:6c:e7 brd ff:ff:ff:ff:ff:ff
    inet 192.168.10.30/24 brd 192.168.10.255 scope global noprefixroute eth1
        valid_lft forever preferred_lft forever
    inet6 fe80::20c:29ff:fe99:6ce7/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
root@kali:~# S
```

Figure 3: Kali Linux network interfaces showing NAT and VMnet1 configuration.

1.6 Windows 10 Client Configuration

Windows 10 was configured as the primary Windows endpoint for the laboratory.

The system was configured with:

NAT Adapter:

- DHCP
- Internet connectivity

VMnet1:

- IP Address: 192.168.10.20
- Subnet Mask: 255.255.255.0

The Windows 10 system will later be joined to the Active Directory domain and will act as an important endpoint for generating authentication, process execution, PowerShell, and Sysmon telemetry.

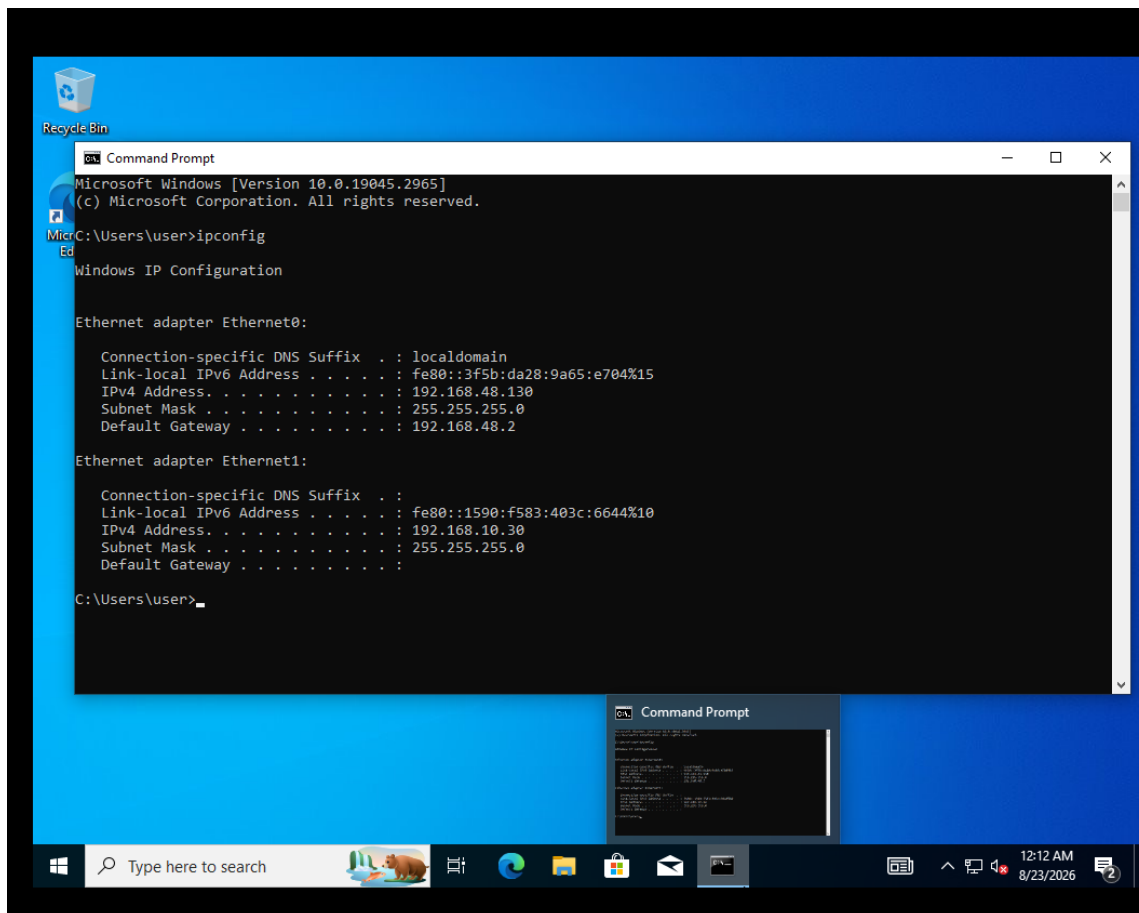


Figure 4: Windows 10 network configuration showing the static VMnet1 address.

1.7 Windows Server 2022 Configuration

Windows Server 2022 was configured as DC01, which will later be promoted to the Active Directory Domain Controller.

The current network configuration is:

NAT Adapter:

- DHCP
- Internet connectivity

VMnet1:

- IP Address: 192.168.10.10
- Subnet Mask: 255.255.255.0

No custom DNS configuration was applied during this phase. DNS configuration will be performed as part of the Active Directory deployment because the Domain Controller will subsequently provide DNS services for the Active Directory environment.

This approach avoids introducing an incorrect DNS configuration before Active Directory Domain Services are installed.

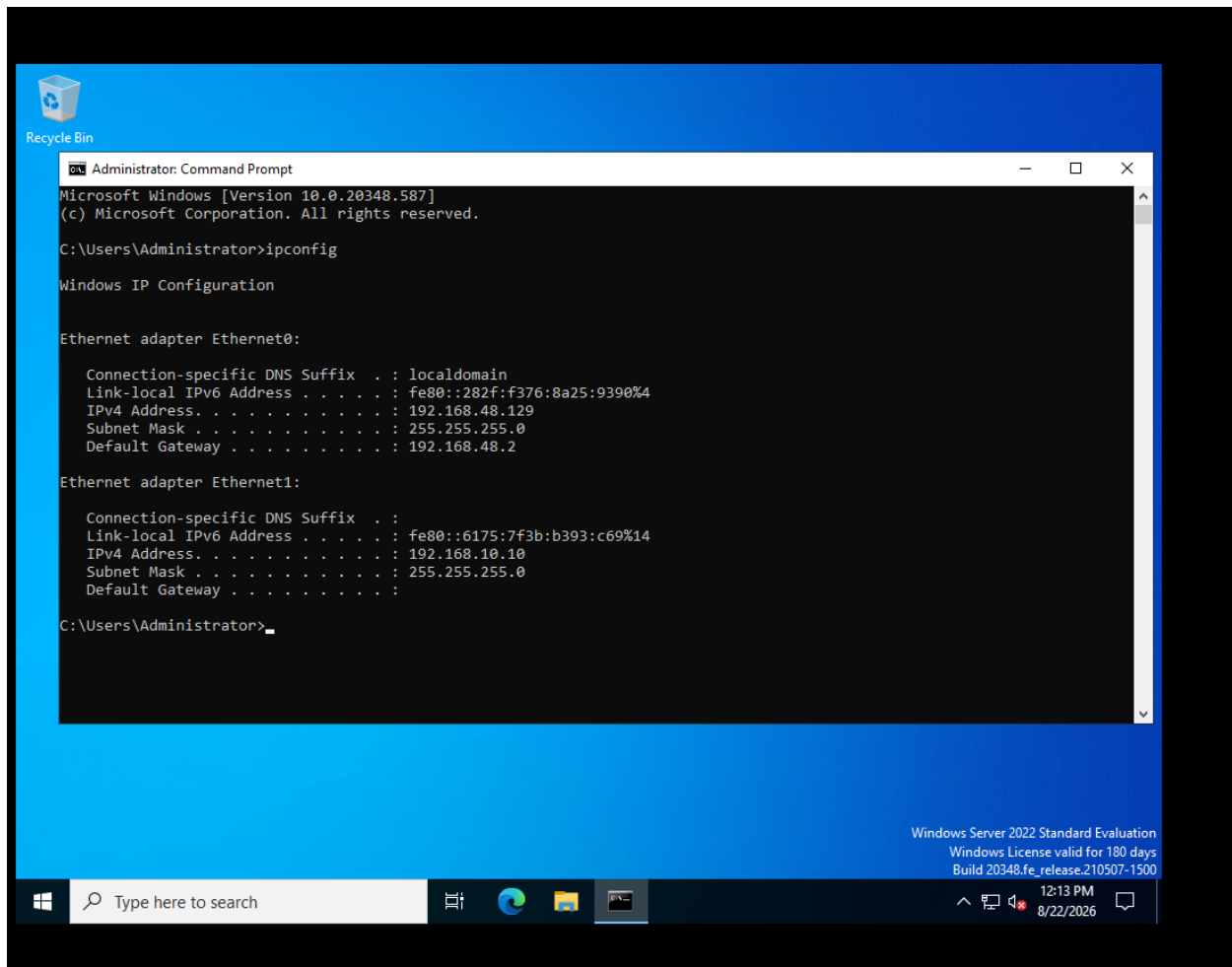


Figure 5: Windows Server 2022 network configuration and static VMnet1 address.

1.8 Ubuntu Desktop / Splunk Server Configuration

Ubuntu Desktop was selected as the dedicated Splunk server for the laboratory.

The system was configured as SPLUNK01 with two network interfaces:

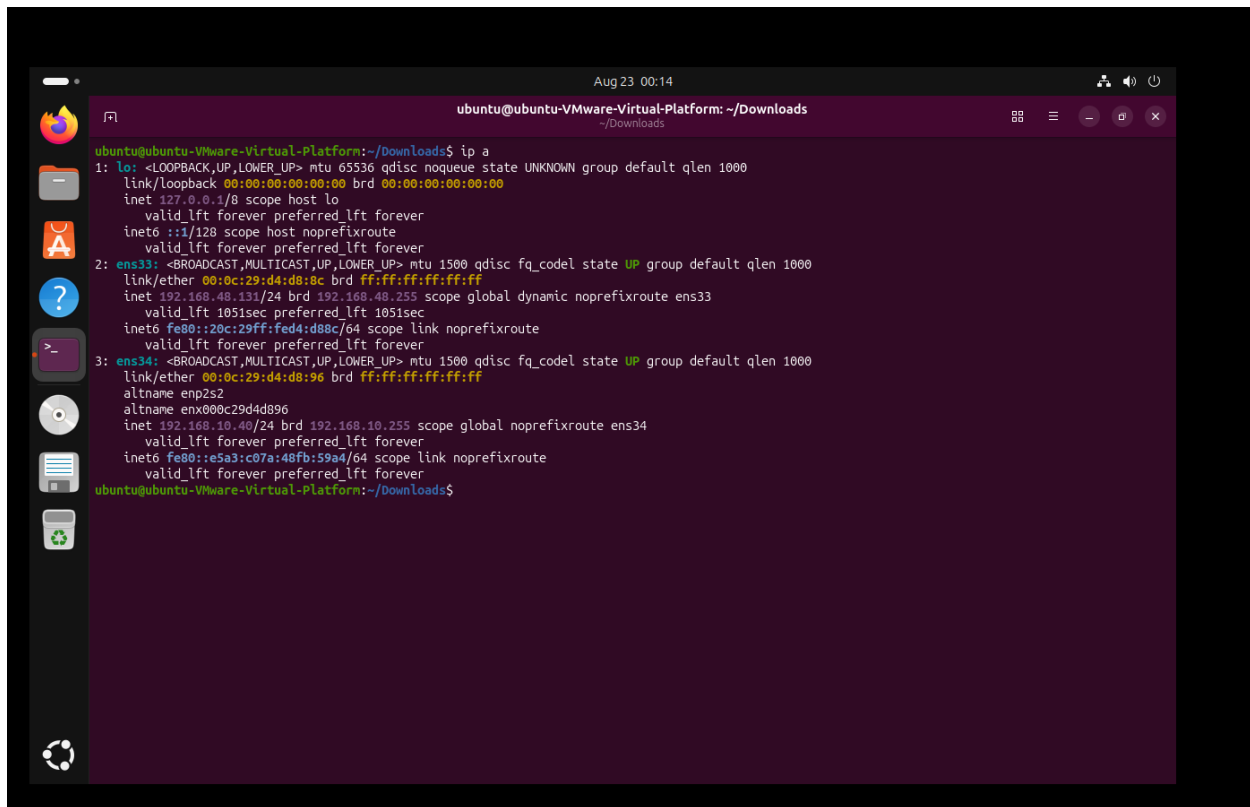
ens33 → NAT

- DHCP
- Internet connectivity

ens34 → VMnet1

- 192.168.10.40/24
- Static

The VMnet1 interface does not use a default gateway. This ensures that internal laboratory communication uses the isolated VMnet1 network while the NAT interface handles Internet connectivity.



```
ubuntu@ubuntu-VMware-Virtual-Platform:~/Downloads$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:d4:d8:8c brd ff:ff:ff:ff:ff:ff
    inet 192.168.48.131/24 brd 192.168.48.255 scope global dynamic noprefixroute ens33
        valid_lft 1051sec preferred_lft 1051sec
    inet6 fe80::20c:29ff:fed4:d88c/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
3: ens34: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:d4:d8:96 brd ff:ff:ff:ff:ff:ff
    altnam enp2s2
    altnam enx00c29d4d896
    inet 192.168.10.255/24 brd 192.168.10.255 scope global noprefixroute ens34
        valid_lft forever preferred_lft forever
    inet6 fe80::e5a3:c07a:48fb:59a4/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
ubuntu@ubuntu-VMware-Virtual-Platform:~/Downloads$
```

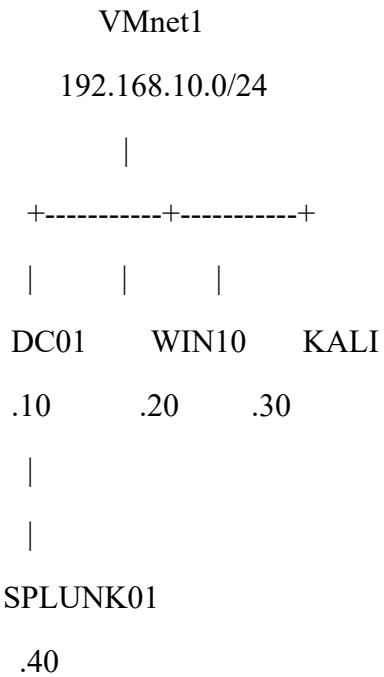
Figure 6: Ubuntu Desktop network configuration showing NAT and VMnet1 interfaces.

1.9 Network Connectivity Verification

After configuring the static addresses, connectivity between the laboratory systems was tested.

Kali Linux was used to verify communication with the other laboratory systems over the VMnet1 network.

The expected communication structure is:



Connectivity testing confirmed communication between the configured laboratory systems.

```
Aug 22 14:18
root@kali: ~

root@kali:~# ping -c 2 192.168.10.10
PING 192.168.10.10 (192.168.10.10) 56(84) bytes of data.
^C
--- 192.168.10.10 ping statistics ---
2 packets transmitted, 0 received, 100% packet loss, time 1016ms

root@kali:~# ping -c 2 192.168.10.20
PING 192.168.10.20 (192.168.10.20) 56(84) bytes of data.
^C
--- 192.168.10.20 ping statistics ---
2 packets transmitted, 0 received, 100% packet loss, time 1008ms

root@kali:~# ping -c 2 192.168.10.40
PING 192.168.10.40 (192.168.10.40) 56(84) bytes of data.
64 bytes from 192.168.10.40: icmp_seq=1 ttl=64 time=5.52 ms
64 bytes from 192.168.10.40: icmp_seq=2 ttl=64 time=3.17 ms

--- 192.168.10.40 ping statistics ---
2 packets transmitted, 2 received, 0% packet loss, time 1009ms
rtt min/avg/max/mdev = 3.168/4.343/5.518/1.175 ms
root@kali:~# ^C
root@kali:~#
```

Figure 7: Connectivity testing from Kali Linux to the internal laboratory systems.

Internet connectivity was also tested independently through the NAT interface to verify that the laboratory systems could access external resources when required.

```
Aug 22 14:19
root@kali: ~

root@kali:~# ping -c 4 8.8.8.8
PING 8.8.8.8 (8.8.8.8) 56(84) bytes of data.
64 bytes from 8.8.8.8: icmp_seq=1 ttl=128 time=46.9 ms
64 bytes from 8.8.8.8: icmp_seq=2 ttl=128 time=47.2 ms
64 bytes from 8.8.8.8: icmp_seq=3 ttl=128 time=44.8 ms
64 bytes from 8.8.8.8: icmp_seq=4 ttl=128 time=46.4 ms

--- 8.8.8.8 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 2998ms
rtt min/avg/max/mdev = 44.800/46.298/47.170/0.911 ms
root@kali:~# ping -c 4 google.com
PING google.com (142.250.202.142) 56(84) bytes of data.
64 bytes from lbg02s04-in-f14.1e100.net (142.250.202.142): icmp_seq=1 ttl=128 time=41.2 ms
64 bytes from lbg02s04-in-f14.1e100.net (142.250.202.142): icmp_seq=2 ttl=128 time=43.0 ms
64 bytes from lbg02s04-in-f14.1e100.net (142.250.202.142): icmp_seq=3 ttl=128 time=46.7 ms
64 bytes from lbg02s04-in-f14.1e100.net (142.250.202.142): icmp_seq=4 ttl=128 time=44.7 ms

--- google.com ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3010ms
rtt min/avg/max/mdev = 41.204/43.897/46.650/2.015 ms
root@kali:~#
```

Figure 8: Internet connectivity verification through the VMware NAT interface.

1.10 Splunk Enterprise Installation

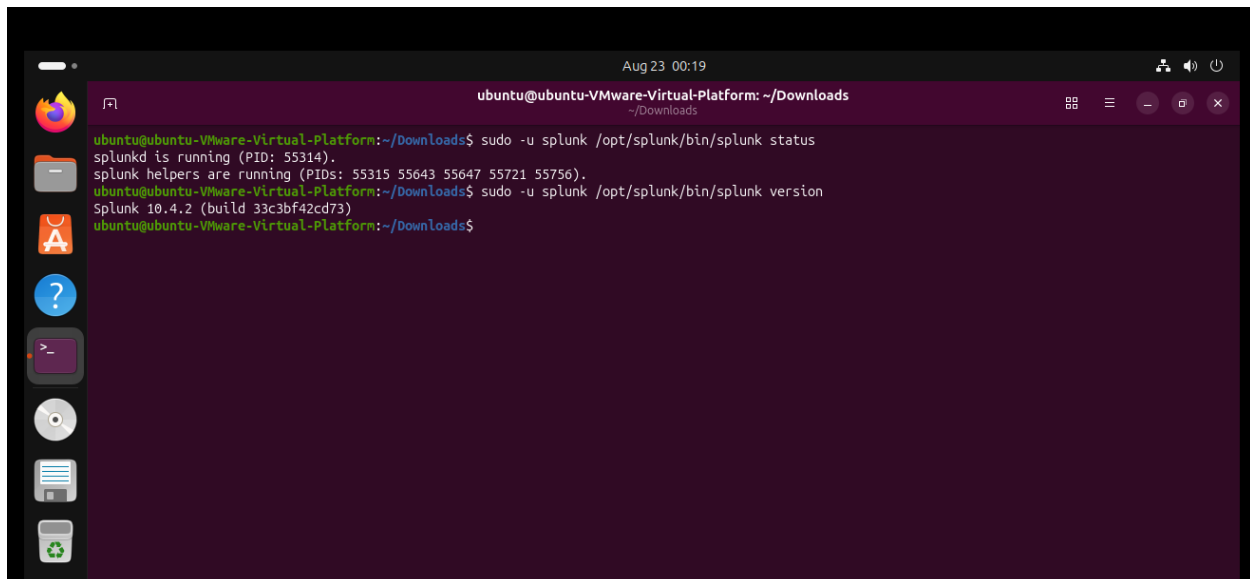
Ubuntu Desktop was selected as the centralized SIEM server for the project.

Splunk Enterprise was installed using the Debian package obtained from the official Splunk distribution. The package was installed on the Ubuntu Desktop system and the Splunk installation directory was established under:

/opt/splunk

The Splunk installation was configured to run under the dedicated splunk user rather than the root account. This follows the principle of least privilege and avoids unnecessarily running the SIEM software with root privileges.

The installed Splunk version was verified after installation.

A screenshot of a terminal window on an Ubuntu system. The window title is 'ubuntu@ubuntu-VMware-Virtual-Platform: ~/Downloads'. The terminal shows the following commands and output:

```
ubuntu@ubuntu-VMware-Virtual-Platform:~/Downloads$ sudo -u splunk /opt/splunk/bin/splunk status
splunkd is running (PID: 55314).
splunk helpers are running (PIDs: 55315 55643 55647 55721 55756).
ubuntu@ubuntu-VMware-Virtual-Platform:~/Downloads$ sudo -u splunk /opt/splunk/bin/splunk version
Splunk 10.4.2 (build 33c3bf42cd73)
ubuntu@ubuntu-VMware-Virtual-Platform:~/Downloads$
```

The terminal window has a dark purple background and a sidebar on the left with various application icons. The top of the window shows the date and time as 'Aug 23 00:19'.

Figure 9: Splunk Enterprise version and running service status.

1.11 Splunk Service Verification

After installation, Splunk was started under the dedicated splunk user.

The service was verified using the Splunk command-line interface.

The verification confirmed that the Splunk daemon and web interface were operational.

The Splunk Web interface was made available through:

<http://192.168.10.40:8000>

Successful access to the Splunk Web interface confirmed that the SIEM server was operational and ready for configuration in subsequent phases.

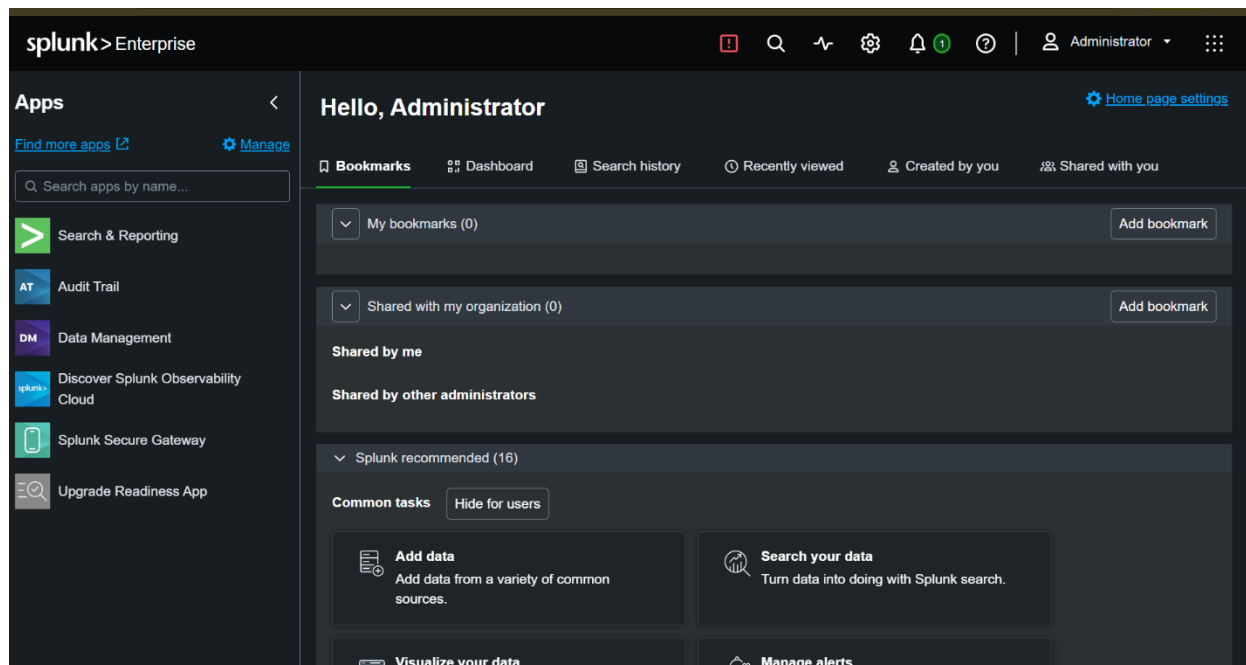


Figure 10: Splunk Web interface successfully accessed through the laboratory network.

1.12 Phase 1 Verification

The following components were successfully established during Phase 1:

Component	Status
VMware virtualization environment	Completed
VMnet1 Host-only network	Completed
VMnet1 DHCP	Disabled
Kali Linux networking	Completed
Windows 10 networking	Completed
Windows Server 2022 networking	Completed
Ubuntu Desktop networking	Completed
Static VMnet1 addressing	Completed
NAT Internet connectivity	Verified
Splunk Enterprise installation	Completed
Splunk service startup	Verified
Splunk Web interface	Verified

The foundational infrastructure required for the subsequent Active Directory and Blue Team monitoring phases is therefore operational.

1.13 Security Considerations

Several configuration decisions were made to maintain an appropriate security boundary for the laboratory.

First, the internal systems communicate through a dedicated Host-only network rather than relying entirely on the physical network. This reduces the possibility of laboratory attack traffic unintentionally reaching external systems.

Second, DHCP was disabled on VMnet1 and static addresses were assigned. This provides predictable addressing and simplifies future Active Directory, DNS, logging, and attack-analysis activities.

Third, the NAT and VMnet1 networks serve different purposes. NAT provides controlled Internet access, while VMnet1 provides internal laboratory communication.

Finally, Splunk was configured to operate using a dedicated splunk account rather than running the application as root.

1.14 Phase 1 Conclusion

Phase 1 successfully established the infrastructure required for the Windows Active Directory Blue Team laboratory.

Four virtual machines were deployed and configured: Windows Server 2022 as the future Domain Controller, Windows 10 as the endpoint, Kali Linux as the attacker system, and Ubuntu Desktop as the Splunk SIEM server.

A dedicated VMware Host-only network using the 192.168.10.0/24 subnet was established with DHCP disabled. Static addresses were assigned to each system to provide predictable internal communication. NAT interfaces were retained separately to provide Internet connectivity when required.

Splunk Enterprise was successfully installed on Ubuntu Desktop and its web interface was verified through the internal laboratory network.

With the infrastructure operational, the laboratory is ready to proceed to Phase 2 — Active Directory Deployment and Configuration, where Windows Server 2022 will be configured as the Domain Controller and the Active Directory environment will be established.

2. Phase 2 — Active Directory Deployment, Security Telemetry and SIEM Integration

2.1 Introduction

Phase 2 focused on transforming the basic virtual network established during Phase 1 into a functional enterprise-style Active Directory environment with centralized security monitoring.

The purpose of this phase was not limited to installing Active Directory Domain Services (AD DS). The objective was to create a controlled Windows domain environment that could support centralized identity management, security policy enforcement, endpoint telemetry collection, authentication monitoring, and later attack-and-detection exercises.

Windows Server 2022 was configured as the Domain Controller for the corp.local domain. A Windows 10 workstation was then integrated into the domain and configured to authenticate using centrally managed Active Directory accounts. Organizational Units (OUs), users, security groups, and a service account were created to represent a simplified corporate identity structure.

Because Active Directory depends heavily on DNS for Domain Controller discovery and authentication services, DNS configuration was also validated and corrected where necessary. Particular attention was given to ensuring that domain clients resolved DC01 through its internal laboratory address rather than through the NAT interface.

After the identity infrastructure was established, the phase shifted toward security monitoring. Group Policy Objects (GPOs) were created to centrally configure advanced Windows auditing, command-line process logging, and PowerShell logging on both the workstation and Domain Controller. Sysmon was additionally deployed to provide enhanced process-level telemetry beyond the standard Windows Security Event Log.

Finally, Splunk Enterprise was used as the centralized Security Information and Event Management (SIEM) platform. Splunk Universal Forwarders were configured on WIN10 and DC01 to transmit Security, PowerShell, and Sysmon telemetry to the Splunk server. The complete monitoring pipeline was validated by observing real process-creation events and Kerberos authentication events generated within the corp.local domain.

By the end of Phase 2, the laboratory had progressed from a collection of connected virtual machines into a monitored Active Directory environment suitable for controlled adversarial testing in Phase 3.

2.2 Objectives

The main objectives of Phase 2 were to:

- Deploy Windows Server 2022 as the Domain Controller for the laboratory.
- Create a new Active Directory forest and domain named corp.local.
- Configure and verify DNS functionality required by Active Directory.
- Design an organized Active Directory structure using Organizational Units.
- Create domain users, security groups, and a dedicated service account.
- Configure WIN10 to use DC01 as its internal DNS server.
- Join the Windows 10 workstation to the corp.local domain.
- Verify domain-user authentication and the workstation's secure channel with the domain.
- Apply centralized security policies through Group Policy.
- Enable Advanced Audit Policy on WIN10 and DC01.
- Enable command-line information in process-creation events.
- Enable PowerShell Script Block Logging.
- Validate Windows Security Event ID 4688 process-creation telemetry.
- Enable Kerberos auditing on the Domain Controller.
- Validate Kerberos Event IDs 4768 and 4769.
- Deploy Sysmon to WIN10 and DC01.
- Validate Sysmon Event ID 1 process-creation telemetry.
- Configure Splunk Enterprise as the centralized SIEM.
- Deploy Splunk Universal Forwarder to the monitored Windows systems.
- Forward Windows Security, Sysmon, and PowerShell logs to Splunk.
- Validate end-to-end Active Directory authentication telemetry within the SIEM.

2.3 Phase 2 Lab Architecture

The Phase 2 laboratory consisted of four virtual machines running within VMware Workstation.

System	Function	Internal IP
DC01	Windows Server 2022 / AD DS / DNS / Domain Controller	192.168.10.10
WIN10	Domain-joined Windows workstation	192.168.10.20
Kali Linux	Attacker system reserved for Phase 3	192.168.10.30
Ubuntu	Splunk Enterprise SIEM	192.168.10.40

The internal laboratory network used VMware Host-Only networking through VMnet1 with the subnet:

192.168.10.0/24

This network provided isolated communication between the Domain Controller, workstation, attacker system, and SIEM server.

NAT connectivity was additionally maintained where Internet access was required for downloading packages, updates, and security tools.

The completed Phase 2 architecture followed the logical flow:

2.4 Active Directory Domain Services Deployment

2.4.1 Windows Server 2022 Baseline Configuration

Before installing Active Directory services, the Windows Server 2022 virtual machine was reviewed to establish its initial baseline configuration.

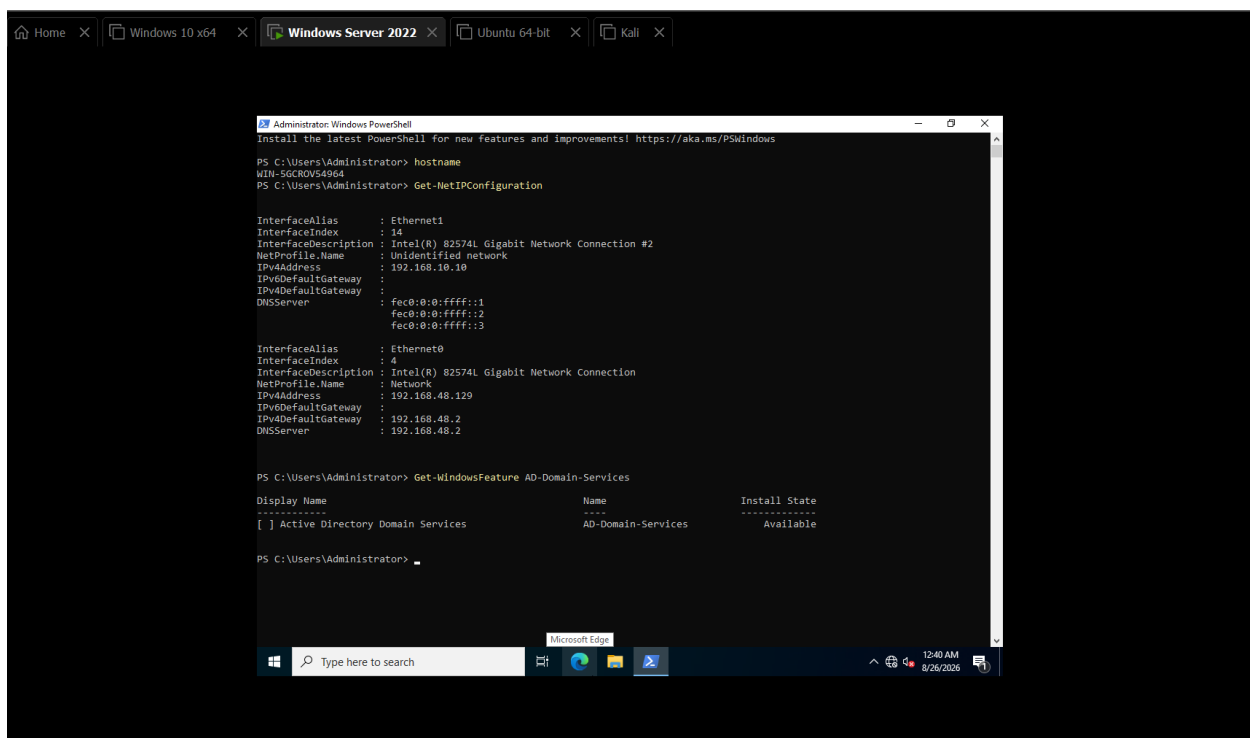
The server contained separate network interfaces for internal laboratory communication and NAT-based Internet connectivity. The internal interface was configured to operate on the 192.168.10.0/24 Active Directory laboratory network.

A stable internal IP address was required because the Domain Controller would later provide authentication, DNS, Group Policy, and other directory services to domain members.

The intended internal address for DC01 was:

192.168.10.10

Using a static address prevents domain clients from losing access to critical services due to an unexpected IP address change.



The screenshot shows a Windows Server 2022 virtual machine environment with several tabs open: Home, Windows 10 x64, Windows Server 2022 (active), Ubuntu 64-bit, and Kali. The PowerShell console is running as Administrator. It displays the following commands and output:

```
PS C:\Users\Administrator> hostname
WIN-SGCR0V54964
PS C:\Users\Administrator> Get-NetIPConfiguration
```

The output of `Get-NetIPConfiguration` shows two network interfaces:

InterfaceAlias	InterfaceIndex	InterfaceDescription	NetProfile.Name	IPv4Address	IPv4DefaultGateway	IPv6Address	IPv6DefaultGateway	DNSServer
Ethernet1	14	Intel(R) 82574L Gigabit Network Connection #2	Unidentified network	192.168.10.10				fec0:0:0:ffff::1 fec0:0:0:ffff::2 fec0:0:0:ffff::3
Ethernet0	4	Intel(R) 82574L Gigabit Network Connection	Network	192.168.48.129				192.168.48.2

Next, the command `Get-WindowsFeature AD-Domain-Services` is executed, showing the following table:

Display Name	Name	Install State
[] Active Directory Domain Services	AD-Domain-Services	Available

The PowerShell prompt is currently at `PS C:\Users\Administrator>`.

Figure 1: Initial Windows Server 2022 / DC01 baseline configuration.

The server hostname and network configuration were then verified before any Active Directory roles were installed. This established that the server was operating with the expected identity and addressing configuration.

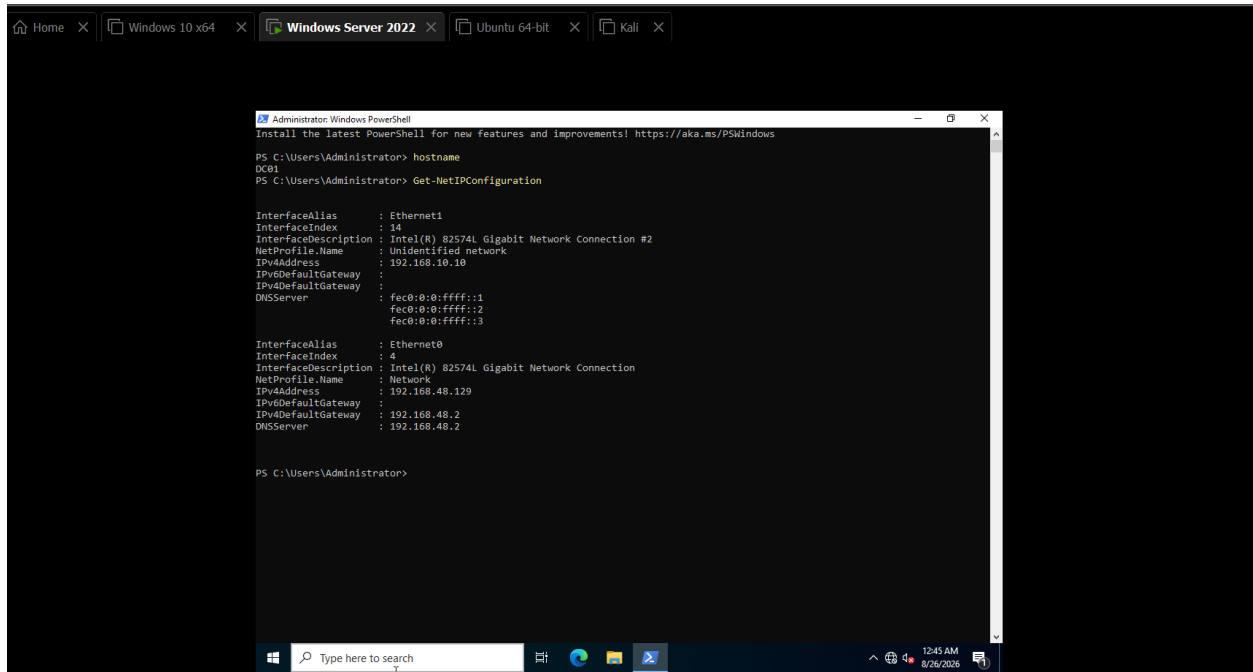


Figure 2: DC01 hostname and network baseline verification.

2.4.2 Installing Active Directory Domain Services

Active Directory Domain Services is the Windows Server role responsible for maintaining directory information about users, computers, groups, and other domain objects.

AD DS provides centralized authentication and authorization, allowing identities and security policies to be managed from the Domain Controller rather than independently on every workstation.

The AD DS role was installed on the Windows Server 2022 system together with the required supporting components.

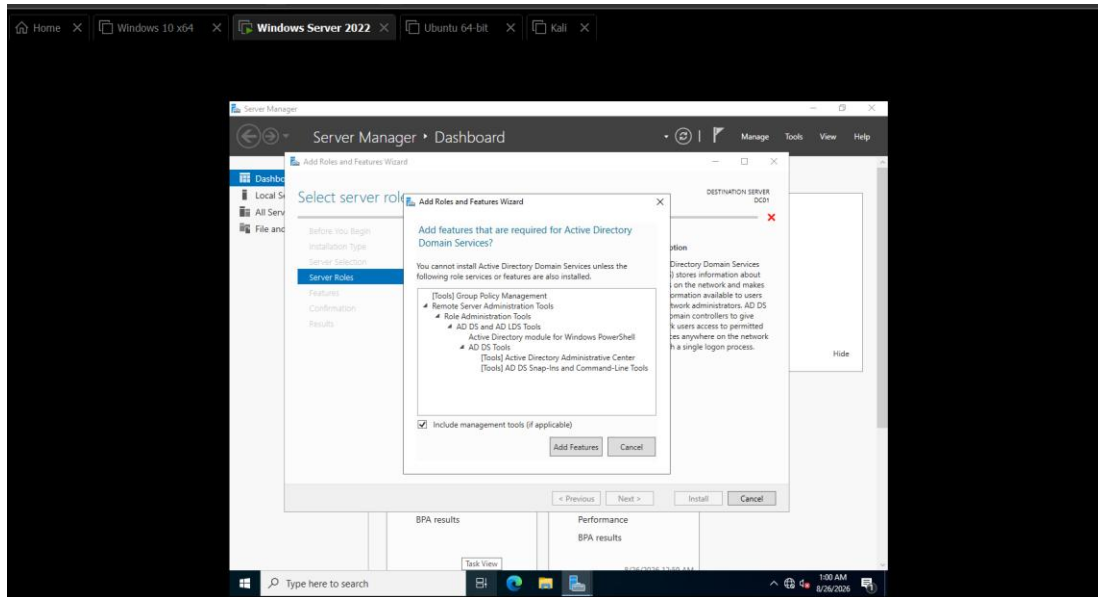


Figure 3: Active Directory Domain Services role installation and configuration.

After the installation completed, the server was checked to verify that the AD DS role was successfully installed and available for Domain Controller promotion.

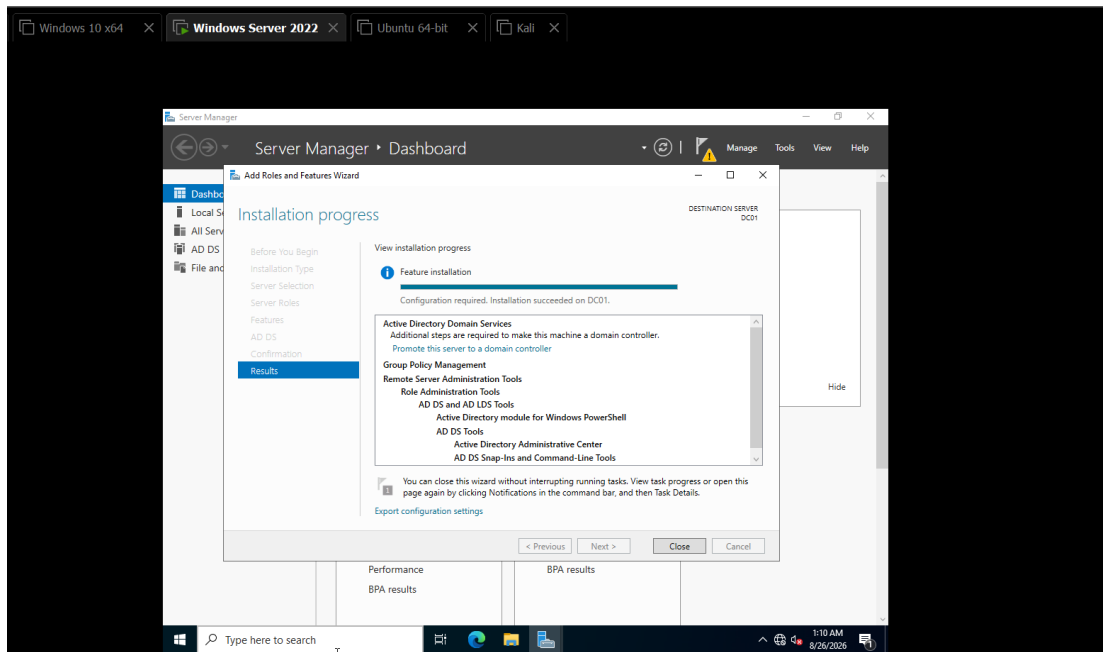


Figure 4: Verification of the Active Directory Domain Services installation.

2.4.3 Creating the corp.local Forest

Installing AD DS does not automatically make a Windows Server a Domain Controller. The server must subsequently be promoted into an existing domain or used to create a new Active Directory forest.

Because this laboratory represented a new organization, a new forest was created using:
corp.local

The forest represents the highest-level logical structure within Active Directory, while corp.local became the root domain for the laboratory.

DC01 was configured as the first Domain Controller within this new forest.

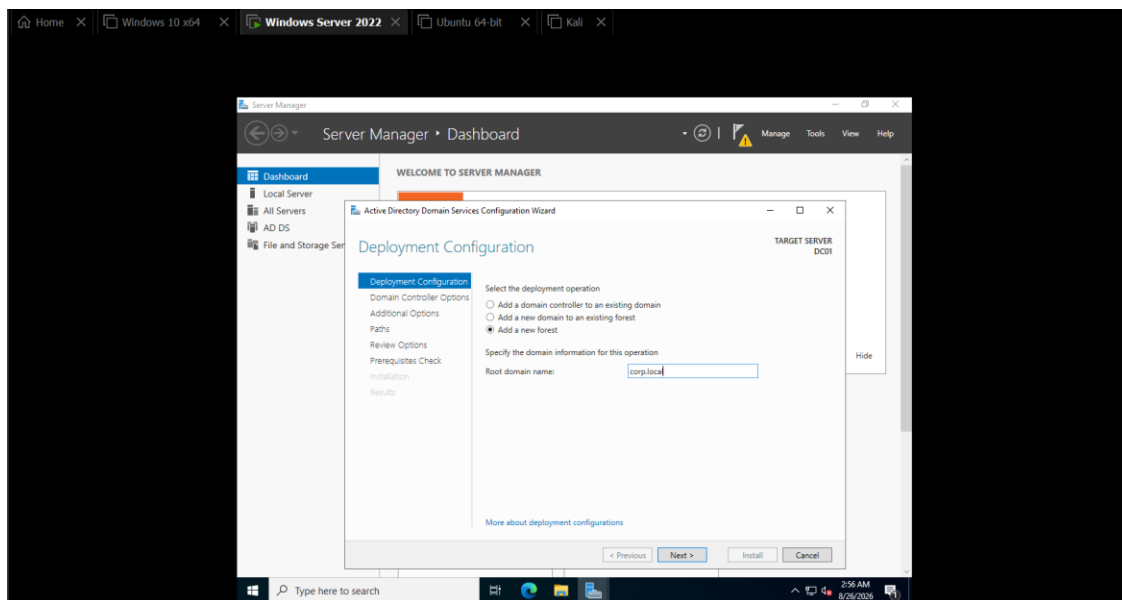


Figure 5: Domain Controller promotion and new forest configuration for corp.local.

The promotion process configured the required Active Directory database, domain configuration, SYSVOL structure, DNS integration, and supporting Domain Controller services.

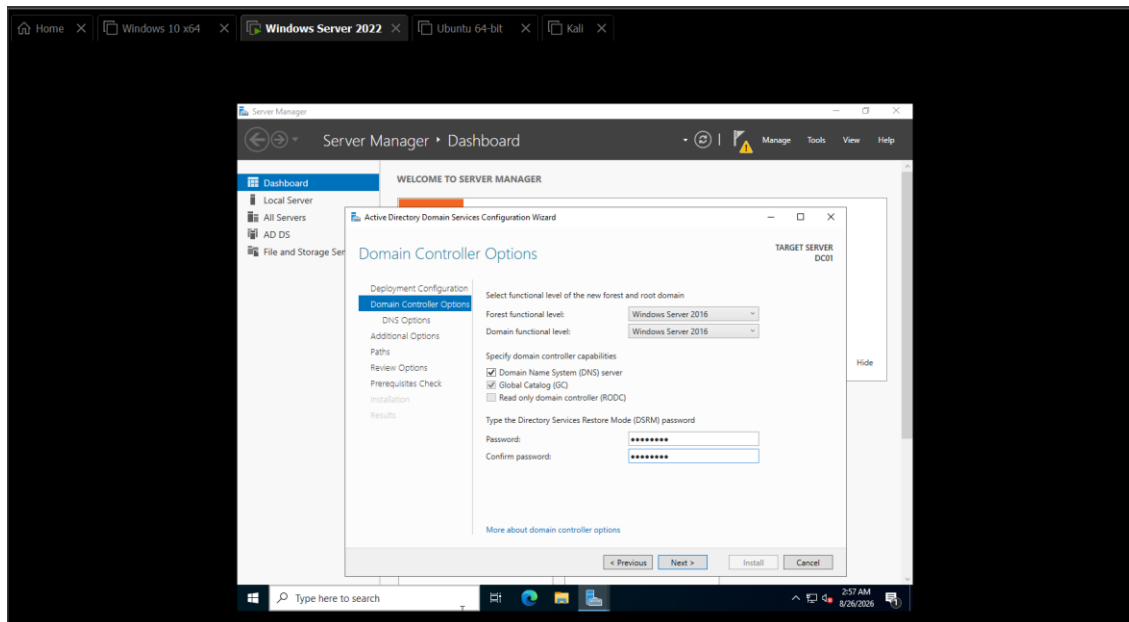


Figure 6: corp.local domain creation and Domain Controller promotion configuration.

After completing the promotion, the server restarted so that the new Active Directory configuration could become fully operational.

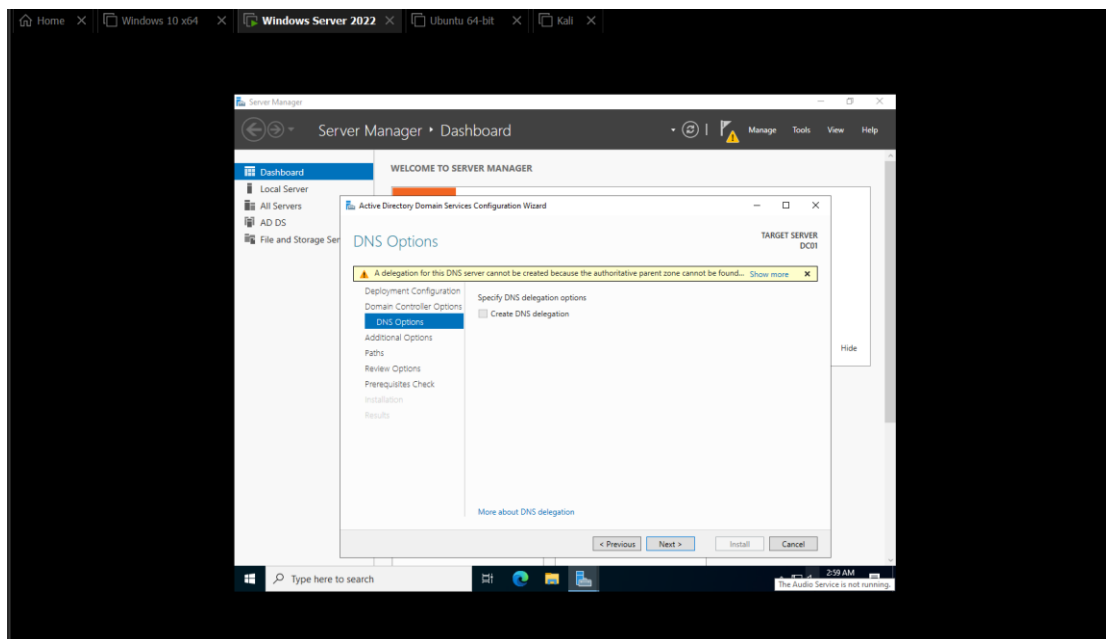


Figure 7: DC01 restart and Domain Controller initialization following promotion.

2.4.4 Domain Authentication and Post-Promotion Verification

Following the restart, authentication was performed within the newly created domain context.

Instead of operating solely as the original local server Administrator, the administrative session was established using the CORP domain.

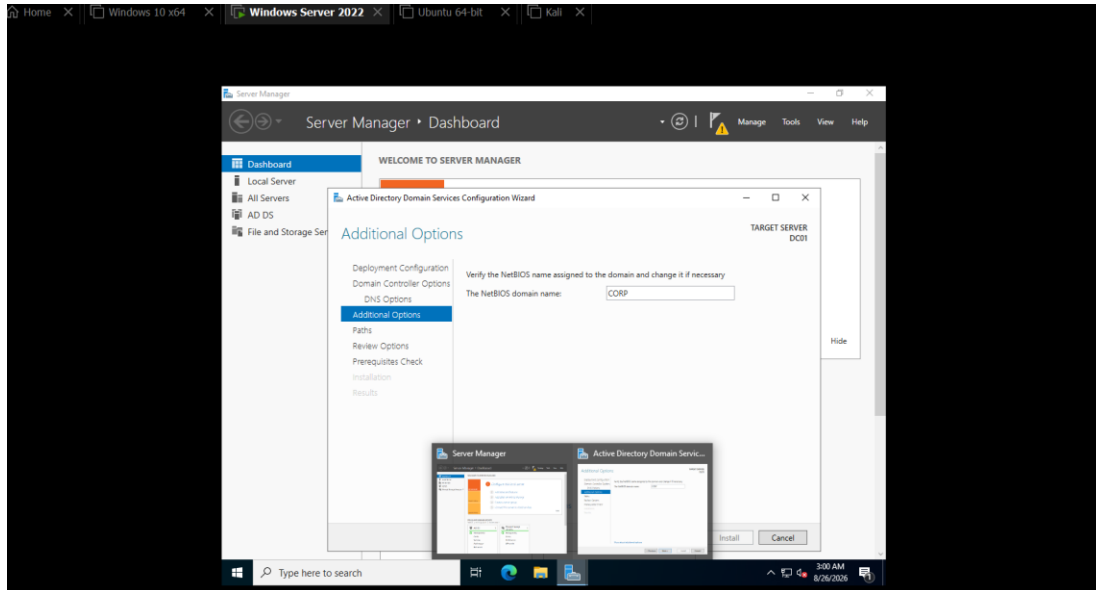


Figure 8: Successful login to the newly created CORP domain as Administrator.

The Active Directory Domain Services configuration was then reviewed to confirm that DC01 had successfully transitioned into its Domain Controller role.

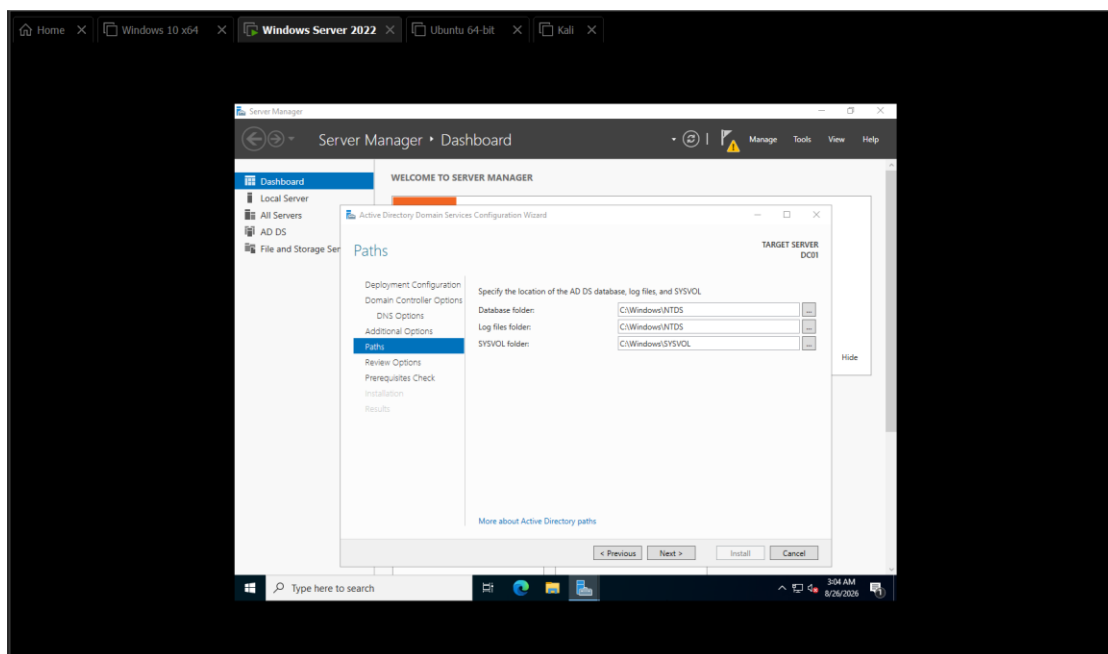


Figure 9: AD DS and Domain Controller post-promotion verification.

The corp.local environment was subsequently inspected using Active Directory management tools. The presence of the newly created domain and its default directory containers confirmed that the forest and domain had initialized successfully.

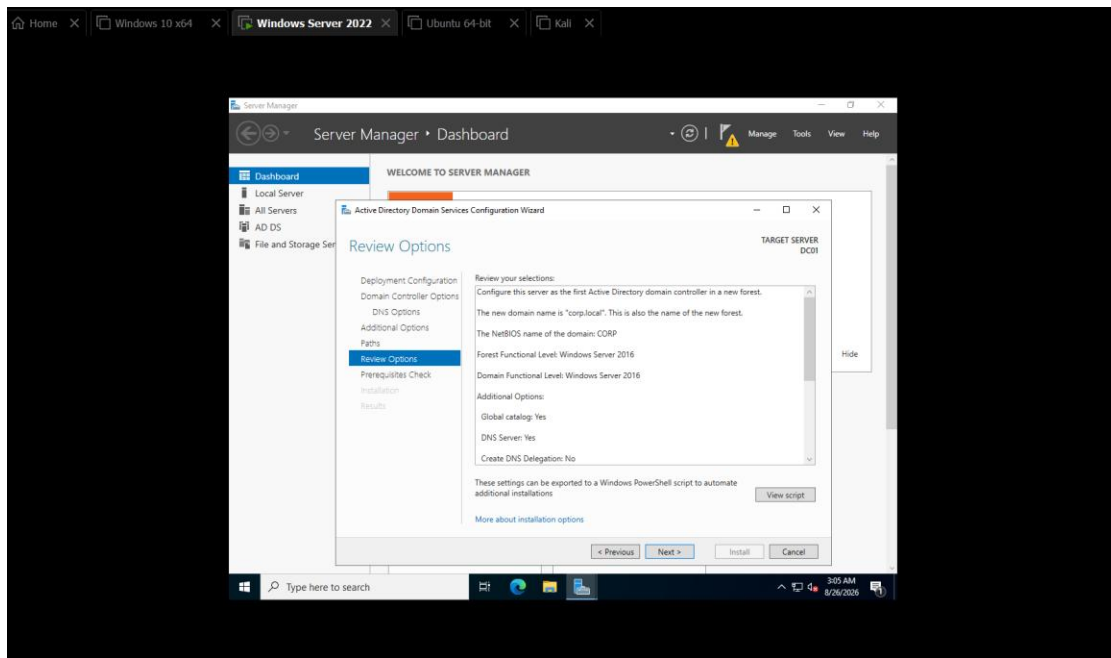


Figure 10: Verification of the corp.local Active Directory environment.

2.5 DNS Configuration and Active Directory Name Resolution

2.5.1 Role of DNS in Active Directory

DNS is a fundamental dependency of Active Directory.

In a traditional network, DNS is commonly associated with converting hostnames into IP addresses. Within Active Directory, however, DNS also allows clients to locate services such as:

- Domain Controllers
- LDAP services
- Kerberos authentication servers
- Global Catalog servers

Active Directory automatically registers special Service Resource Records (SRV records) within DNS. Domain members query these records when locating authentication and directory services.

For example:

_ldap._tcp.dc._msdcs.corp.local

can be used to locate Domain Controllers providing LDAP services for the domain.

Incorrect DNS configuration can therefore cause domain joins, authentication, Group Policy, and Kerberos communication to fail even when basic network connectivity remains functional.

2.5.2 DC01 DNS Verification

Following Domain Controller promotion, DNS configuration was inspected on DC01.

Because the first Domain Controller also hosted DNS services for the laboratory, the DNS role and associated configuration were verified before connecting domain clients.

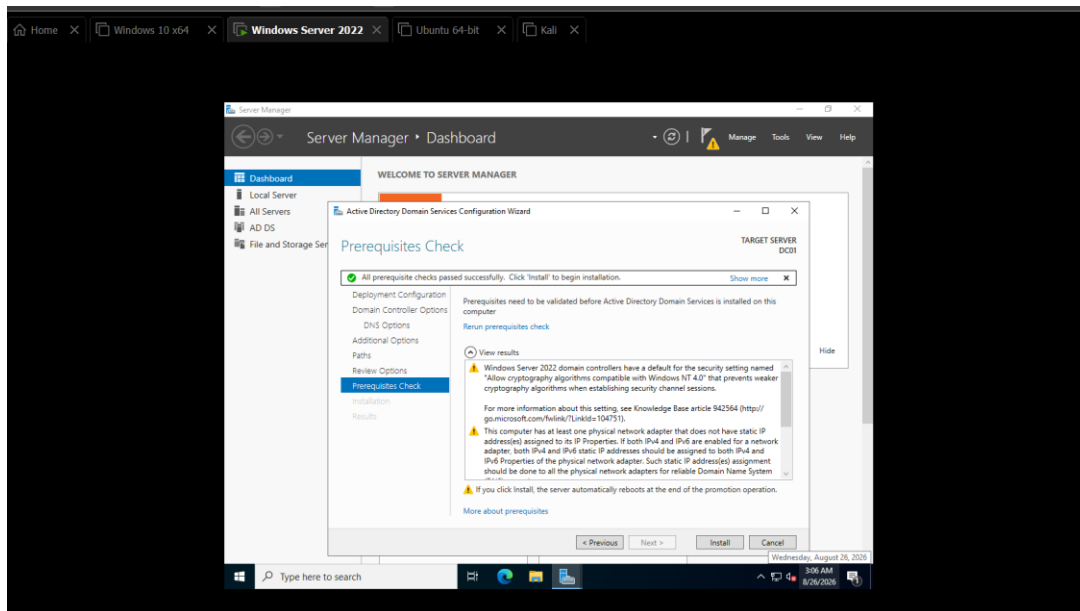


Figure 11: DC01 DNS configuration and DNS role verification.

The DNS Manager was then examined to verify the presence of the corp.local zone and the records automatically generated during Active Directory deployment.

2.6 Active Directory Organizational Structure

2.6.1 Initial Domain Structure

The Active Directory Users and Computers console was opened to inspect the default structure created during Domain Controller promotion.

Active Directory initially provides containers such as Users, Computers, Builtin, and Domain Controllers. Although these locations are sufficient for basic operation, placing every custom object inside default containers would not represent a well-organized enterprise environment.

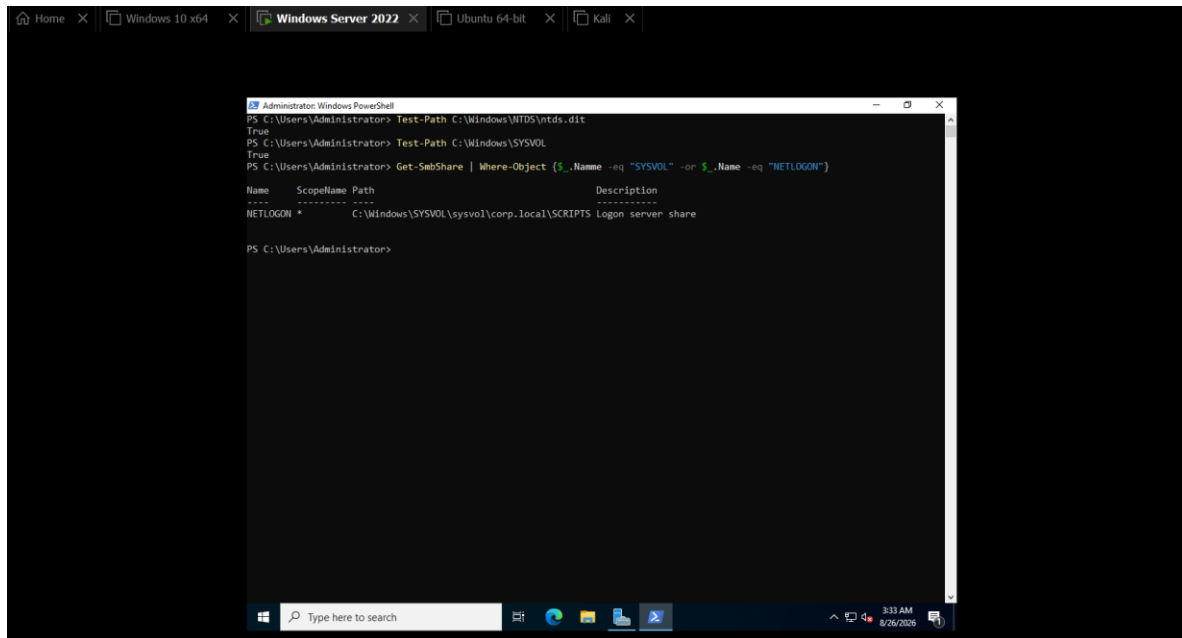


Figure 14: Initial corp.local structure displayed in Active Directory Users and Computers.

2.6.2 Creating Organizational Units

Organizational Units are logical containers used to organize Active Directory objects.

They are particularly useful for:

- Organizing users and computers.
- Delegating administrative responsibilities.
- Applying targeted Group Policy Objects.
- Separating different types of systems and identities.

A custom Organizational Unit structure was therefore created for the corp.local laboratory.

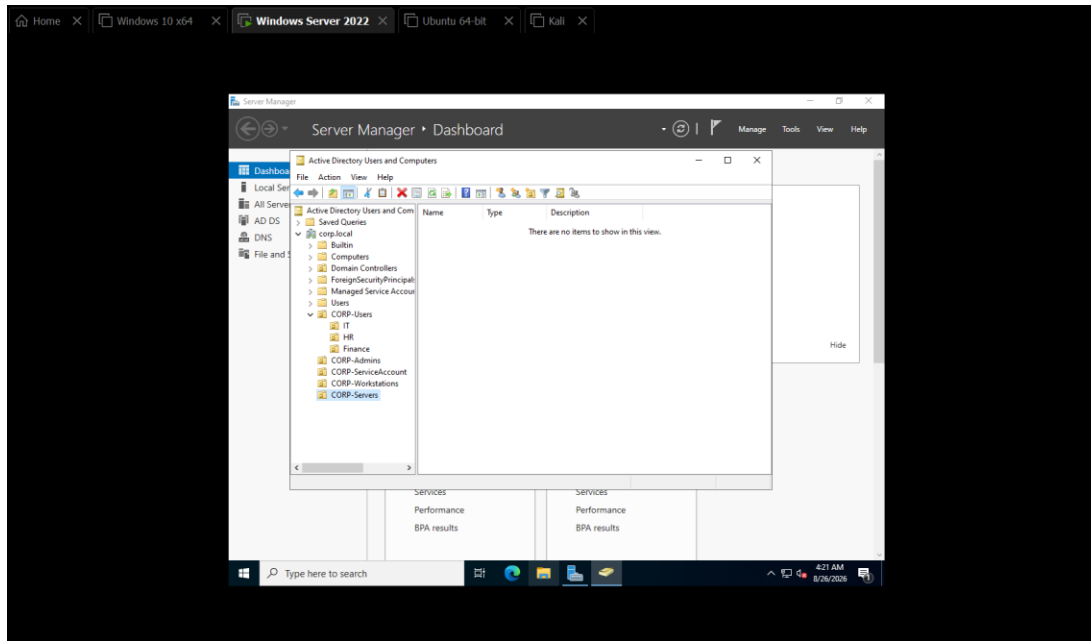


Figure 15: Creation of the custom Active Directory Organizational Unit structure.

The resulting hierarchy included dedicated locations for workstations, users, groups, and service accounts rather than storing all objects inside the default containers.

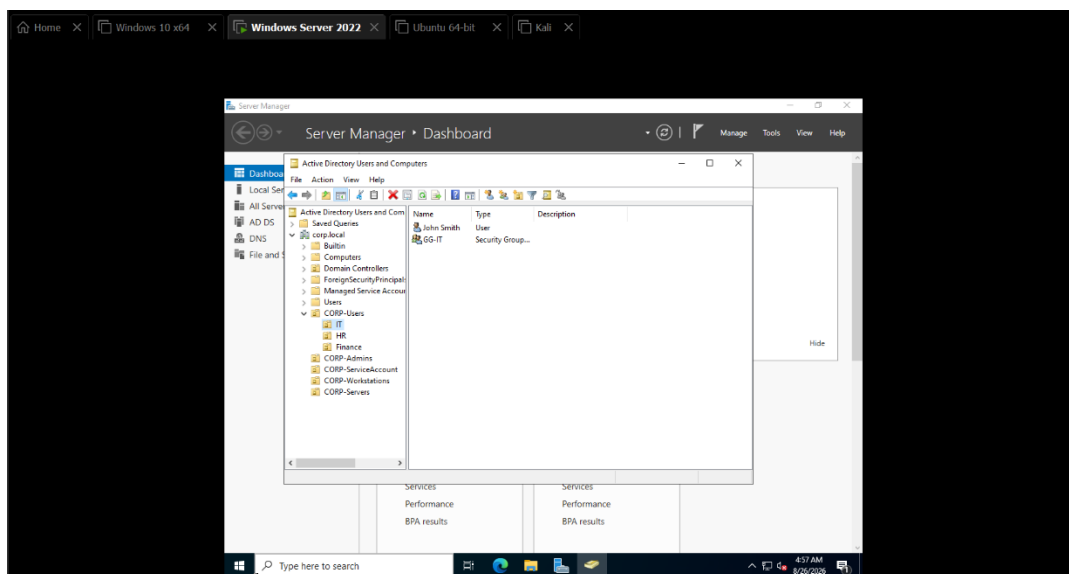


Figure 16: Verification of the completed CORP Organizational Unit hierarchy.

This structure later became particularly important when different security policies were applied to workstations and Domain Controllers.

2.7 Domain Users, Groups, and Service Accounts

2.7.1 Creating Domain Users

Domain user accounts provide centrally managed identities that can authenticate to domain-joined systems and access authorized resources.

Initial domain users were created and placed within the appropriate Organizational Units.

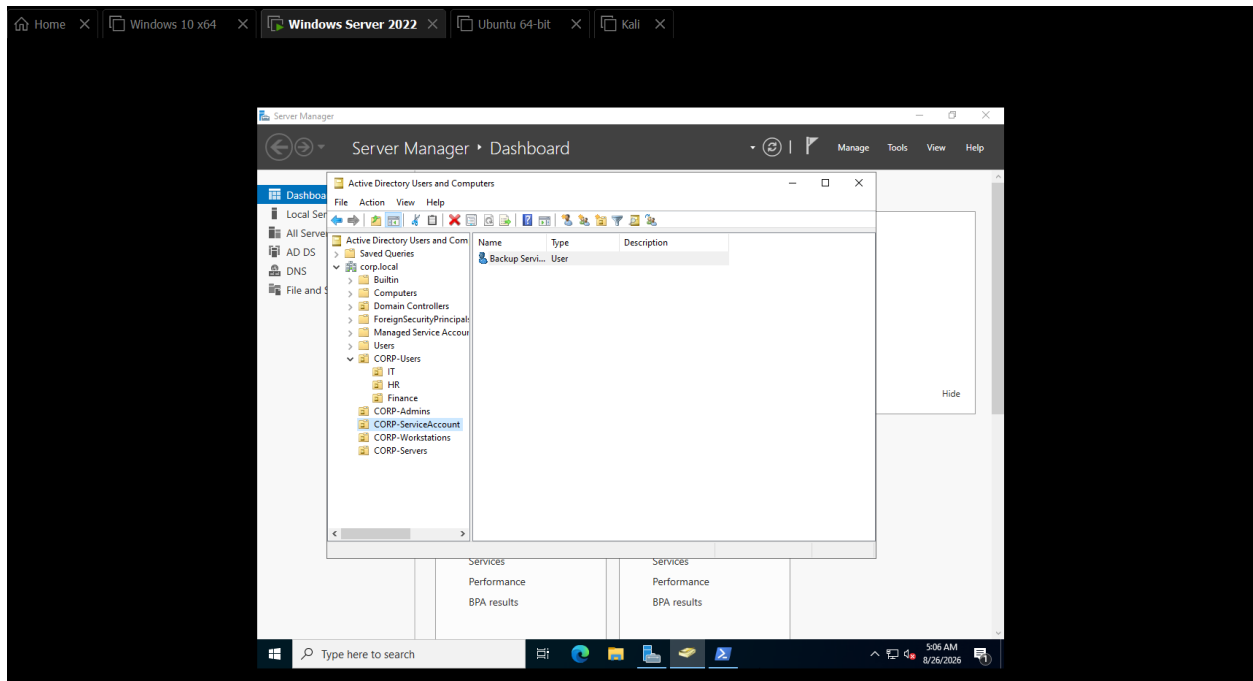


Figure 17: Initial domain user and account creation.

The created accounts and their OU placement were then reviewed through Active Directory Users and Computers.

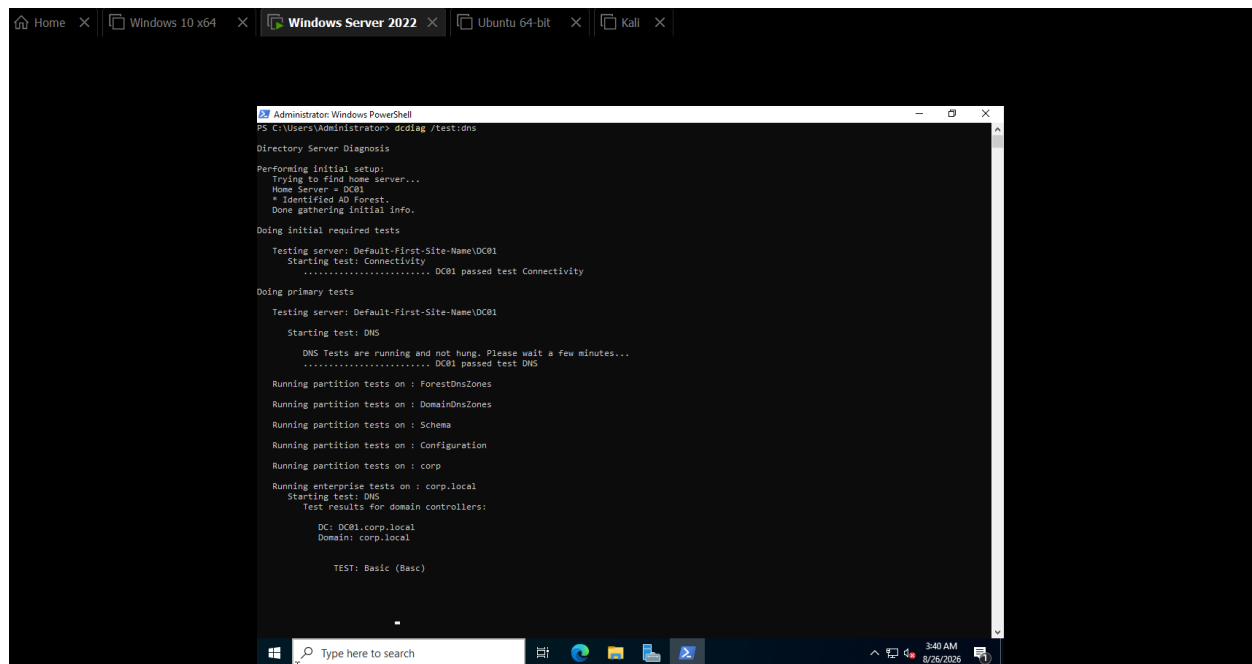


Figure 18: Verification of Active Directory users and OU configuration.

2.7.2 DNS Record Verification and Multi-NIC Considerations

During configuration, additional attention was required because DC01 contained both an internal Host-Only interface and a NAT interface.

The two addresses included:

Internal VMnet1: 192.168.10.10

NAT interface: 192.168.48.x

For Active Directory communication, domain clients needed to resolve DC01 to its internal address:

192.168.10.10

Allowing the NAT interface to register itself in the corp.local DNS zone could cause clients to receive an inappropriate address when attempting to locate the Domain Controller.

The DNS registration was therefore reviewed and corrected so that the appropriate internal address represented DC01 within the Active Directory environment.

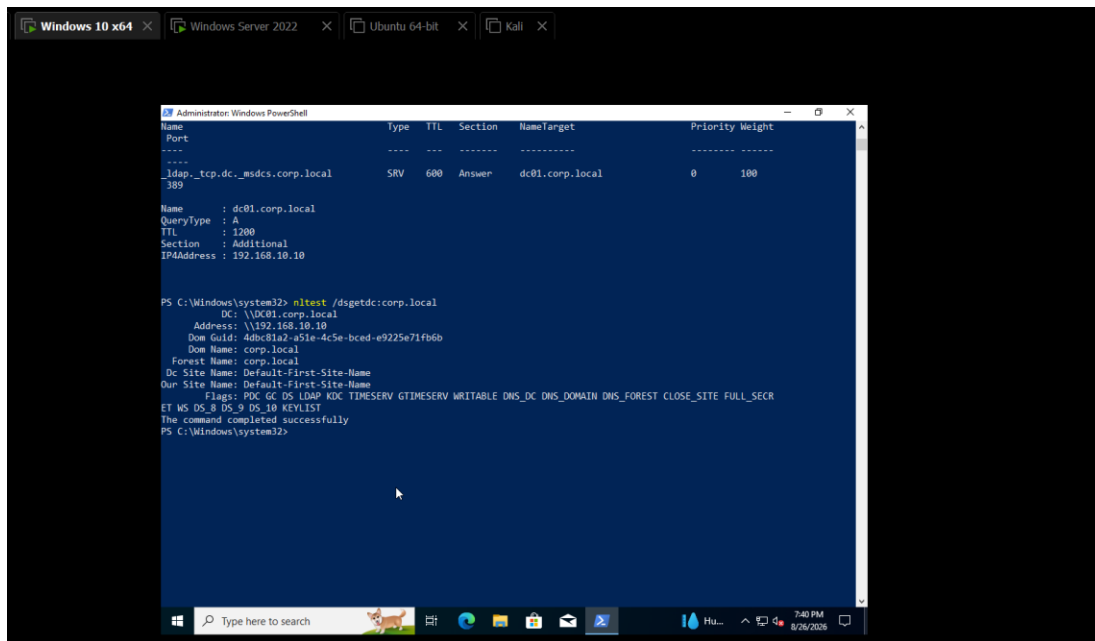


Figure 19: DNS record verification for DC01 within the corp.local domain.

The resulting AD and DNS configuration was then tested to ensure that domain services continued operating correctly.

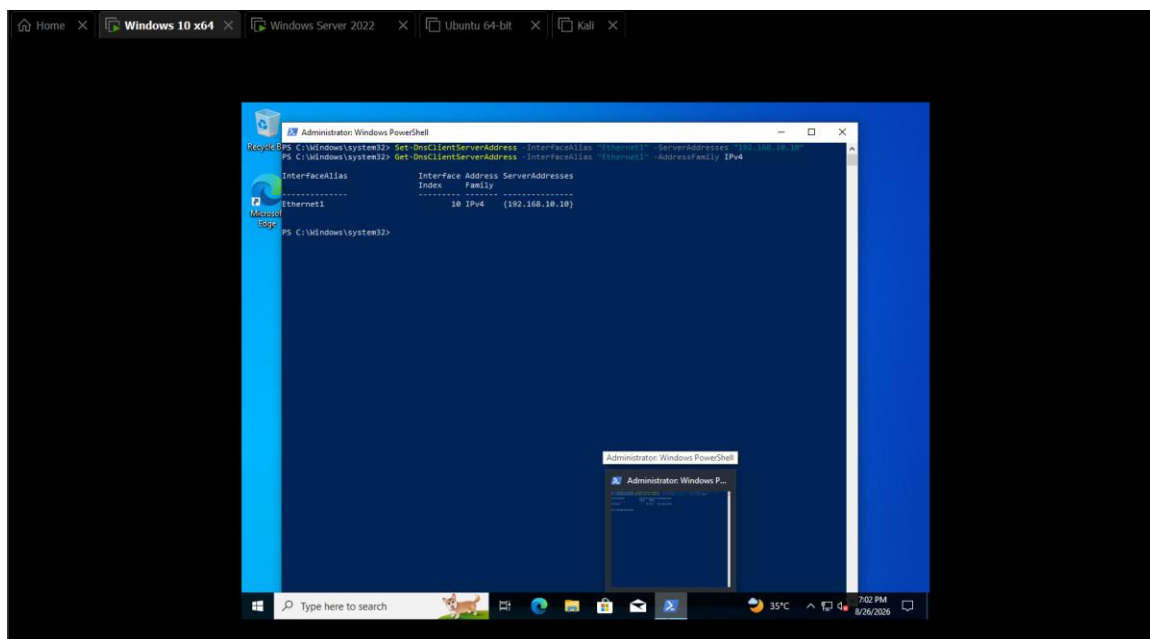


Figure 20: Active Directory and DNS configuration verification following network/DNS configuration.

This configuration allowed Internet connectivity to remain available through NAT without allowing the NAT interface to interfere with internal Active Directory service discovery.

2.7.3 Additional Directory Configuration

Following the DNS correction, additional Active Directory objects and organizational settings were completed.

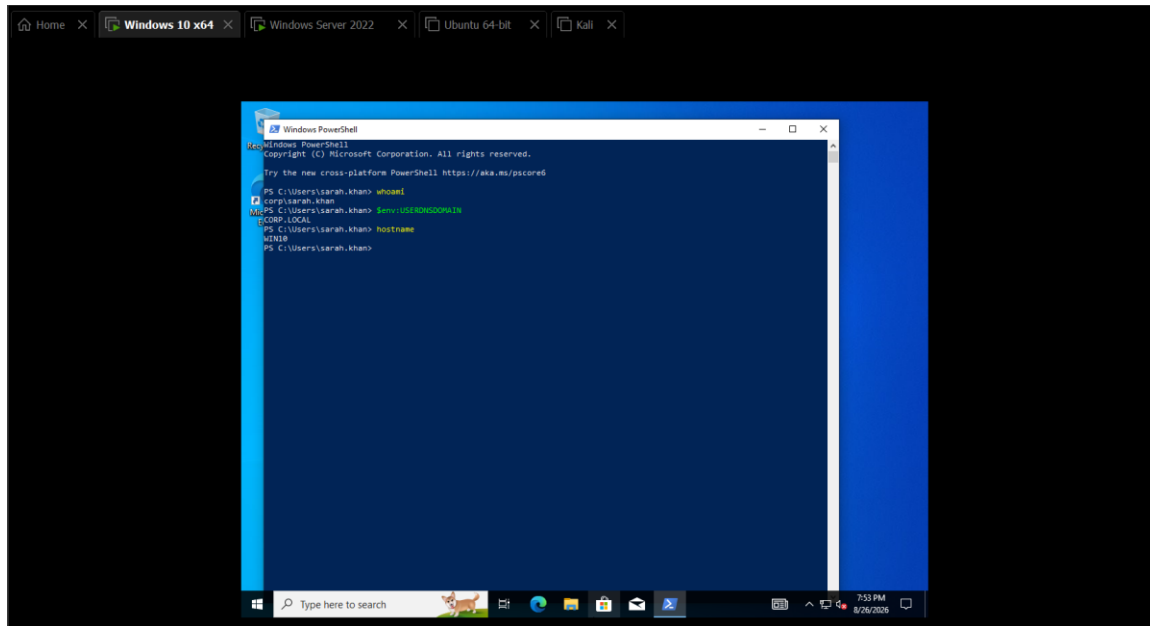


Figure 21: Additional Active Directory organizational and user configuration.

The created domain users were then verified within their intended Organizational Units.

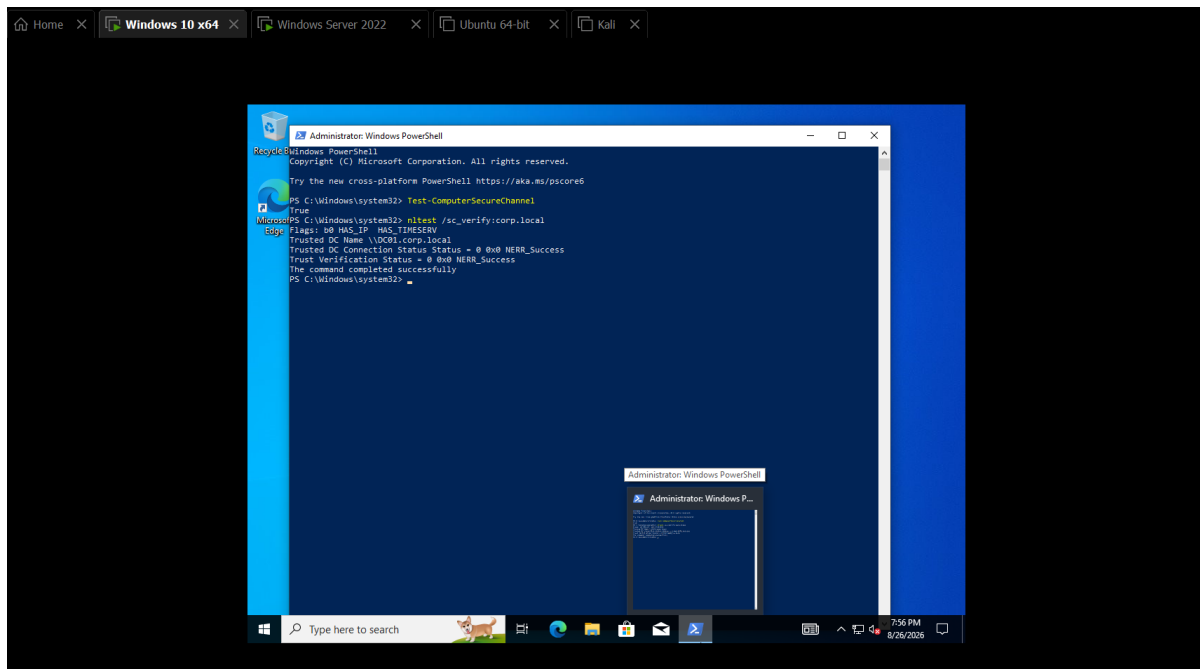


Figure 22: Domain user accounts configured within the appropriate Organizational Units.

2.7.4 Security Groups

Security groups provide a scalable method for controlling permissions.

Instead of assigning permissions individually to every user, accounts can be added to security groups and permissions can subsequently be granted to the group.

Dedicated security groups were created within the laboratory and appropriate users were assigned as members.

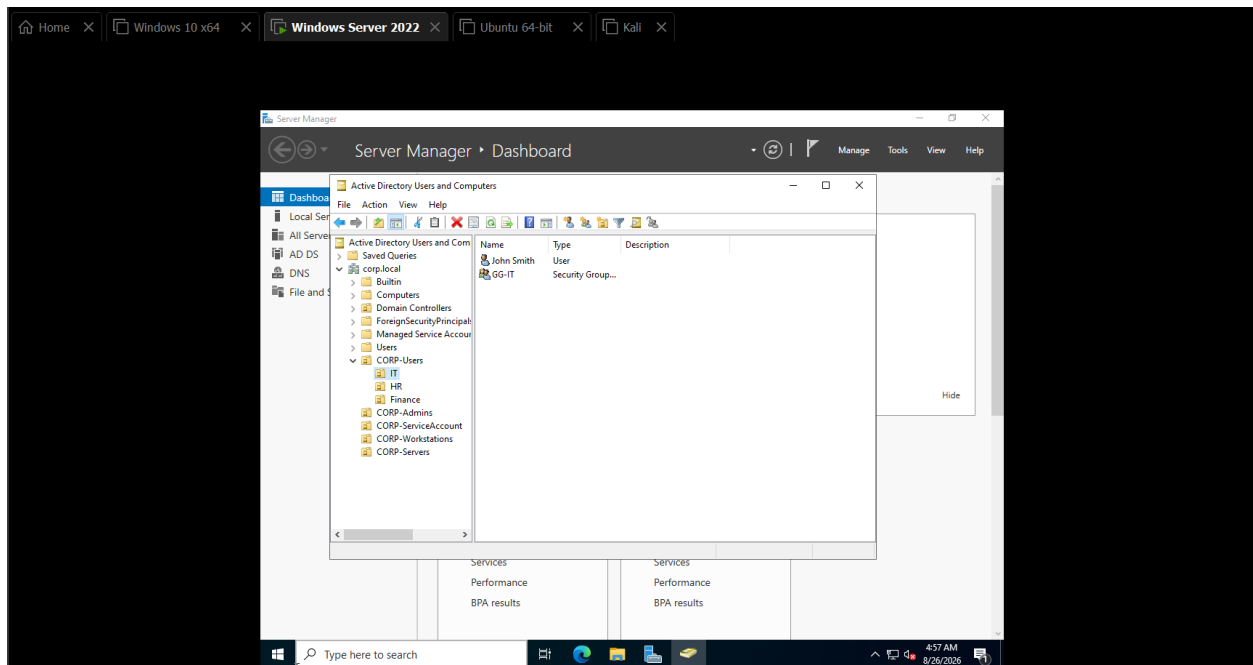


Figure 23: Active Directory security groups and group-membership configuration.

This structure also created useful security telemetry for later phases because modifications to privileged or sensitive groups can be monitored as potential indicators of unauthorized privilege changes.

2.7.5 Service Account

A dedicated service account was created to represent a non-human identity used by applications or background services.

Service accounts are important from a defensive perspective because they commonly possess predictable usage patterns. Interactive authentication or unusual activity involving a service identity can therefore warrant investigation.

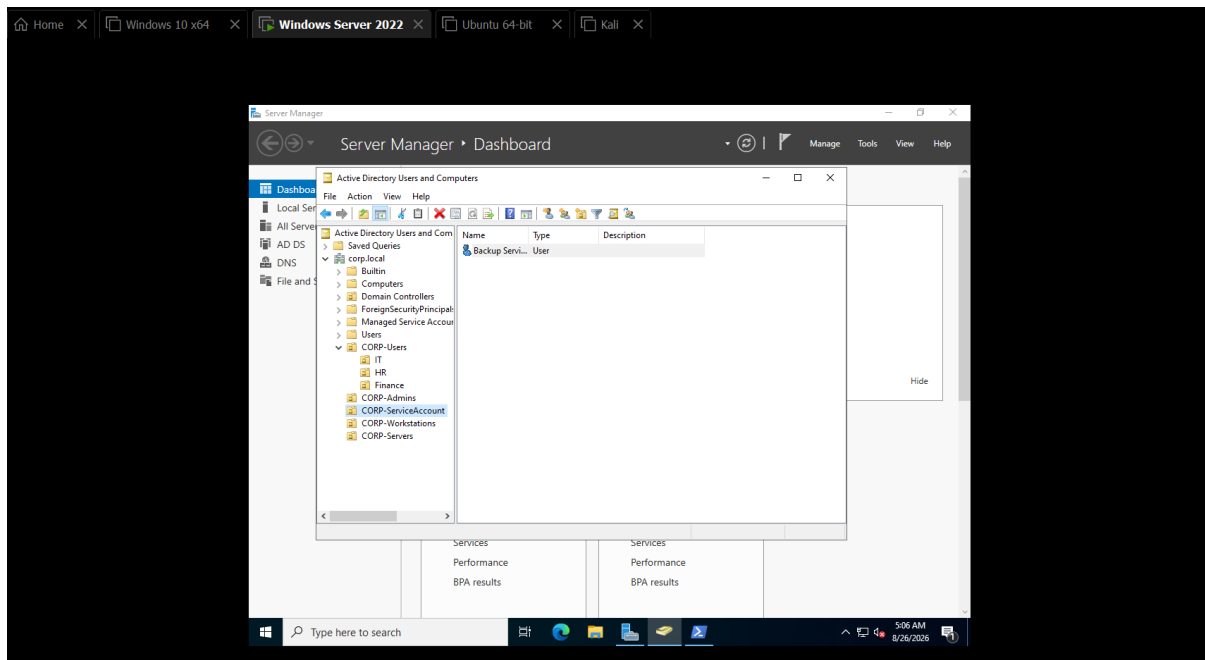


Figure 24: Active Directory service-account configuration.

Following the creation of users, groups, and the service identity, the overall directory structure was reviewed.

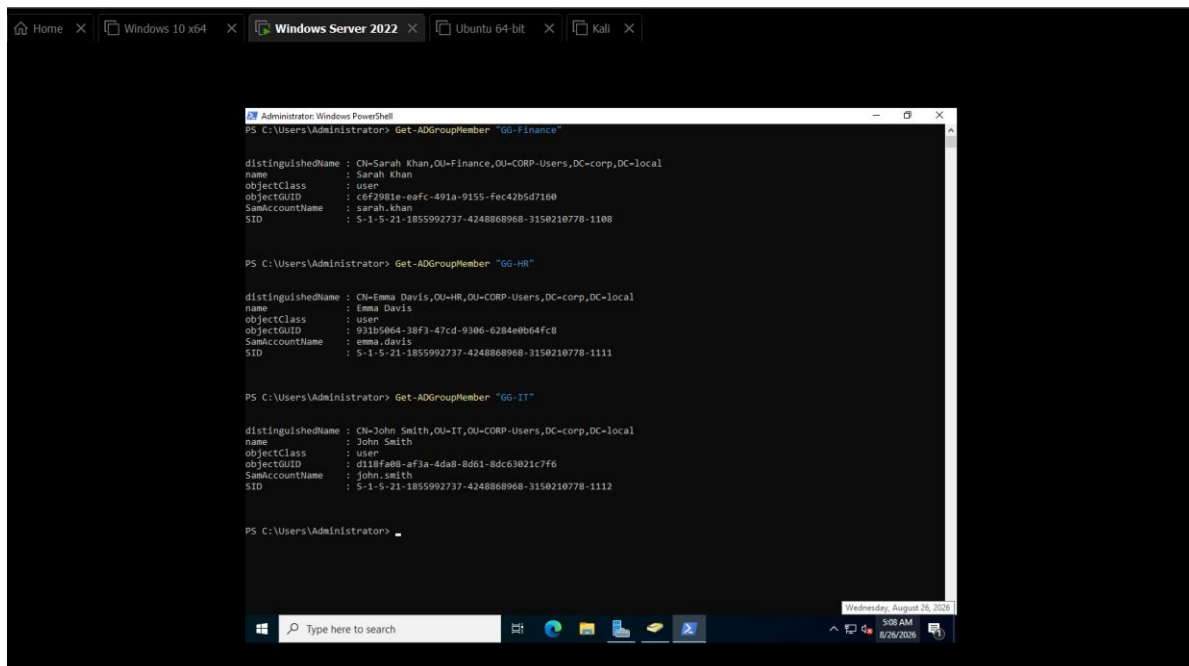


Figure 25: Final Active Directory users, groups, and service-account structure verification.

At this stage, the server-side identity infrastructure was complete and the Windows workstation could be integrated into the domain.

2.8 Windows 10 Domain Integration

2.8.1 Configuring WIN10 DNS

Before WIN10 could join the domain, its internal DNS configuration needed to point toward the Domain Controller.

WIN10 was configured with the internal address:

192.168.10.20

and its DNS server was configured as:

192.168.10.10

This ensured that queries for corp.local and its Active Directory service records were handled by DC01.

Ip conf

Figure 26: WIN10 DNS configured to use DC01 at 192.168.10.10.

2.8.2 Domain Discovery

Before performing the domain join, connectivity and Active Directory discovery were verified from WIN10.

The workstation successfully discovered:

DC01.corp.local

corp.local

confirming that DNS and network communication between the workstation and Domain Controller were functioning correctly.

(ping)

Figure 27: WIN10 successfully discovering DC01.corp.local and the corp.local domain.

2.8.3 Joining the Workstation to the Domain

WIN10 was then joined to:

corp.local

During this process, the Domain Controller authenticated the administrative credentials and established a computer identity for WIN10 within Active Directory.

A domain-joined computer is itself an Active Directory security principal and maintains a machine account password that is used as part of its trusted relationship with the domain.

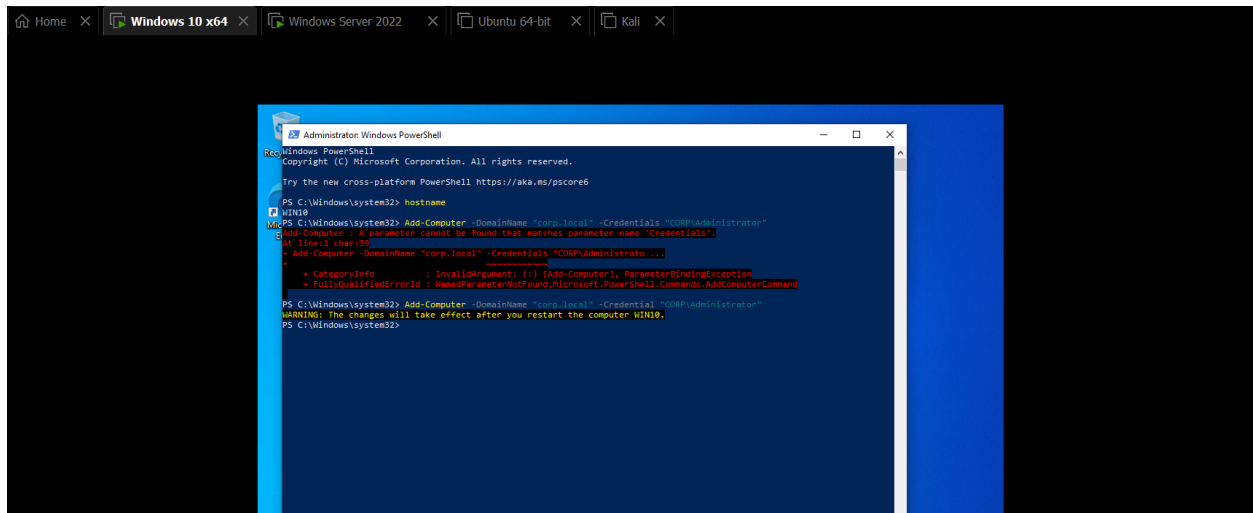


Figure 28: WIN10 successfully joined to the corp.local domain.

2.8.4 Domain User Authentication

Following the domain join and workstation restart, a standard domain user account was used to authenticate to WIN10.

The account:

CORP\sarah.khan

successfully logged into the workstation.

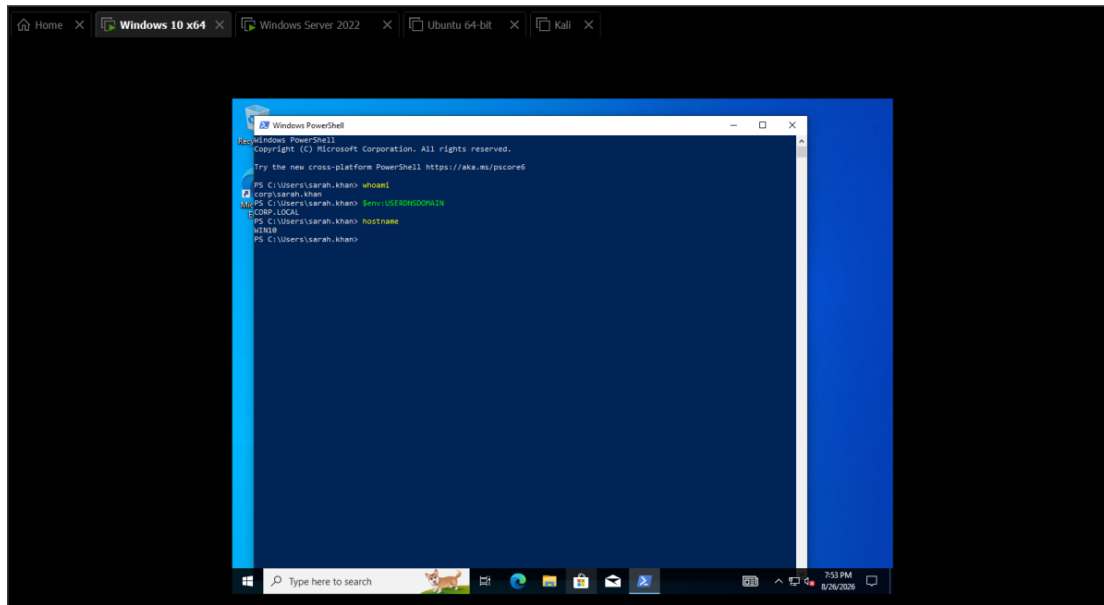


Figure 29: Successful domain-user authentication as CORP\sarah.khan on WIN10.

This demonstrated that the workstation could use identities centrally managed by DC01 rather than relying solely on local Windows accounts.

2.8.5 Secure Channel Verification

Domain-joined workstations maintain a secure channel with their Active Directory domain.

This secure channel allows the workstation and Domain Controller to establish trusted communication using the workstation's machine account.

The WIN10 secure channel was tested and successfully verified.

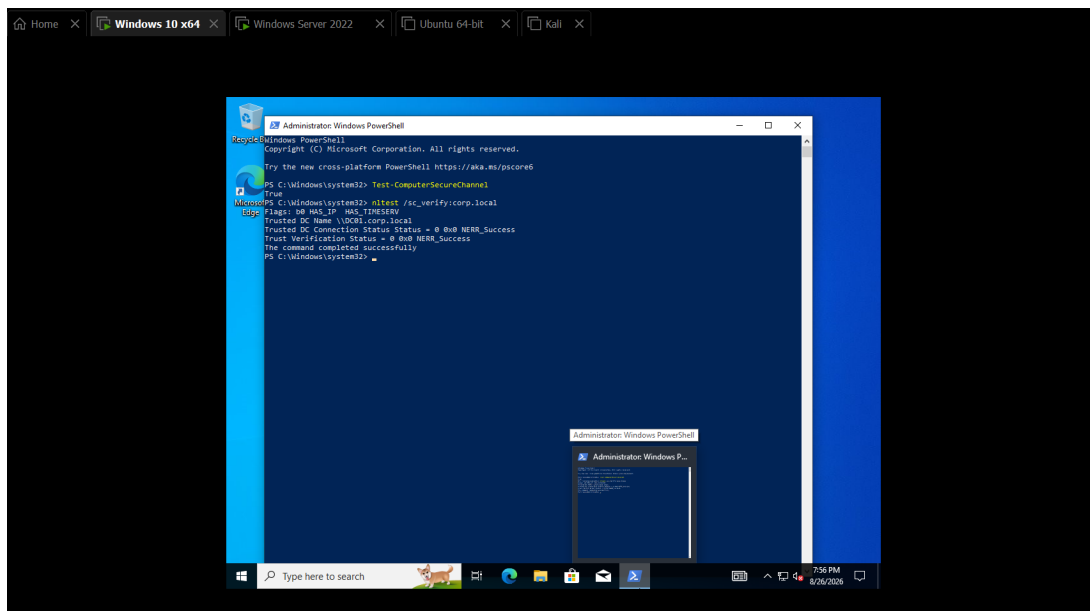


Figure 30: Successful secure-channel verification between WIN10 and corp.local.

2.8.6 Moving WIN10 to the Workstation OU

After the domain join, the WIN10 computer object was moved into:

CORP-Workstations

Placing the workstation inside the dedicated OU allowed workstation-specific Group Policy Objects to be centrally targeted at the endpoint.

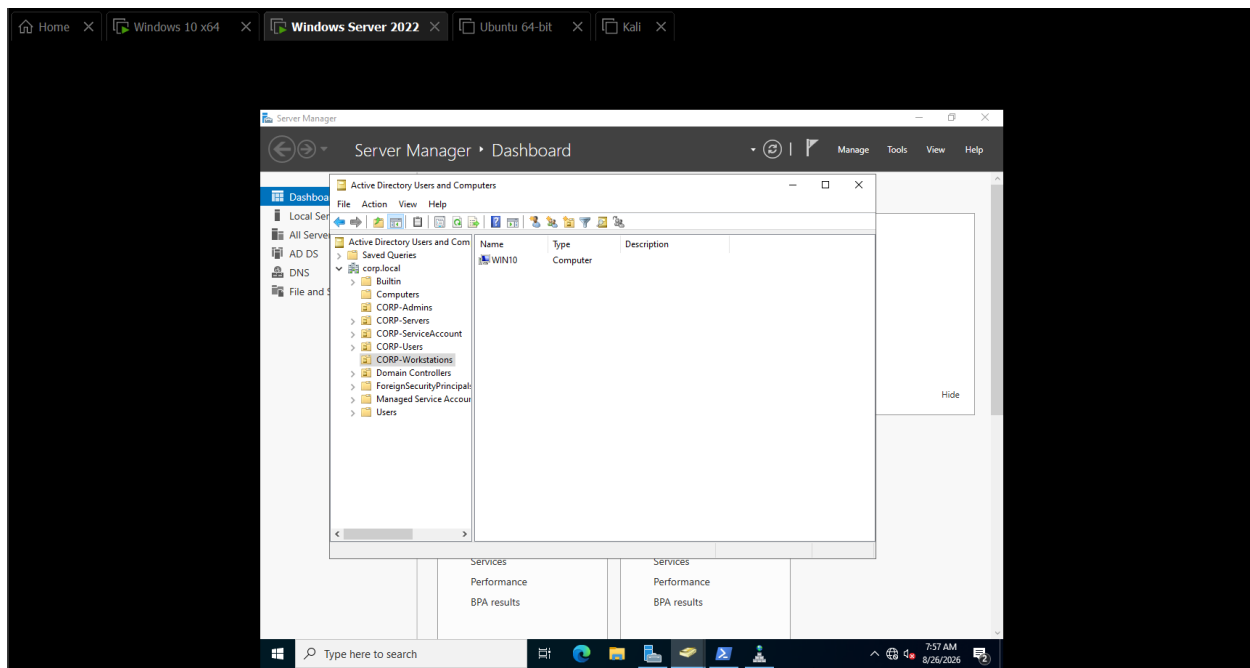


Figure 31: WIN10 computer object moved into the CORP-Workstations Organizational Unit.

2.9 Workstation Security Auditing and Group Policy

2.9.1 Purpose of Centralized Security Auditing

Once WIN10 was successfully integrated into Active Directory, the next objective was to improve endpoint visibility.

Default Windows logging provides useful information, but a SOC environment requires more detailed telemetry to investigate authentication activity, process execution, PowerShell usage, and other security-relevant actions.

Rather than manually configuring each workstation, Group Policy was used to centrally enforce the required audit configuration.

A dedicated Group Policy Object named:

CORP-BlueTeam-Audit-Policy

was created and linked to:

CORP-Workstations

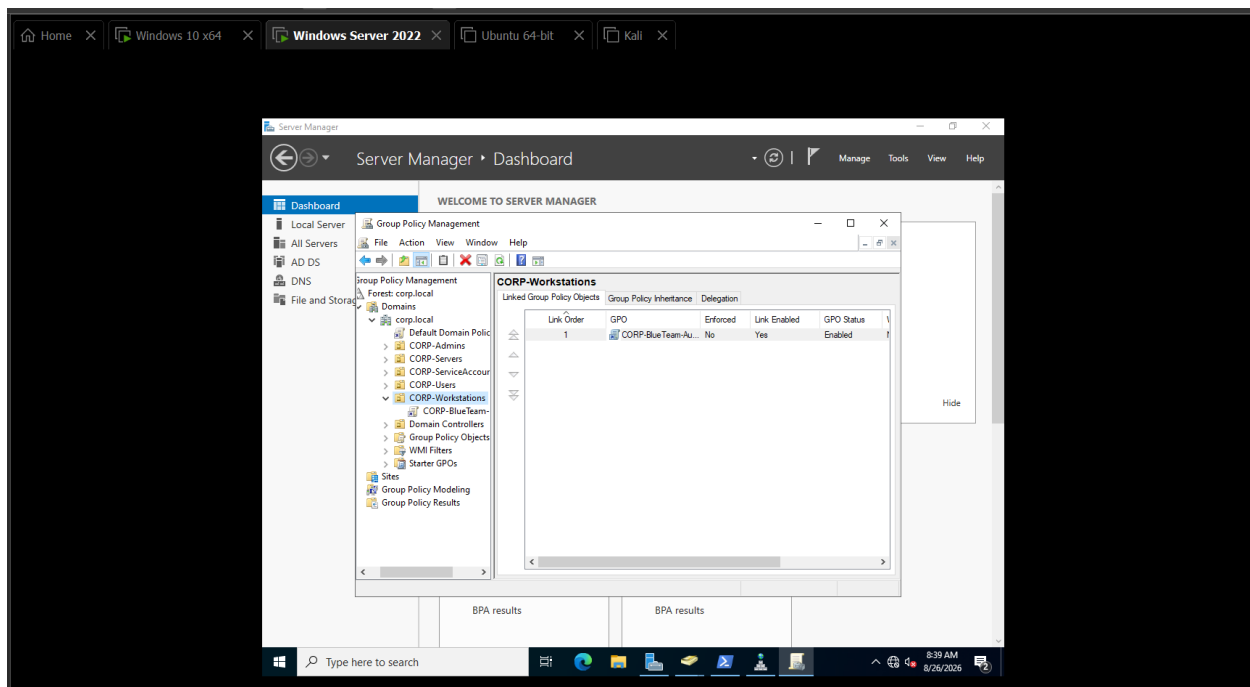


Figure 32: CORP-BlueTeam-Audit-Policy created and linked to the CORP-Workstations OU.

2.9.2 Advanced Audit Policy

Windows Advanced Audit Policy provides granular control over which security events are recorded.

Relevant categories were configured so that the workstation would record security events required for later attack detection and forensic analysis.

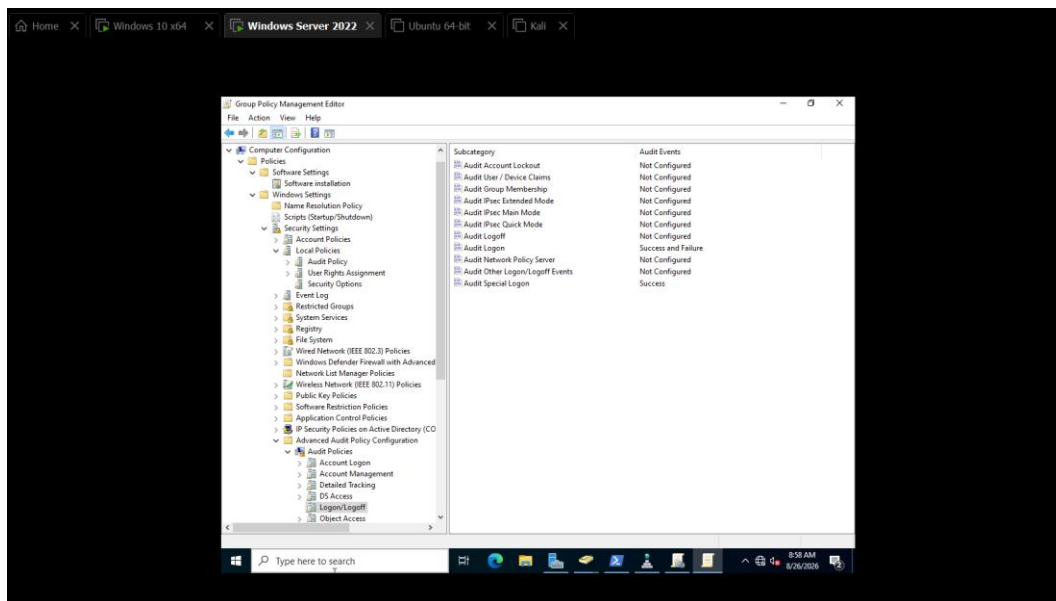


Figure 33: WIN10 Advanced Audit Policy configured through Group Policy.

These audit categories establish the telemetry required to observe activities such as authentication attempts and process execution.

2.9.3 Command-Line Process Auditing

Process creation auditing becomes significantly more useful when the command line associated with the process is also recorded.

Without command-line visibility, an analyst may only observe that:

powershell.exe

or:

cmd.exe

executed.

With command-line auditing enabled, the analyst can also observe what the process was instructed to execute.

The Group Policy setting:

Include command line in process creation events

was therefore enabled.

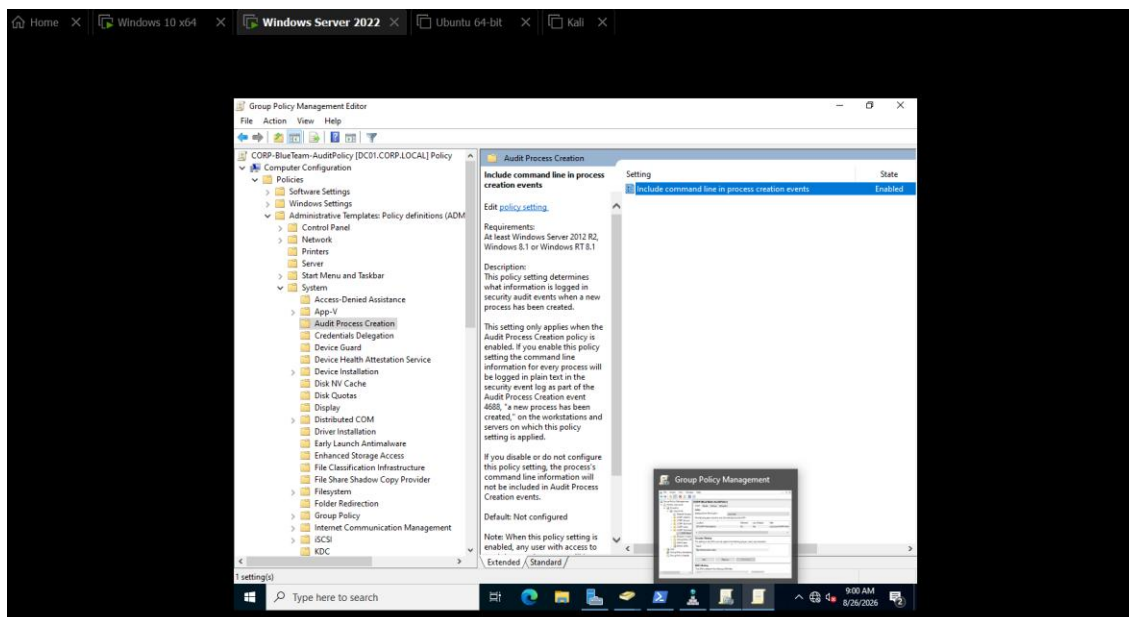


Figure 34: Command-line information enabled for Windows process-creation events.

This configuration significantly improves investigation quality because process arguments often reveal attacker intent.

2.9.4 PowerShell Script Block Logging

PowerShell is widely used for legitimate Windows administration but is also frequently abused during post-exploitation.

PowerShell Script Block Logging was enabled through Group Policy to increase visibility into PowerShell activity.

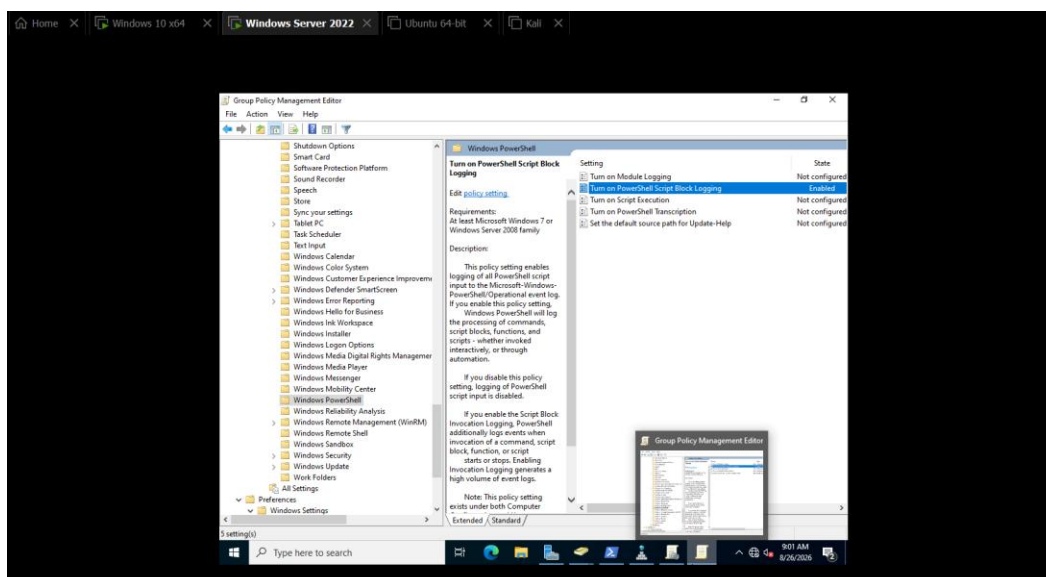


Figure 35: PowerShell Script Block Logging enabled through Group Policy.

This provides an additional telemetry source that can later be correlated with Windows Security and Sysmon events.

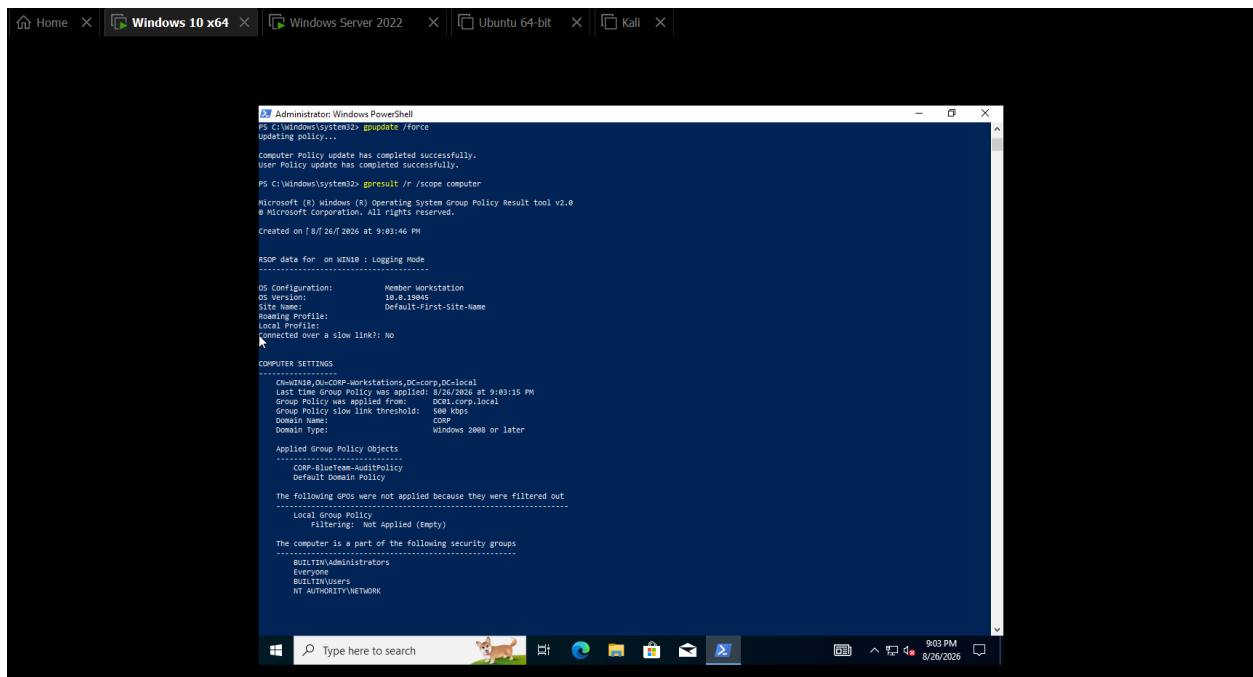
2.9.5 Group Policy Application Verification

Creating a Group Policy Object does not prove that the endpoint actually received it.

The gpresult utility was therefore used on WIN10 to verify that:

CORP-BlueTeam-Audit-Policy

was successfully applied to the workstation.



```
PS C:\Windows\system32> gpupdate /force
Updating policy....
Computer Policy update has completed successfully.
User Policy update has completed successfully.
PS C:\Windows\system32> gpresult /r /scope:computer

Microsoft (R) Windows (R) Operating System Group Policy Result tool v2.0
© Microsoft Corporation. All rights reserved.
Created on 8/26/2026 at 9:03:46 PM

RDP data for: 00000000-00000000-00000000-00000000
-----
OS Configuration:      Member Workstation
OS Version:             10.0.19045
Site Name:              Default-First-Site-Name
Roaming Profile:        Local Profile
Local Profile:          Connected over a slow link?: No
-----

COMPUTER SETTINGS
-----
Computer: 00000000-CORP-Workstations,DC=corp,DC=local
Last time Group Policy was applied: 8/26/2026 at 9:03:15 PM
Group Policy was applied from: 00000000-CORP-Workstations,DC=corp,DC=local
Group Policy slow link threshold: 500 kbps
Domain Name:            CORP
Domain Type:             Windows 2008 or later

Applied Group Policy Objects
-----
CORP-BlueTeam-AuditPolicy
Default domain Policy

The following GPOs were not applied because they were filtered out
-----
Local Group Policy
Filtering: Not Applied (Empty)

The computer is a part of the following security groups
-----
BUILTIN\Administrators
Everyone
BUILTIN\Users
NT AUTHORITY\NETWORK
```

Figure 36: gpresult confirming that CORP-BlueTeam-Audit-Policy was applied to WIN10.

The effective Advanced Audit Policy was additionally verified using:

Auditpol

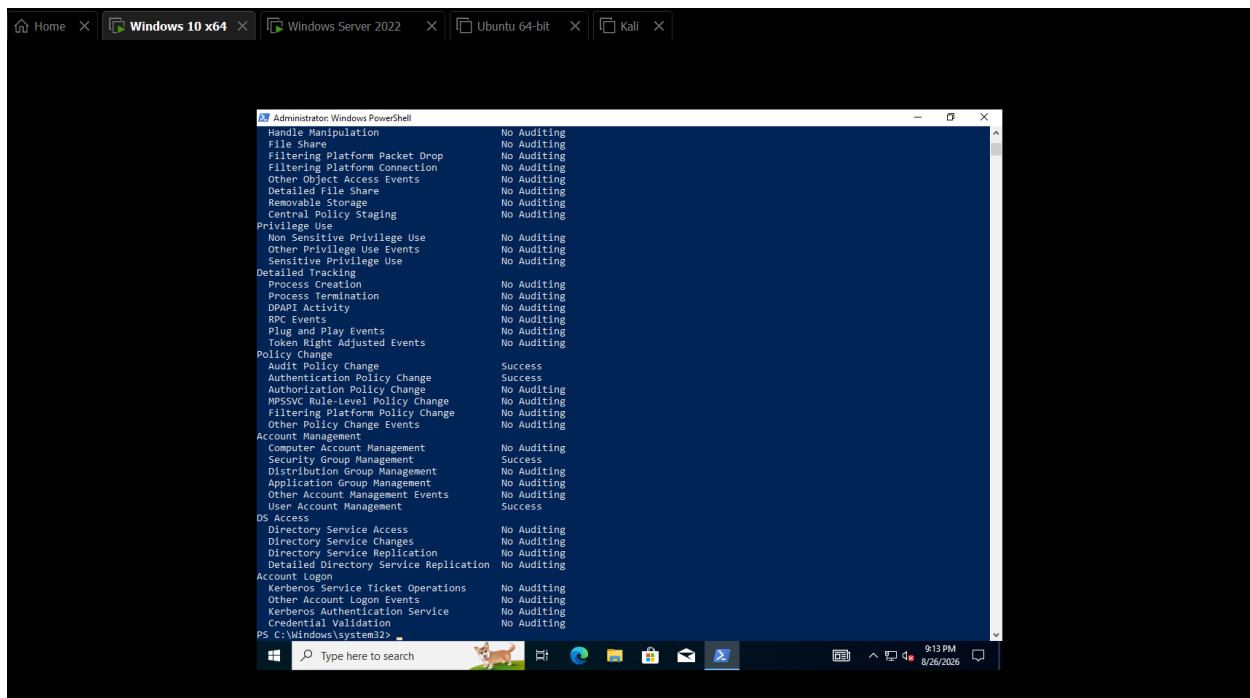


Figure 37: auditpol verification of WIN10 Advanced Audit Policy settings.

These checks confirmed that the security configuration was not merely present on the Domain Controller but was actively enforced on the workstation.

2.9.6 Validating Windows Event ID 4688

A controlled process-execution test was performed to verify that the configured auditing generated useful security telemetry.

Windows Security Event ID:

4688 — A new process has been created

records process-creation activity when the appropriate audit policy is enabled.

A controlled command containing:

BLUE_TEAM_TEST

was executed through cmd.exe from PowerShell.

The resulting Event ID 4688 recorded the process, its parent process, and the associated command-line information.

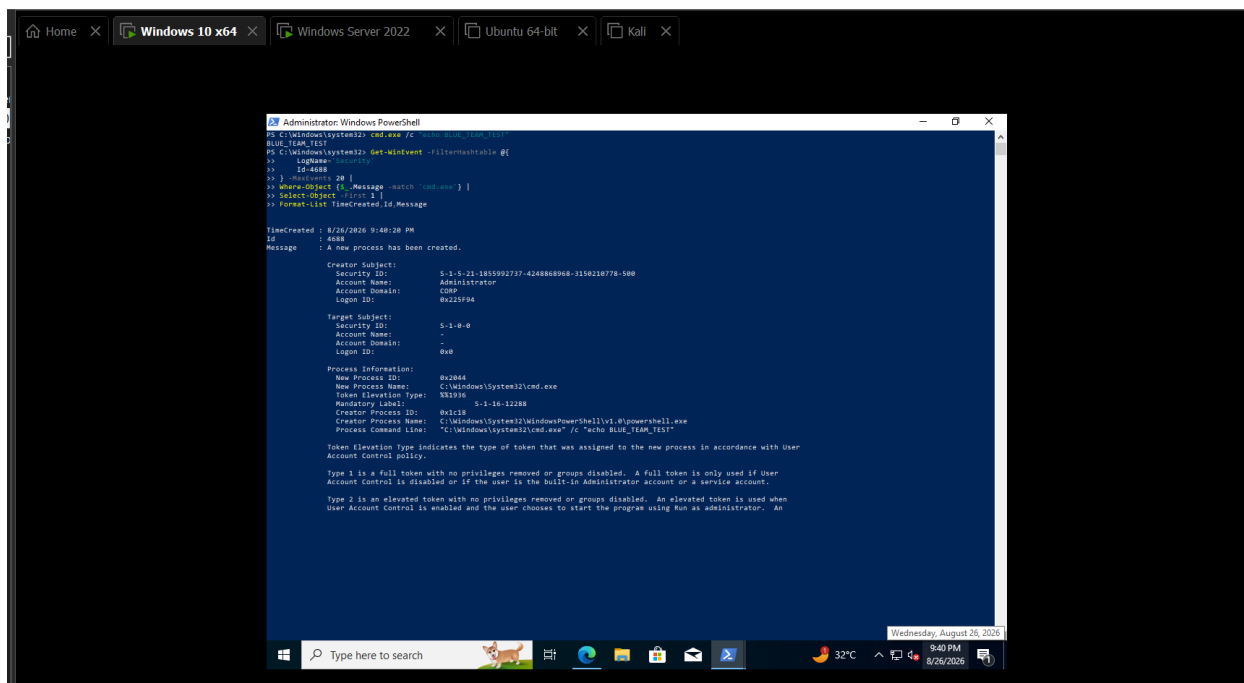


Figure 38: Real Windows Security Event ID 4688 showing cmd.exe, parent PowerShell process, and the BLUE_TEAM_TEST command line.

This provided practical evidence that the workstation auditing policy was functioning as intended.

2.10 Domain Controller Security Auditing and Kerberos Telemetry

2.10.1 Dedicated Domain Controller Audit Policy

The Domain Controller requires different security visibility from a normal workstation because it processes authentication, account management, Kerberos ticketing, and other domain-wide security operations.

A separate Group Policy Object was therefore created:

CORP-DC-Audit-Policy

and linked to the built-in:

Domain Controllers

Organizational Unit.

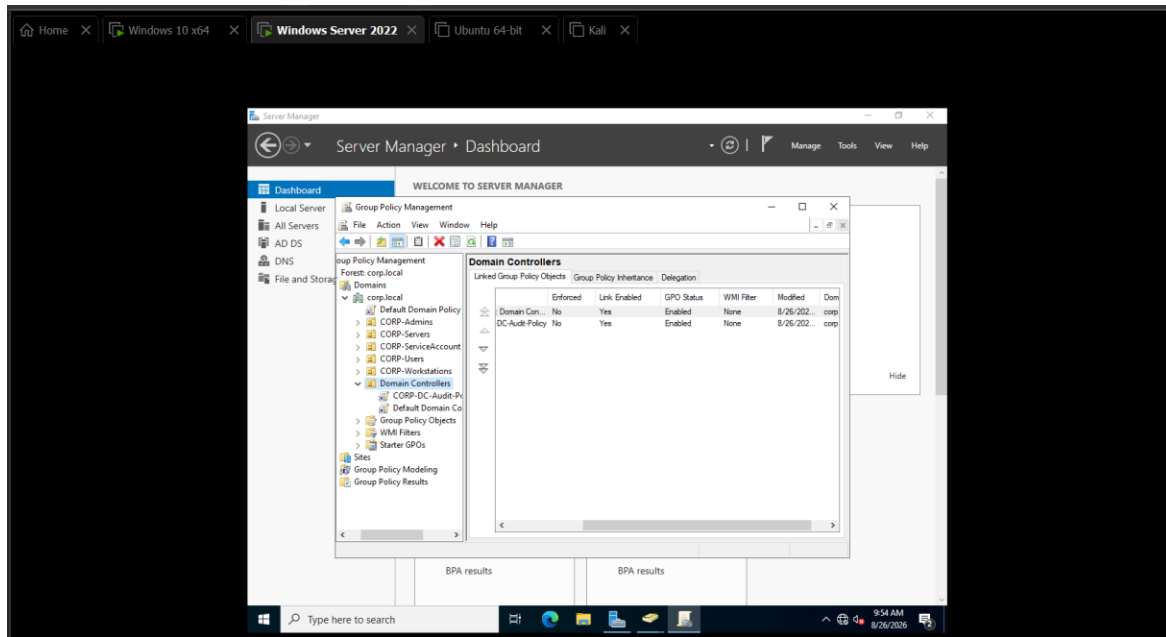


Figure 39: CORP-DC-Audit-Policy created and linked to the Domain Controllers OU.

Separating workstation and Domain Controller policies allowed more appropriate auditing settings to be applied to each system type.

2.10.2 Advanced Domain Controller Auditing

Advanced Audit Policy was configured on DC01 with additional emphasis on authentication and account-related events.

This included telemetry relevant to Kerberos and account-management activity.

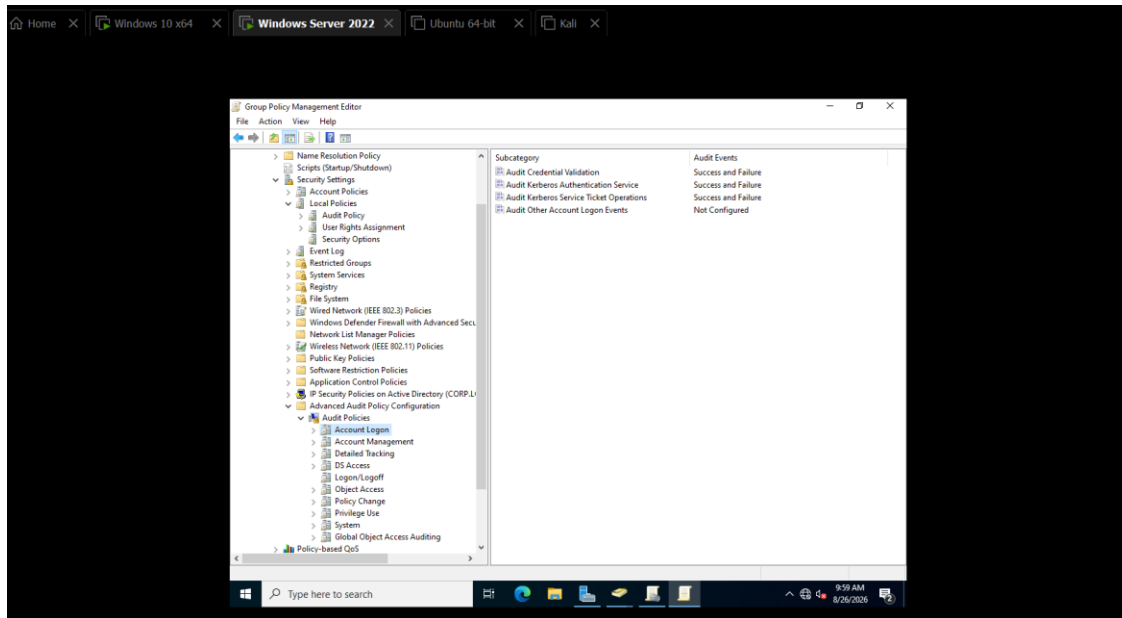


Figure 40: DC01 Advanced Audit Policy configured with Kerberos and account-management auditing.

Such telemetry becomes especially valuable when detecting credential attacks, ticket abuse, account modification, and unauthorized privilege changes.

2.10.3 Command-Line and PowerShell Visibility

Command-line information for process-creation events was also enabled on DC01.

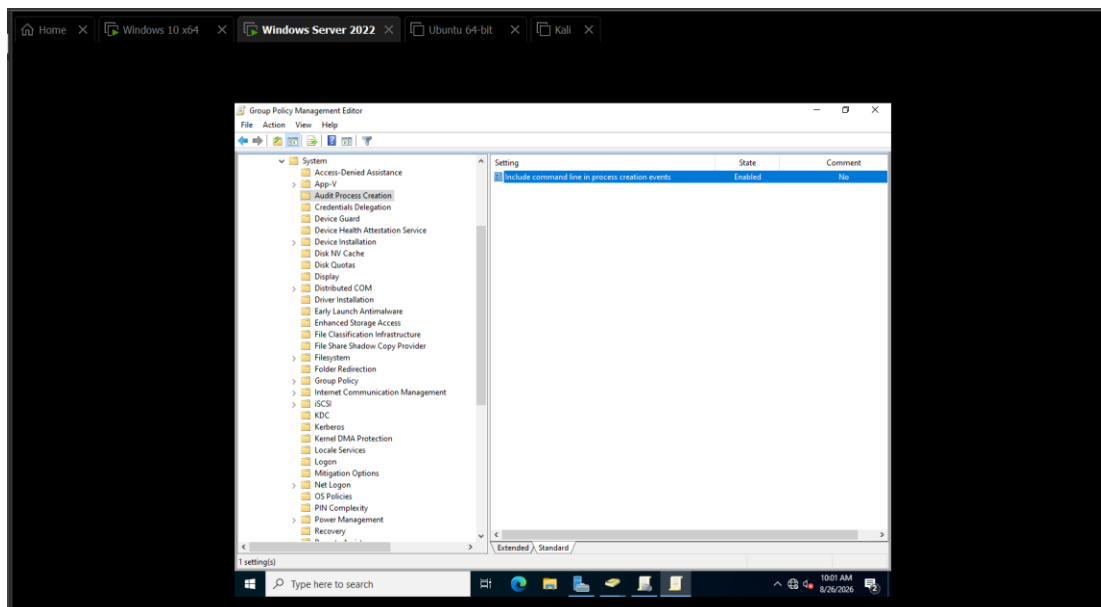


Figure 41: DC01 command-line process auditing enabled.

PowerShell Script Block Logging was additionally configured for the Domain Controller.

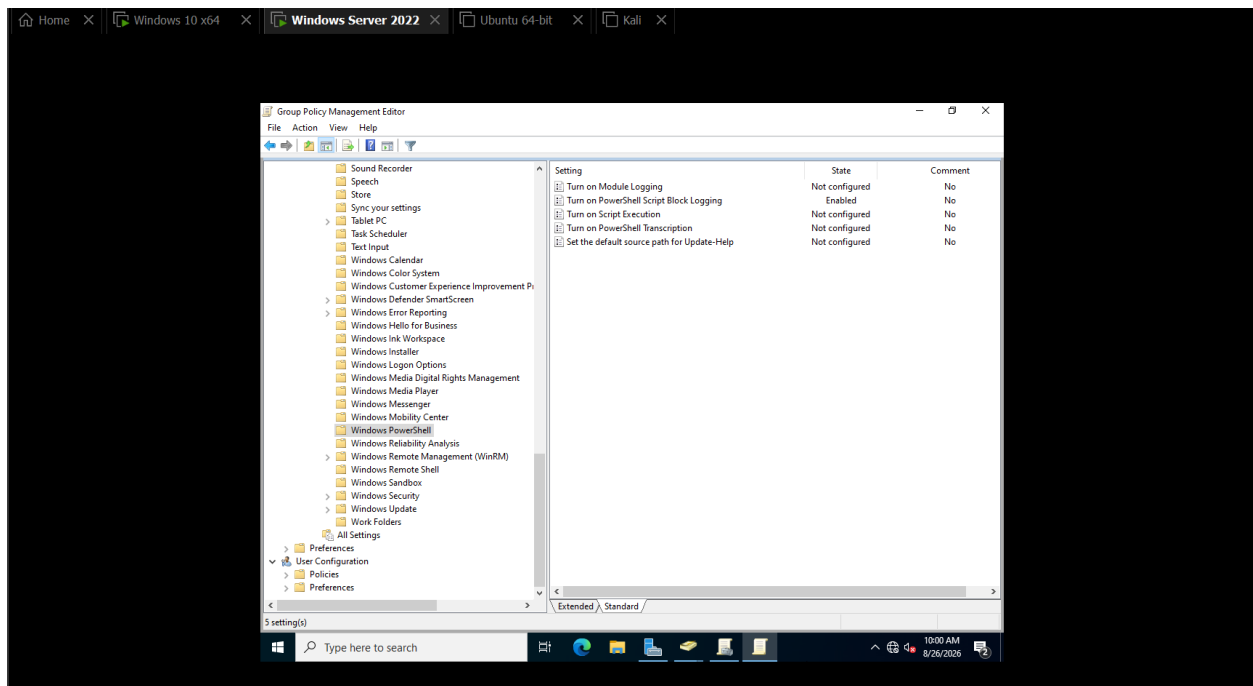


Figure 42: DC01 PowerShell Script Block Logging enabled.

This ensured that administrative or suspicious PowerShell activity executed directly on the Domain Controller could later be investigated.

2.10.4 Verifying the Domain Controller GPO

The effective policy on DC01 was validated using gresult.

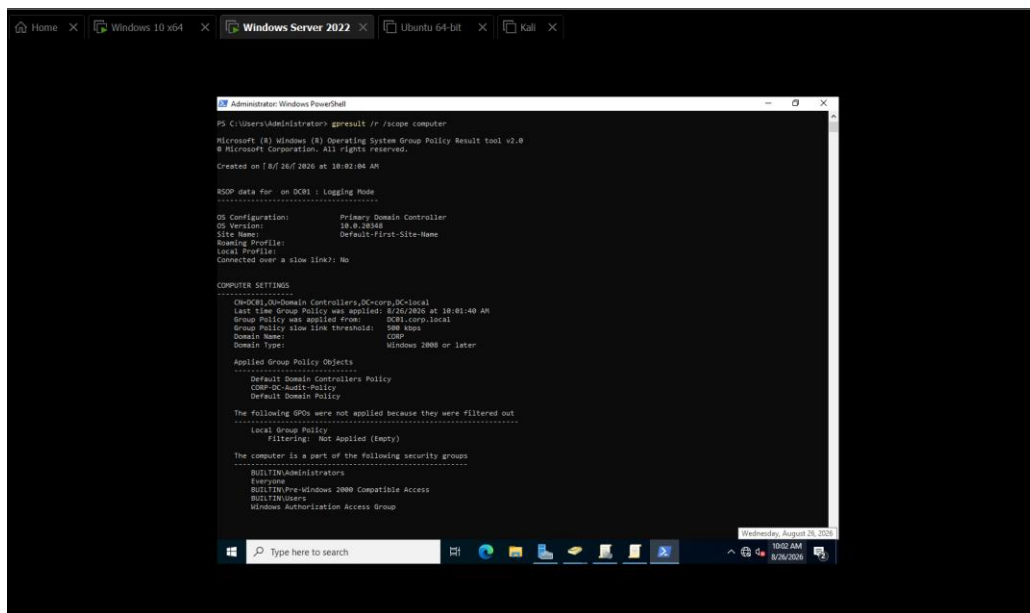


Figure 43: gpresult confirming that CORP-DC-Audit-Policy was successfully applied to DC01.

The resulting Advanced Audit Policy was then verified using auditpol.

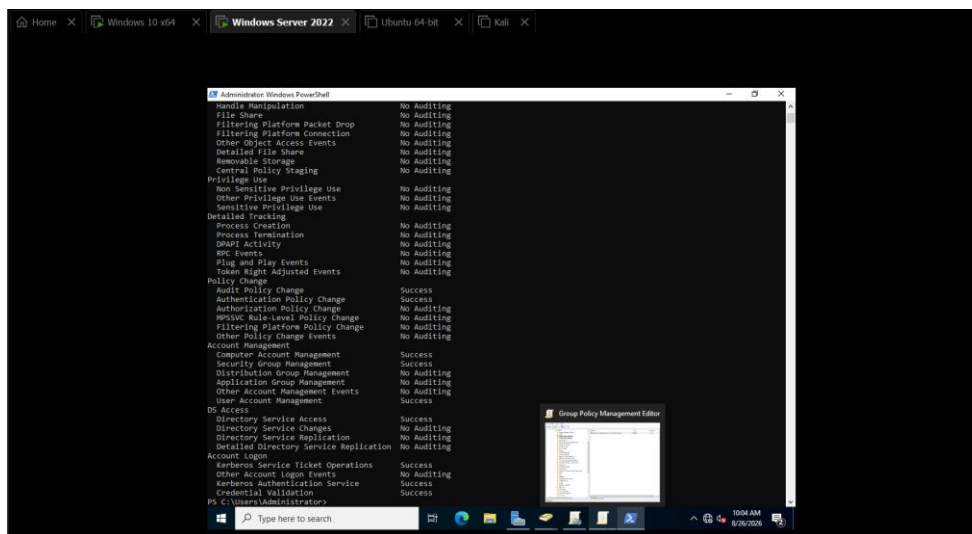


Figure 44: auditpol verification of DC01 Advanced Audit Policy settings.

At this point, the Domain Controller was configured to generate the authentication telemetry required for the later attack-and-detection phases.

2.10.5 Kerberos Event ID 4768

Kerberos is the primary authentication protocol used within modern Active Directory domains.

When a domain user initially authenticates and requests a Ticket Granting Ticket (TGT), the Domain Controller can generate:

Event ID 4768

A Kerberos authentication ticket (TGT) was requested

A real authentication operation involving:

sarah.khan

was observed on DC01.

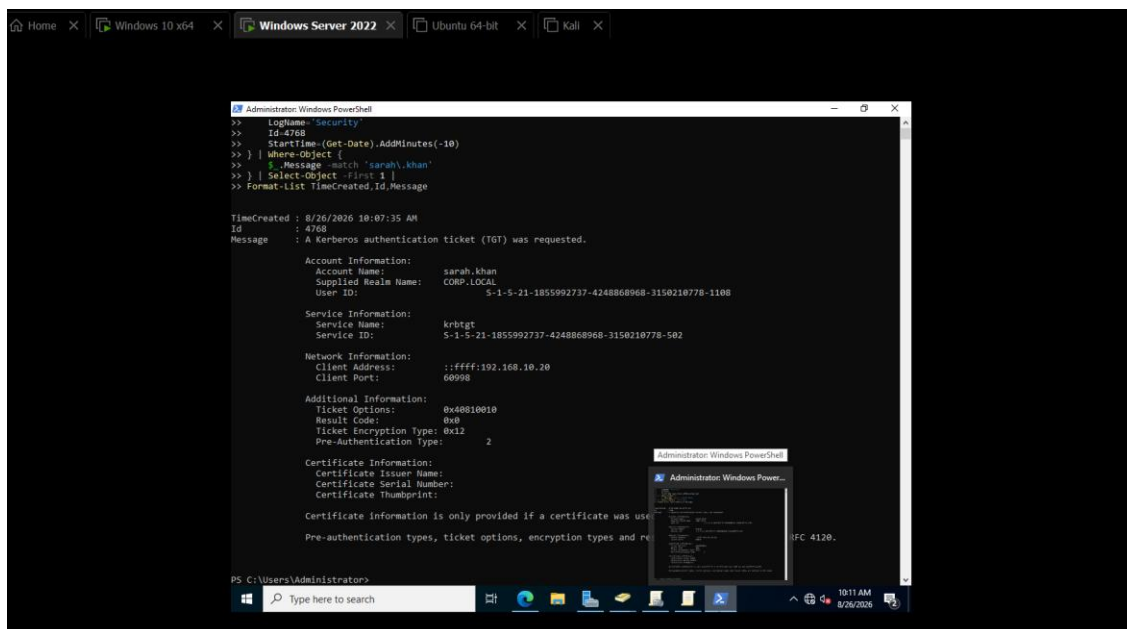


Figure 45: Real DC01 Kerberos Event ID 4768 showing a TGT request for sarah.khan.

This event is important during security monitoring because abnormal TGT requests can provide evidence of suspicious authentication behavior.

2.10.6 Kerberos Event ID 4769

After receiving a TGT, a domain user can request service tickets to access specific domain resources.

These requests can generate:

Event ID 4769

A Kerberos service ticket was requested

The laboratory successfully captured a real service-ticket request associated with sarah.khan.

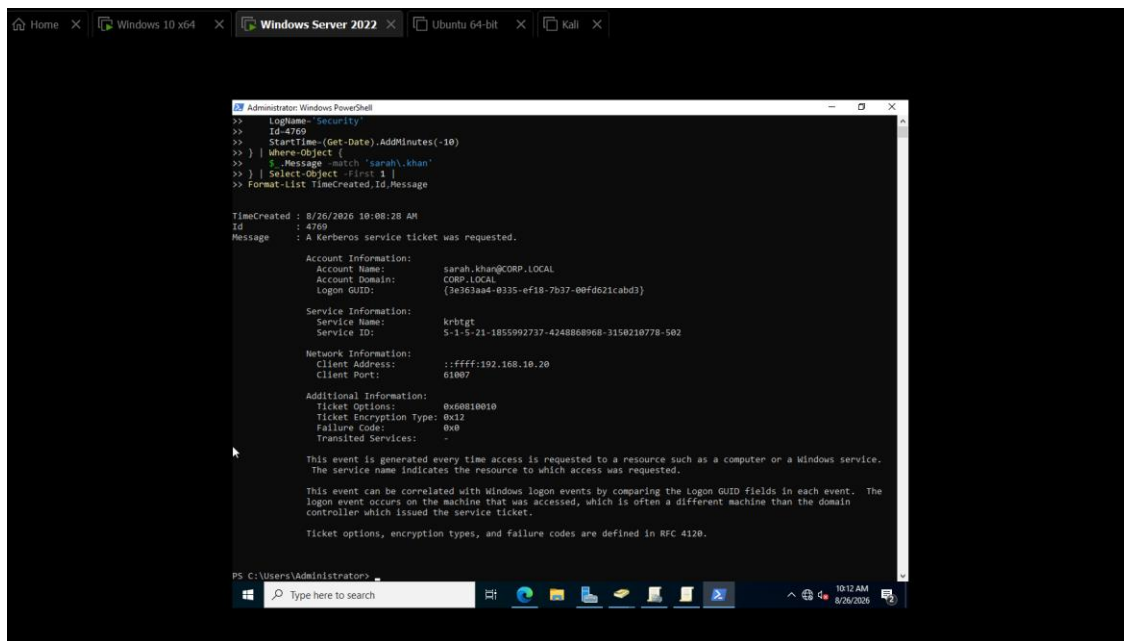


Figure 46: Real DC01 Kerberos Event ID 4769 showing a service-ticket request for sarah.khan.

The successful generation of Events 4768 and 4769 confirmed that Kerberos auditing was operational before any attack simulation was performed.

2.11 Sysmon Deployment and Validation

2.11.1 Why Sysmon Was Added

Although Windows Security auditing provides valuable information, it does not provide every piece of endpoint telemetry useful to a SOC analyst.

Microsoft Sysmon extends Windows visibility by recording detailed system activity, including process creation, network connections, image loading, registry operations, and other security-relevant behavior depending on its configuration.

Sysmon was therefore deployed alongside standard Windows auditing rather than being treated as a replacement for it.

The monitoring model became:

Windows Security Logs

+

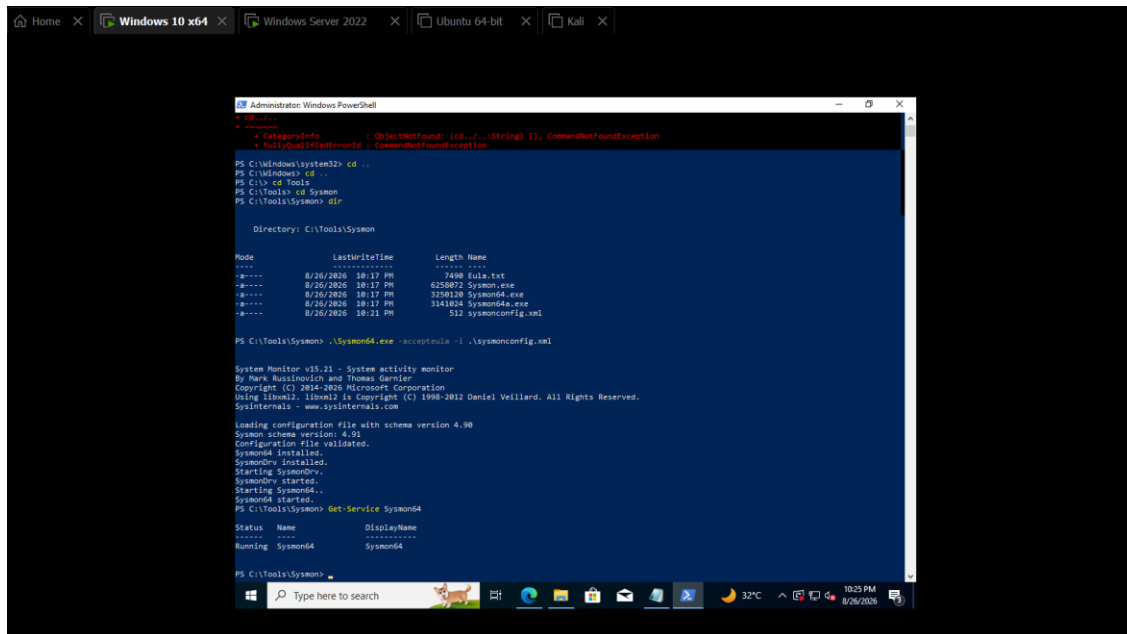
PowerShell Operational Logs

+
Sysmon
↓
Higher-fidelity endpoint telemetry

2.11.2 Sysmon Deployment on WIN10

Sysmon was installed and configured on the Windows 10 workstation.

Following installation, its service status was checked to confirm that Sysmon was actively monitoring the endpoint.



The screenshot shows a Windows 10 desktop with a taskbar at the bottom. A PowerShell terminal window is open, displaying the following commands and output:

```
PS C:\Windows\system32> cd ..
PS C:\Windows> cd
PS C:\> cd Tools
PS C:\Tools> cd Sysmon
PS C:\Tools\Sysmon> dir

Directory: C:\Tools\Sysmon

Mode                LastWriteTime         Length Name
----                -
-a-----      8/26/2026 18:17 PM             7408 Eula.txt
-a-----      8/26/2026 18:17 PM          6250872 Sysmon.exe
-a-----      8/26/2026 18:17 PM          3250128 Sysmon4.exe
-a-----      8/26/2026 18:17 PM          3141824 Sysmon4a.exe
-a-----      8/26/2026 18:21 PM             512 Sysmonconfig.xml

PS C:\Tools\Sysmon> .\Sysmon4.exe -accepteula -i .\sysmonconfig.xml

System Monitor v5.21 - System activity monitor
By Mark Russinovich and Thomas Garnier
Copyright (C) 2014-2026 Microsoft Corporation
Using libwinl2. libwinl2 is Copyright (C) 1998-2012 Daniel Veillard. All Rights Reserved.
Sysinternals - www.sysinternals.com

Loading configuration file with schema version 4.90
Sysmon schema version: 4.91
Configuration file validated.
Sysmon4 installed.
SysmonDrv installed.
Starting SysmonDrv.
SysmonDrv started.
Starting Sysmon4.
Sysmon4 started.
PS C:\Tools\Sysmon> Get-Service Sysmon4

Status Name      DisplayName
-----
Running Sysmon4 Sysmon4

PS C:\Tools\Sysmon>
```

Figure 47: Sysmon installed and service running on WIN10.

A controlled process execution was subsequently performed and the Sysmon Operational log was examined.

Sysmon Event ID:

1 — Process Create

provides detailed information about newly created processes.

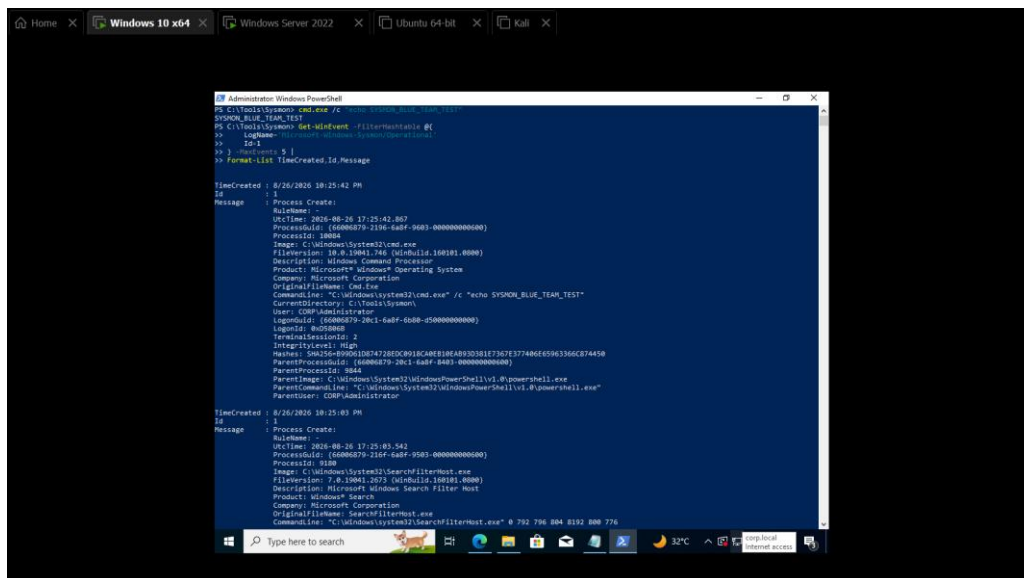


Figure 48: Real Sysmon Event ID 1 process-creation telemetry generated on WIN10.

This confirmed that Sysmon was successfully producing endpoint telemetry before integration with Splunk.

2.11.3 Sysmon Deployment on DC01

Sysmon was also installed on the Domain Controller to provide enhanced visibility into processes executing on the most security-critical Windows system in the environment.

The service was verified after installation.

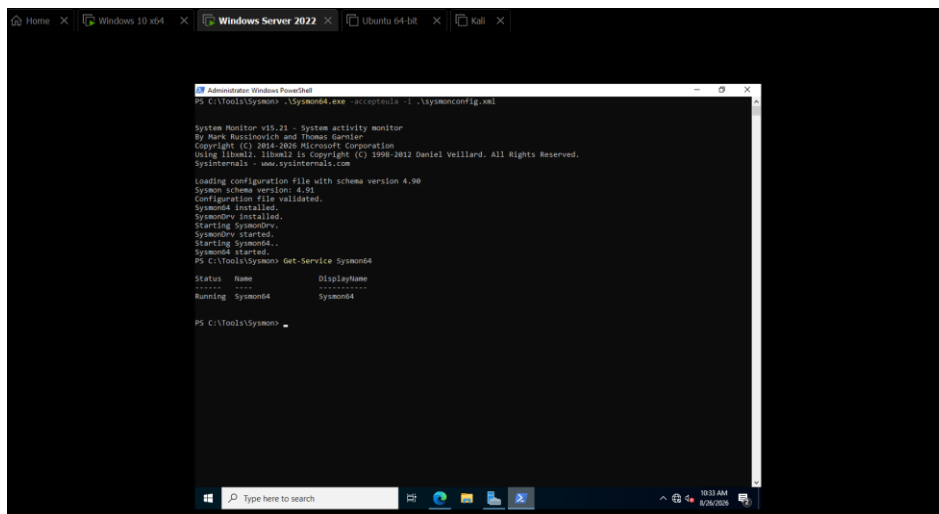


Figure 49: Sysmon installed and service running on DC01.

A controlled process was then executed and successfully recorded as Sysmon Event ID 1.

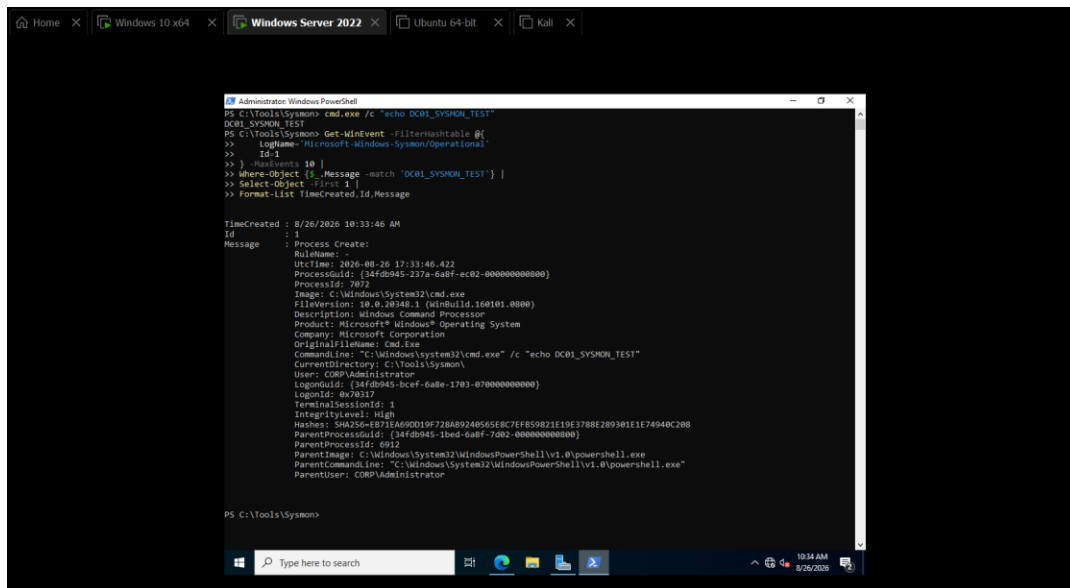


Figure 50: Real Sysmon Event ID 1 process-creation telemetry generated on DC01.

At this stage, both Windows systems were producing three major categories of telemetry:

Security Event Logs

PowerShell Operational Logs

Sysmon Operational Logs

The remaining objective was to centralize those events within the SIEM.

2.12 Splunk SIEM Integration

2.12.1 Centralized Log Collection Architecture

Generating security events locally is useful, but manually checking Event Viewer on every endpoint is not scalable.

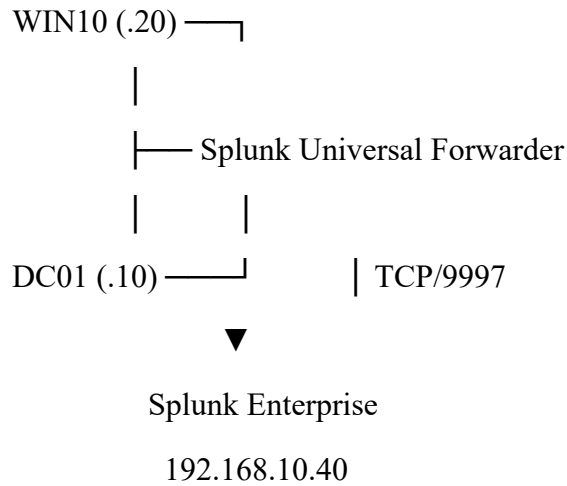
A Security Information and Event Management platform centralizes telemetry from multiple systems so that analysts can search, correlate, and investigate events from a single location.

Splunk Enterprise was deployed on the Ubuntu system at:

192.168.10.40

The Windows systems used Splunk Universal Forwarder to transmit selected telemetry to the Splunk server.

The resulting architecture was:



Splunk was configured to receive forwarded events using TCP port:

9997

Connectivity from both Windows systems to the receiving service was successfully validated.

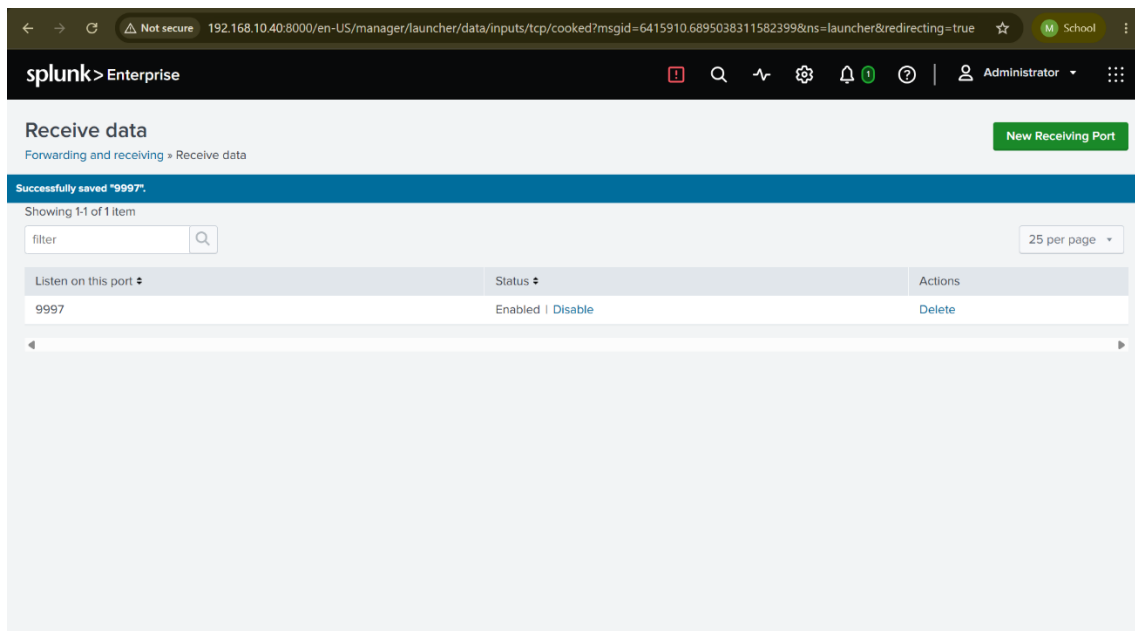


Figure 51: Splunk configured to receive Universal Forwarder traffic on TCP port 9997, with connectivity from WIN10 and DC01 verified.

2.12.2 WIN10 Universal Forwarder

Splunk Universal Forwarder was installed on WIN10 and configured to send telemetry to:

192.168.10.40:9997

The service was verified as running, and the forwarding configuration showed the Splunk server under active forwards.

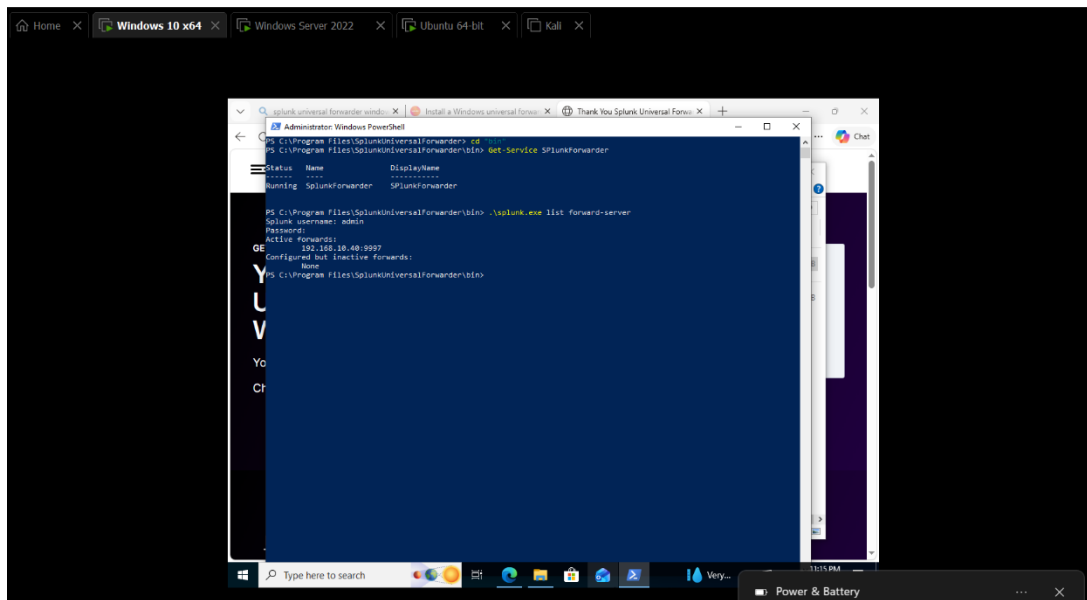


Figure 52: WIN10 Splunk Universal Forwarder running with an active forwarding connection to 192.168.10.40:9997.

The forwarder was configured to collect three important Windows channels:

Security

Microsoft-Windows-Sysmon/Operational

Microsoft-Windows-PowerShell/Operational

During testing, Security and PowerShell events initially reached Splunk while Sysmon telemetry did not.

Troubleshooting confirmed that Sysmon itself was functioning, the input configuration was correct, and the forwarder had successfully loaded the Sysmon input. Inspection of the Splunk forwarder logs revealed that the service was unable to query the Sysmon channel.

On WIN10, the Universal Forwarder service was operating as:

NT SERVICE\SplunkForwarder

The service identity was therefore granted the required Event Log Reader access. Following the permission correction and service restart, Sysmon events began arriving successfully.

This troubleshooting exercise demonstrated the importance of validating the entire telemetry pipeline rather than assuming that a running forwarding service guarantees access to every configured event channel.

Splunk subsequently displayed all three WIN10 telemetry sources.

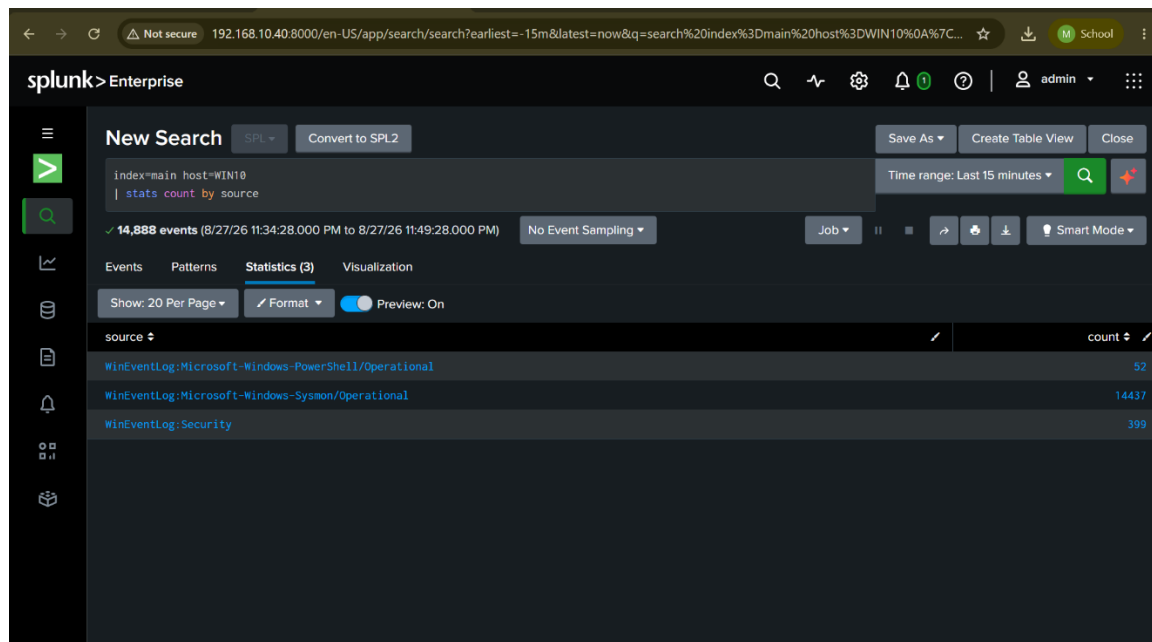


Figure 53: Splunk receiving Windows Security, PowerShell Operational, and Sysmon Operational telemetry from WIN10.

2.12.3 End-to-End WIN10 Sysmon Validation

After confirming general ingestion, a controlled process was generated on WIN10 using a recognizable test string:

BLUE_TEAM_SPLUNK_TEST

Sysmon recorded the process as Event ID 1, the Universal Forwarder transmitted the event, and the resulting telemetry was successfully located through Splunk.

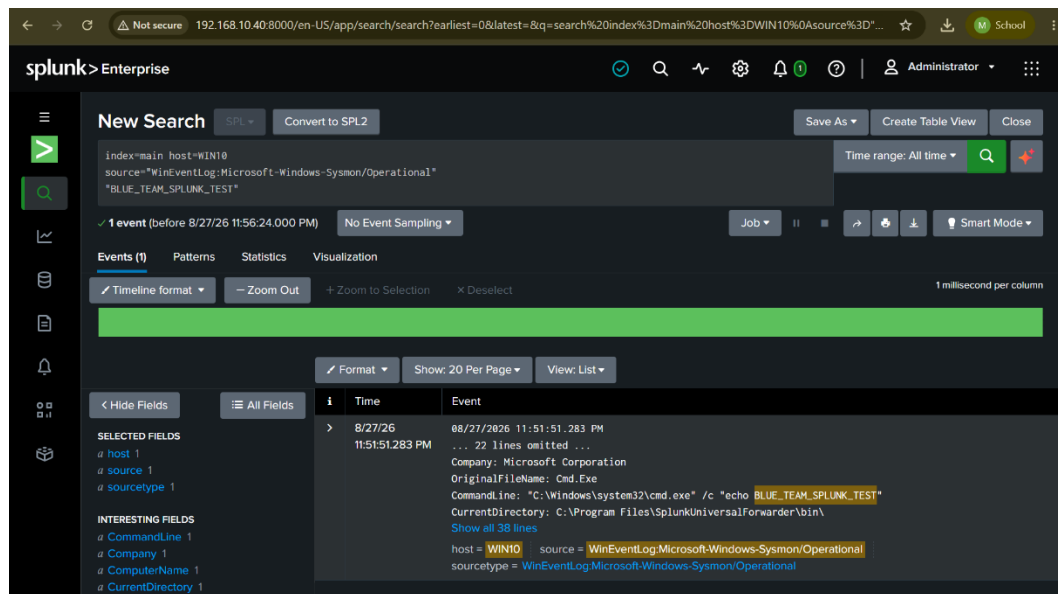


Figure 54: End-to-end WIN10 Sysmon Event ID 1 process telemetry successfully searchable within Splunk.

This validation proved the complete endpoint monitoring chain:

Process Execution



Sysmon Event ID 1



Windows Event Channel



Splunk Universal Forwarder



TCP/9997



Splunk Enterprise



SOC Search and Investigation

2.12.4 DC01 Universal Forwarder

The same centralized logging architecture was then extended to DC01.

Splunk Universal Forwarder was installed on the Domain Controller and configured to send telemetry to:

192.168.10.40:9997

The service was confirmed to be running and the Splunk server appeared as an active forwarding destination.

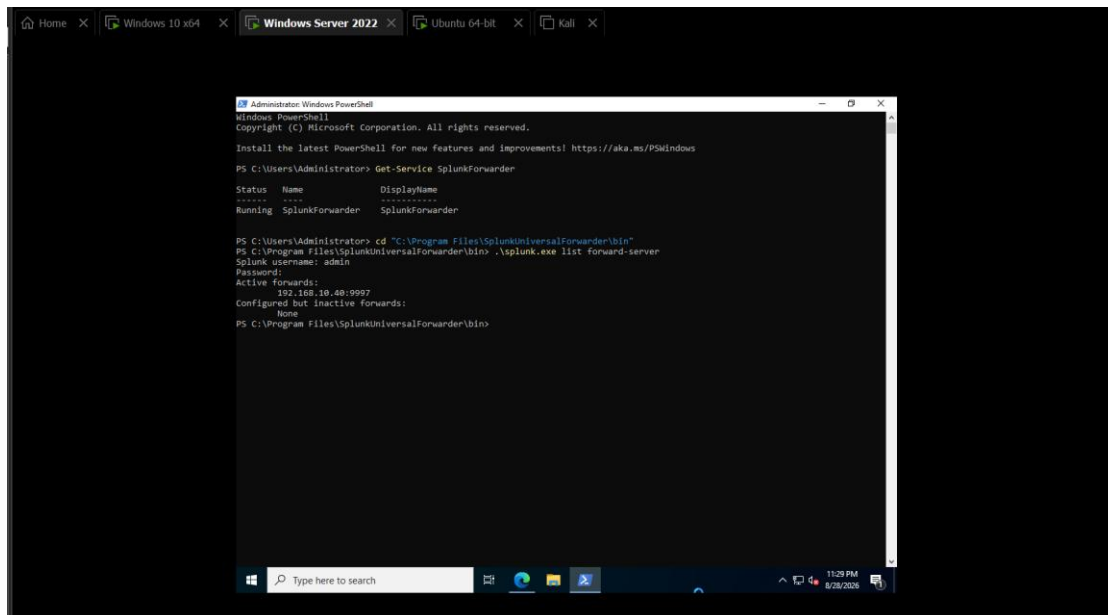


Figure 55: DC01 Splunk Universal Forwarder running with an active connection to 192.168.10.40:9997.

Unlike WIN10, the Universal Forwarder service on DC01 operated under:

LocalSystem

and did not require the same Event Log Readers permission correction encountered on the workstation.

2.12.5 DC01 Telemetry Ingestion

DC01 was configured to forward:

Security

Microsoft-Windows-Sysmon/Operational

Microsoft-Windows-PowerShell/Operational

Splunk successfully received events from all three sources.

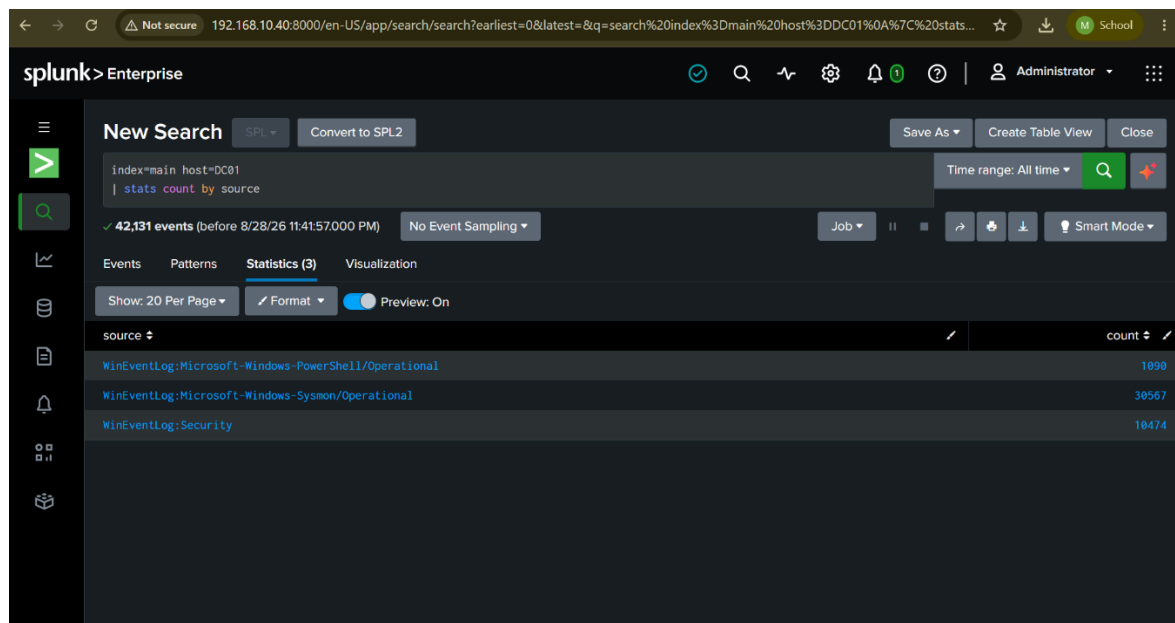


Figure 56: Splunk receiving Security, Sysmon Operational, and PowerShell Operational telemetry from DC01.

At the time of verification, tens of thousands of Domain Controller events had already been indexed, demonstrating that centralized monitoring was functioning continuously rather than only during isolated test events.

2.12.6 Kerberos Authentication Monitoring in Splunk

The final validation focused on real Active Directory authentication activity.

Authentication activity was generated from WIN10 using the sarah.khan domain account. Kerberos ticket activity was then searched centrally within Splunk using DC01's Security events.

Splunk successfully displayed:

4768 → Kerberos TGT request

4769 → Kerberos service-ticket request

The events included activity associated with:

User: sarah.khan

Domain: CORP.LOCAL

Client IP: 192.168.10.20

along with Kerberos services such as krbtgt and other domain resources.

The screenshot shows the Splunk Enterprise interface with a search results table. The search query is `index=main host=DC01 (EventCode=4768 OR EventCode=4769)`. The results are sorted by `_time` in descending order. The table displays 31 events, with the first 10 rows visible. The columns are `_time`, `EventCode`, `Account_Name`, `Service_Name`, and `Client_Address`.

_time	EventCode	Account_Name	Service_Name	Client_Address
2026-08-28 23:47:04.353	4769	WIN10\$@CORP.LOCAL	WIN10\$	::ffff:192.168.10.20
2026-08-28 23:47:04.346	4768	WIN10\$	krbtgt	::ffff:192.168.10.20
2026-08-28 23:47:02.246	4769	sarah.khan@CORP.LOCAL	DC01\$	::ffff:192.168.10.20
2026-08-28 23:47:01.697	4769	sarah.khan@CORP.LOCAL	WIN10\$	::ffff:192.168.10.20
2026-08-28 23:47:01.691	4768	sarah.khan	krbtgt	::ffff:192.168.10.20
2026-08-28 23:45:20.747	4769	WIN10\$@CORP.LOCAL	krbtgt	::ffff:192.168.10.20
2026-08-28 23:45:20.741	4769	WIN10\$@CORP.LOCAL	DC01\$	::ffff:192.168.10.20
2026-08-28 23:45:19.655	4769	WIN10\$@CORP.LOCAL	WIN10\$	::ffff:192.168.10.20
2026-08-28 23:45:06.159	4769	WIN10\$@CORP.LOCAL	krbtgt	::ffff:192.168.10.20
2026-08-28 23:45:06.153	4769	WIN10\$@CORP.LOCAL	DC01\$	::ffff:192.168.10.20

Figure 57: Kerberos Event IDs 4768 and 4769 from DC01 successfully monitored within Splunk for domain authentication activity originating from WIN10.

This was the final Phase 2 validation because it demonstrated more than simple log forwarding. It proved that real Active Directory authentication activity generated on a domain workstation could be captured by the Domain Controller and investigated centrally from the SIEM.

2.13 Phase 2 Verification and Results

Phase 2 successfully established and validated the complete Active Directory and security-monitoring environment.

The major technical outcomes achieved during the phase included:

- Successful deployment of the corp.local Active Directory domain.
- Correct operation of DC01 as the Domain Controller and DNS server.
- Structured Active Directory organization using OUs, users, groups, and service accounts.
- Successful integration of WIN10 into the domain.
- Successful authentication using the CORP\sarah.khan account.
- Verified secure channel between WIN10 and the domain.
- Centralized workstation auditing through Group Policy.
- Dedicated Domain Controller auditing through Group Policy.
- Successful Windows Event ID 4688 process-creation auditing with command-line visibility.
- Successful Kerberos Event IDs 4768 and 4769 generation.
- Successful Sysmon deployment and Event ID 1 process telemetry on WIN10 and DC01.
- Successful forwarding of Security, PowerShell, and Sysmon telemetry from both monitored Windows systems.
- Successful centralized searching of endpoint and Domain Controller events within Splunk.
- Successful end-to-end visibility of real Kerberos authentication activity from WIN10 through DC01 into Splunk.

These results confirmed that the environment was technically ready for controlled attack simulation.

2.14 Skills and Learning Outcomes

Phase 2 provided practical experience across Active Directory administration, Windows security monitoring, endpoint telemetry, and SIEM engineering.

Key skills developed during this phase included:

Active Directory Administration: Deployment of AD DS, forest and domain creation, Organizational Unit design, domain-user management, security-group configuration, service-account creation, workstation domain joining, and computer-object management.

DNS and Network Troubleshooting: Configuration of Active Directory DNS, validation of domain service discovery, troubleshooting of multi-NIC DNS registration, and management of internal versus NAT interfaces.

Group Policy Management: Creation and targeting of workstation and Domain Controller GPOs, configuration of Advanced Audit Policy, process command-line auditing, and PowerShell Script Block Logging.

Windows Security Monitoring: Analysis and validation of important Windows Security Event IDs, including 4688, 4768, and 4769.

Sysmon: Deployment and verification of Sysmon on workstations and servers and analysis of Event ID 1 process-creation telemetry.

Splunk: Configuration of Splunk receiving services, deployment of Universal Forwarders, configuration of Windows event inputs, troubleshooting forwarder permissions, and centralized searching of security telemetry.

Blue-Team Troubleshooting: Systematic troubleshooting of DNS registration, Group Policy application, Windows event generation, Sysmon collection, forwarding permissions, and SIEM ingestion.

Most importantly, the phase demonstrated that security monitoring requires validation at every layer. A configured policy does not guarantee that an event is generated, a generated event does not guarantee that a forwarder can read it, and a running forwarder does not guarantee that the SIEM has indexed the event correctly. Each component must therefore be independently tested and then validated as part of the complete telemetry pipeline.

2.15 Conclusion

Phase 2 successfully transformed the virtual laboratory into a monitored Active Directory environment suitable for realistic Blue-Team security testing.

Windows Server 2022 was deployed as the Domain Controller for the corp.local domain, while DNS, Organizational Units, users, security groups, service accounts, and a domain-joined Windows 10 workstation were configured to create a simplified enterprise identity infrastructure.

Security visibility was then strengthened through centralized Group Policy. Advanced Audit Policy, command-line process auditing, PowerShell Script Block Logging, and Sysmon were implemented to provide detailed telemetry from both the workstation and Domain Controller.

Splunk Enterprise provided centralized log collection and analysis. Universal Forwarders on WIN10 and DC01 successfully transmitted Windows Security, Sysmon, and PowerShell telemetry to the SIEM. Controlled process execution and Kerberos authentication activity were successfully observed within Splunk, proving that the complete monitoring pipeline was operational.

The completion of Phase 2 therefore established the defensive visibility required for the next stage of the project.

Phase 3 — Active Directory Attack Simulation, Detection and Mitigation

1. Password Spraying Attack

1.1 Overview

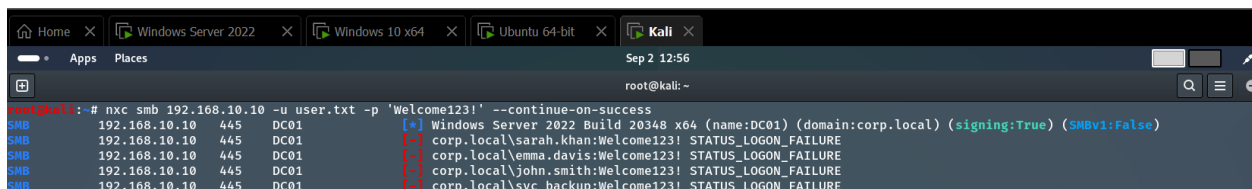
Password spraying is an authentication attack where an attacker attempts a small number of commonly used passwords against **multiple user accounts**. Unlike traditional brute-force attacks, which test many passwords against one account, password spraying distributes attempts across several accounts to reduce the chance of triggering account lockout policies.

The objective of this test was to simulate password spraying against the corp.local Active Directory environment and determine whether the resulting authentication activity could be detected through Splunk.

1.2 Attack Execution

The password spraying attack was performed from the Kali Linux attacker machine against multiple domain accounts in the corp.local environment. The attempts were directed toward the Active Directory infrastructure hosted on DC01.

The attack generated multiple authentication attempts across different domain accounts, producing Windows Security events on the Domain Controller.



```
root@kali: ~  
root@kali:~# nxc smb 192.168.10.10 -u user.txt -p 'Welcome123!' --continue-on-success  
[*] Windows Server 2022 Build 20348 x64 (name:DC01) (domain:corp.local) (signing:True) (SMBv1:False)  
SMB 192.168.10.10 445 DC01 [-] corp.local\sarah.khan:Welcome123! STATUS_LOGON_FAILURE  
SMB 192.168.10.10 445 DC01 [-] corp.local\emma.davis:Welcome123! STATUS_LOGON_FAILURE  
SMB 192.168.10.10 445 DC01 [-] corp.local\john.smith:Welcome123! STATUS_LOGON_FAILURE  
SMB 192.168.10.10 445 DC01 [-] corp.local\svc_backup:Welcome123! STATUS_LOGON_FAILURE
```

Figure 1: Password Spraying Attack Against the corp.local Domain

Figure 1 shows the password spraying attack being performed from the Kali Linux attacker machine against multiple Active Directory accounts.

1.3 Detection in Splunk

The resulting authentication activity was investigated using the Windows Security logs forwarded from DC01 to Splunk. Failed Windows authentication attempts can be identified through events such as **Event ID 4625 (An account failed to log on)**.

The important detection indicator was not a single failed login, but the pattern of **multiple accounts receiving authentication attempts within a short period of time**. This behavior is significantly more suspicious than an occasional failed login from a legitimate user.

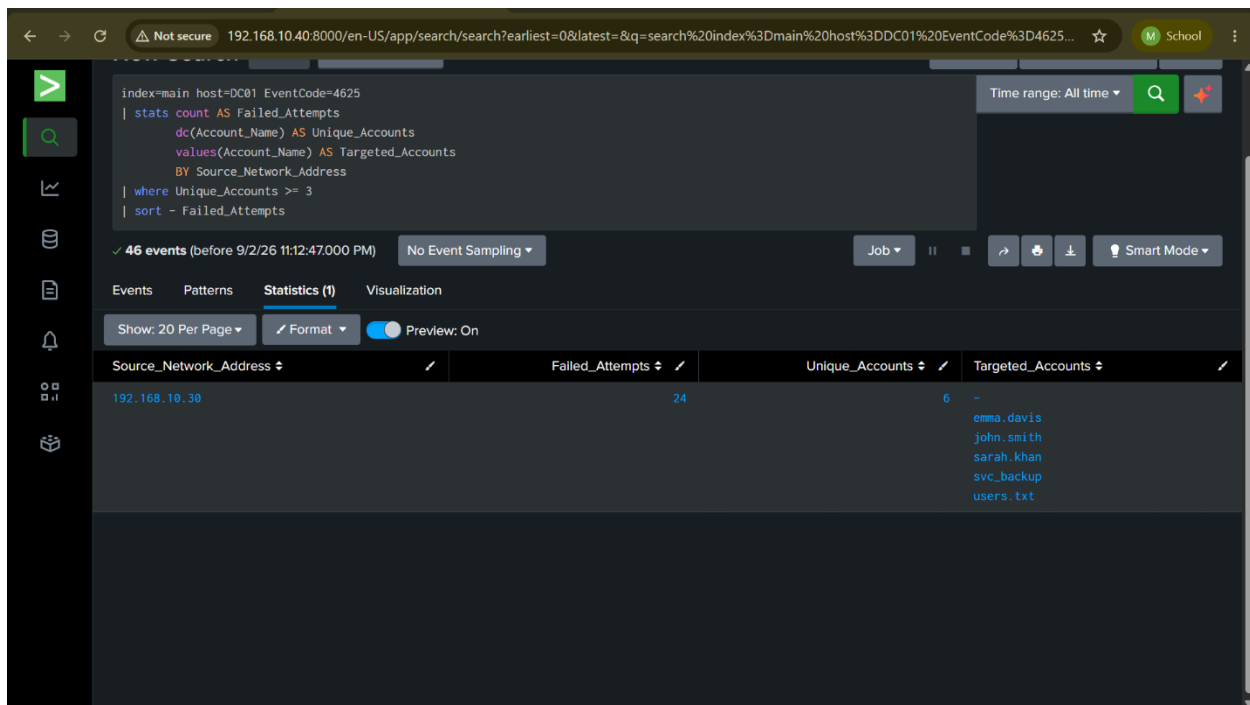


Figure 2: Password Spraying Activity Detected in Splunk

Figure 2 shows the authentication failures generated by the password spraying attack and observed through centralized logging in Splunk.

1.4 Mitigation

Password spraying can be mitigated through a combination of **multi-factor authentication (MFA)**, **strong password policies**, **compromised-password blocking**, **appropriate account lockout controls**, and **continuous authentication monitoring**.

From a Blue Team perspective, SIEM detection rules should identify a source generating failed authentication attempts against multiple accounts within a short time window. Analysts should also investigate cases where repeated failures are followed by a successful authentication, as this may indicate that a valid password was discovered.

1.5 Result

The password spraying simulation was successfully performed against the corp.local environment, and the resulting authentication activity was successfully identified in Splunk. This validated the ability of the Phase 2 monitoring infrastructure to detect suspicious Active Directory authentication behavior.

2. AS-REP Roasting

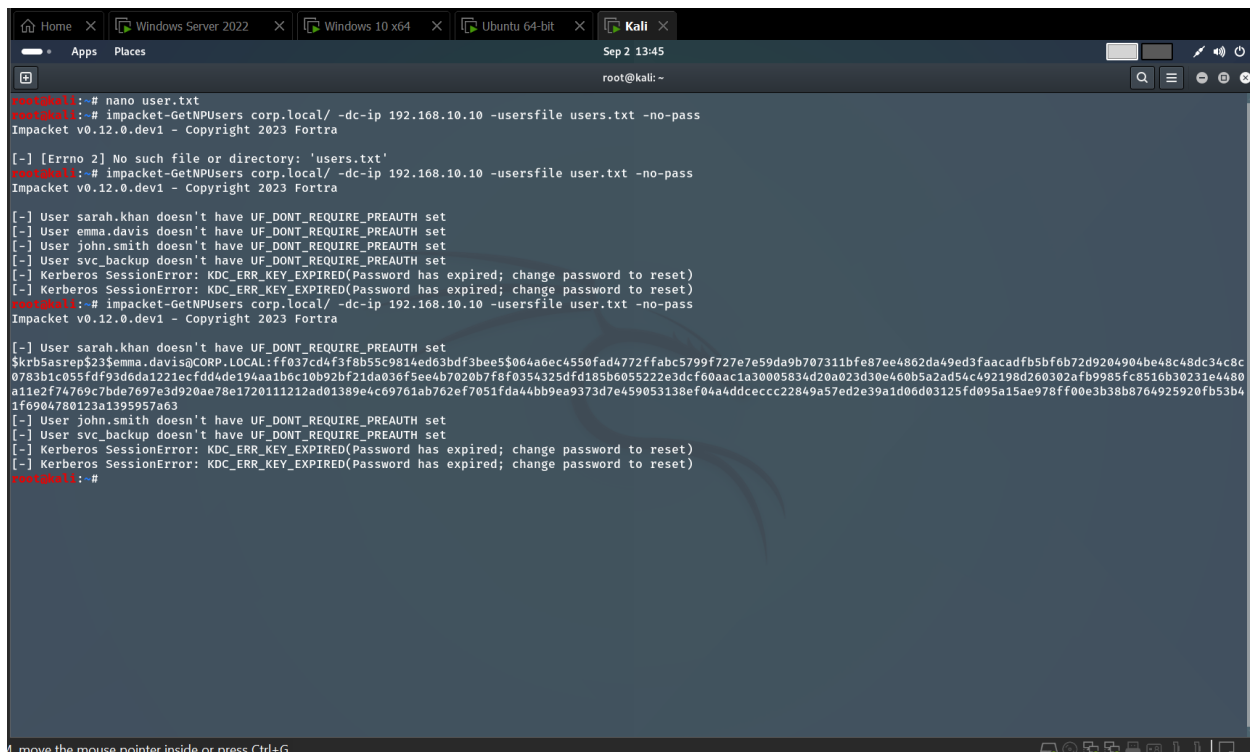
AS-REP Roasting is a Kerberos-based Active Directory attack that targets user accounts for which **Kerberos pre-authentication has been disabled**. Under normal circumstances, a user must prove knowledge of their password before the Key Distribution Center (KDC) provides authentication material. If pre-authentication is disabled, however, an attacker can request an AS-REP response for the affected account without first supplying valid credentials.

The returned AS-REP contains data encrypted using a key derived from the user's password. This information can be extracted and subjected to offline password cracking. Because the cracking process occurs locally on the attacker's system, it does not continuously interact with the Domain Controller, making strong password policies particularly important in reducing the effectiveness of this attack.

2.1 Enumeration and AS-REP Extraction

The first stage of the simulation involved enumerating the Active Directory environment to identify accounts that could be susceptible to AS-REP Roasting. The enumeration identified an intentionally vulnerable account configured without Kerberos pre-authentication.

After identifying the account, a Kerberos authentication request was made to the Domain Controller. Because pre-authentication was disabled for the target account, the Domain Controller returned the corresponding AS-REP authentication material, which could then be saved for offline analysis.



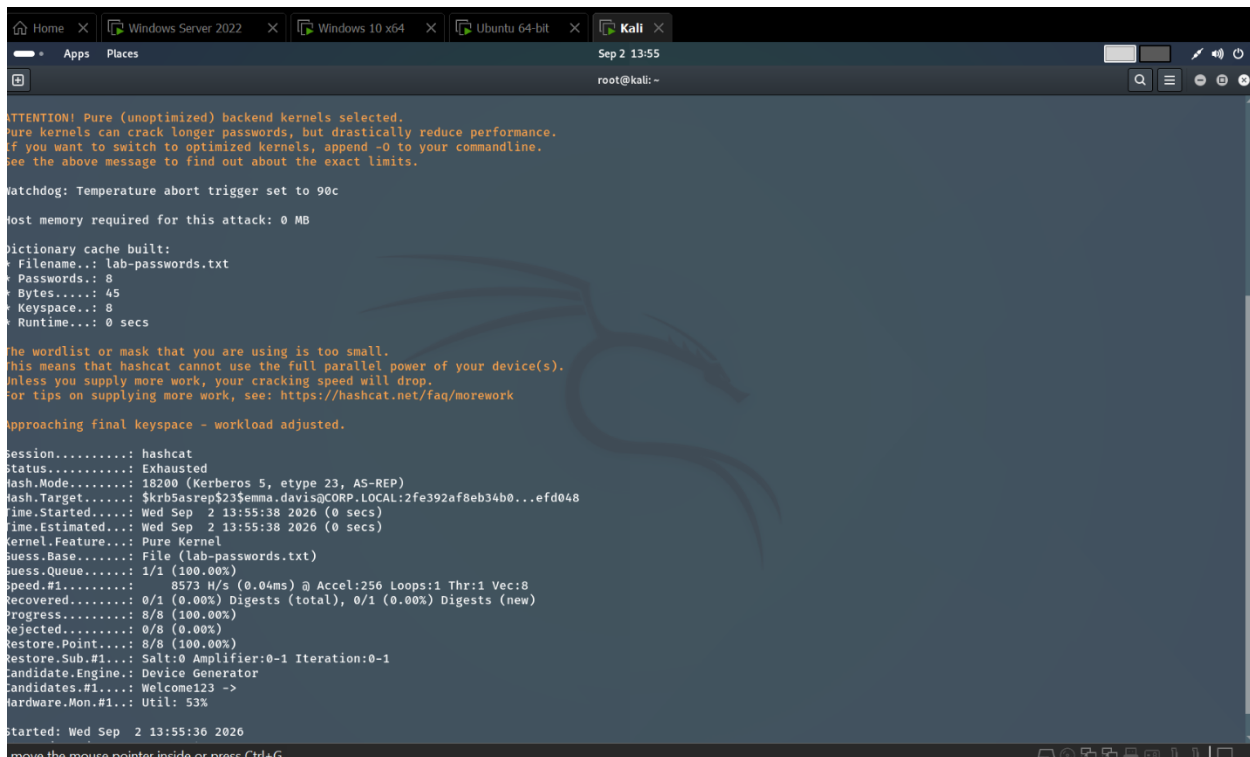
```
root@kali: ~  
root@kali:~# nano user.txt  
root@kali:~# impacket-GetNPUsers corp.local/ -dc-ip 192.168.10.10 -usersfile users.txt -no-pass  
Impacket v0.12.0.dev1 - Copyright 2023 Fortra  
[!] [Errno 2] No such file or directory: 'users.txt'  
root@kali:~# impacket-GetNPUsers corp.local/ -dc-ip 192.168.10.10 -usersfile user.txt -no-pass  
Impacket v0.12.0.dev1 - Copyright 2023 Fortra  
[~] User sarah.khan doesn't have UF_DONT_REQUIRE_PREAUTH set  
[~] User emma.davis doesn't have UF_DONT_REQUIRE_PREAUTH set  
[~] User john.smith doesn't have UF_DONT_REQUIRE_PREAUTH set  
[~] User svc_backup doesn't have UF_DONT_REQUIRE_PREAUTH set  
[~] Kerberos SessionError: KDC_ERR_KEY_EXPIRED(Password has expired; change password to reset)  
[~] Kerberos SessionError: KDC_ERR_KEY_EXPIRED(Password has expired; change password to reset)  
root@kali:~# impacket-GetNPUsers corp.local/ -dc-ip 192.168.10.10 -usersfile user.txt -no-pass  
Impacket v0.12.0.dev1 - Copyright 2023 Fortra  
[~] User sarah.khan doesn't have UF_DONT_REQUIRE_PREAUTH set  
$krb5asrep$23$emma.davis@CORP.LOCAL:ff037cd4f3f8b55c9814ed63bdf3bee55064a6ec4550fad4772f7fab5799f727e7e59da9b707311bfe87ee4862da49ed3faacadfb5bf6b72d9204904be48c48dc34c8c  
0783b1c055fdf93d6da1221ecfdd4de194aa1b6c10b92bf21da036f5ee4b7020b7f8f0354325dfd185b605522e3dcf60aac1a30005834d20a023d30e460b5a2ad54c492198d260302afb9985fc8516b30231e4480  
a11e2f74769c7bde7697e3d920ae78e1720111212ad01389e4c69761ab762ef7051fda44bb9ea9373d7e459053138ef04a4ddceccc22849a57ed2e39a1d06d03125fd095a15ae978ff00e3b38b8764925920fb53b4  
1f6904780123a1395957a63  
[~] User john.smith doesn't have UF_DONT_REQUIRE_PREAUTH set  
[~] User svc_backup doesn't have UF_DONT_REQUIRE_PREAUTH set  
[~] Kerberos SessionError: KDC_ERR_KEY_EXPIRED(Password has expired; change password to reset)  
[~] Kerberos SessionError: KDC_ERR_KEY_EXPIRED(Password has expired; change password to reset)  
root@kali:~#
```

Figure 3: Enumeration and Identification of an Account Vulnerable to AS-REP Roasting

2.2 Offline Password Cracking

The captured AS-REP authentication material was subsequently analyzed through an offline password-cracking process. Password candidates were tested locally against the encrypted Kerberos data rather than repeatedly authenticating against the Domain Controller.

The password of the intentionally vulnerable laboratory account was successfully recovered, demonstrating the security risk created by combining disabled Kerberos pre-authentication with a weak or guessable password.



```
ATTENTION! Pure (unoptimized) backend kernels selected.
Pure kernels can crack longer passwords, but drastically reduce performance.
If you want to switch to optimized kernels, append -O to your commandline.
See the above message to find out about the exact limits.

Watchdog: Temperature abort trigger set to 90c
Host memory required for this attack: 0 MB

Dictionary cache built:
* Filename..: lab-passwords.txt
* Passwords.: 8
* Bytes.....: 45
* Keyspace...: 8
* Runtime...: 0 secs

The wordlist or mask that you are using is too small.
This means that hashcat cannot use the full parallel power of your device(s).
Unless you supply more work, your cracking speed will drop.
For tips on supplying more work, see: https://hashcat.net/faq/morework

Approaching final keyspace - workload adjusted.

Session.....: hashcat
Status.....: Exhausted
Hash.Mode.....: 16200 (Kerberos 5, etype 23, AS-REP)
Hash.Target.....: $krb5asrep$23$emma.davis@CORP.LOCAL:2fe392af8eb34b0...efd048
Time.Started.....: Wed Sep 2 13:55:38 2026 (0 secs)
Time.Estimated...: Wed Sep 2 13:55:38 2026 (0 secs)
Kernel.Feature...: Pure Kernel
Guess.Base.....: File (lab-passwords.txt)
Guess.Queue.....: 1/1 (100.00%)
Speed.#1.....: 8573 H/s (0.04ms) @ Accel:256 Loops:1 Thr:1 Vec:8
Recovered.....: 0/1 (0.00%) Digests (total), 0/1 (0.00%) Digests (new)
Progress.....: 8/8 (100.00%)
Rejected.....: 0/8 (0.00%)
Restore.Point...: 8/8 (100.00%)
Restore.Sub.#1...: Salt:0 Amplifier:0-1 Iteration:0-1
Candidate.Engine.: Device Generator
Candidates.#1...: Welcome123 ->
Hardware.Mon.#1..: Util: 53%

Started: Wed Sep 2 13:55:36 2026
move the mouse pointer inside or press Ctrl+G.
```

Figure 4: Offline Cracking of the Captured Kerberos AS-REP Hash

2.3 Detection in Splunk

Following the attack simulation, Domain Controller authentication telemetry was investigated through Splunk. **Windows Security Event ID 4768**, which records Kerberos Ticket Granting Ticket requests, provided visibility into the authentication activity generated during the AS-REP Roasting simulation.

The relevant event was correlated with the targeted account and originating client information. This demonstrated that although the password-cracking stage occurred offline, the initial Kerberos request still generated valuable Domain Controller telemetry that could be centrally collected and investigated by the SIEM.

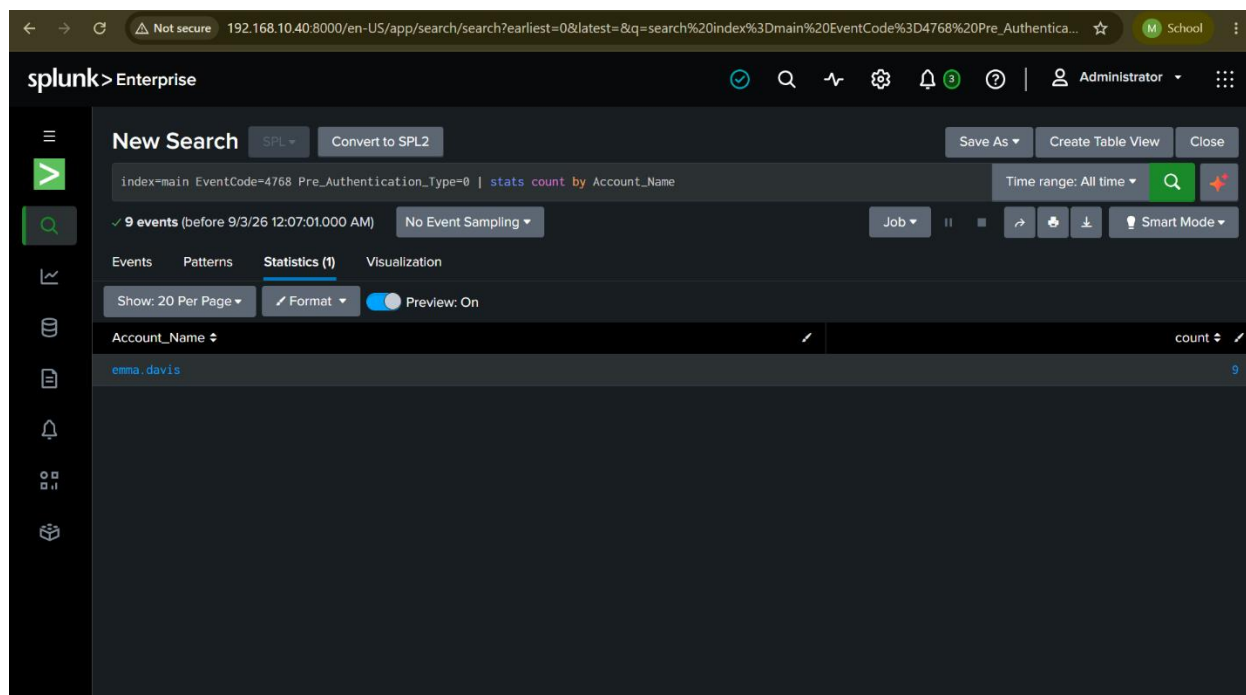


Figure 5: Detection of AS-REP Roasting Activity Using Kerberos Event ID 4768 in Splunk

2.4 Mitigation

AS-REP Roasting can primarily be mitigated by ensuring that **Kerberos pre-authentication remains enabled for all user accounts** unless there is a specific and justified operational requirement to disable it. Administrators should regularly review Active Directory accounts for the “**Do not require Kerberos preauthentication**” setting and remove it wherever it is unnecessary.

Strong password policies should also be enforced, particularly for privileged and service accounts. Even if AS-REP authentication material is obtained, a sufficiently long and complex password significantly increases the difficulty of successful offline cracking. Organizations should additionally monitor Kerberos authentication activity, especially unusual **Event ID 4768** requests, and investigate unexpected authentication requests originating from unfamiliar systems.

In this lab, the vulnerable configuration was intentionally introduced for attack simulation and should be removed after testing to return the account to a secure state.

2.5 Result

The AS-REP Roasting simulation was successfully completed within the controlled Active Directory environment. Enumeration identified an account with Kerberos pre-authentication disabled, and the Domain Controller returned AS-REP authentication material for the vulnerable account. The captured material was subsequently subjected to offline password cracking, successfully demonstrating the security impact of the misconfiguration.

From the defensive perspective, the initial Kerberos request was successfully recorded on **DC01** and forwarded to **Splunk**, where the relevant **Event ID 4768** activity could be investigated. This confirmed that the monitoring infrastructure established during Phase 2 provided visibility into the authentication activity associated with the attack.

Overall, the simulation demonstrated both the **offensive impact of disabled Kerberos pre-authentication** and the **defensive value of centralized Kerberos monitoring**, completing the full attack lifecycle of **enumeration → exploitation → offline cracking → detection → mitigation**.

3. Credential Dumping and Pass-the-Hash

Credential dumping and Pass-the-Hash represent a common post-compromise attack chain in Windows and Active Directory environments. After an attacker obtains privileged access to a system, credential material stored by Windows may become accessible. This can include **NTLM password hashes**, cached credentials, LSA secrets, and Kerberos keys.

An NTLM hash is derived from a user's password and is used by Windows authentication mechanisms. A major security concern is that an attacker does not necessarily need to recover the original plaintext password after obtaining a valid NTLM hash. In certain authentication scenarios, the hash itself can be reused to authenticate as the compromised user. This technique is known as **Pass-the-Hash (PtH)**.

In this simulation, credential dumping was performed against the Domain Controller, followed by reuse of the recovered Administrator NTLM hash. The resulting authentication activity was then investigated using the telemetry collected by Splunk.

3.1 Credential Dumping from the Domain Controller

The first stage of the simulation involved extracting credential material from **DC01 (192.168.10.10)**. The attack was performed from the Kali Linux system after privileged credentials had been obtained within the controlled laboratory environment.

Impacket's credential-dumping functionality was used to access Windows credential information stored by the Domain Controller. The operation successfully exposed several types of sensitive authentication material, including local SAM hashes, LSA secrets, domain account credential hashes, and Kerberos keys.

The results demonstrated the significant impact of privileged compromise on a Domain Controller. Because a Domain Controller maintains authentication information for the domain, unauthorized privileged access can potentially expose credentials belonging to multiple users rather than only the account initially compromised.

Figure 7: Successful Pass-the-Hash Authentication to DC01 Using the Administrator NTLM Hash

3.3 Detection in Splunk

Following the attack simulation, the authentication telemetry generated by DC01 was investigated through Splunk.

The investigation focused on **Windows Security Event ID 4624**, which records successful logon activity. The events were filtered to examine network authentication characteristics associated with NTLM and to exclude routine machine and anonymous authentication where possible.

The resulting Splunk data showed authentication activity originating from **192.168.10.30**, which was the IP address assigned to the Kali Linux attacker system. This provided a direct correlation between the attack source and authentication events recorded by the Domain Controller.

Although successful NTLM authentication is not inherently malicious, its context is important. Authentication involving a privileged account from an unexpected system, particularly an attacker workstation such as Kali Linux in this laboratory, represents behavior that should be investigated. Additional surrounding authentication, process, and endpoint telemetry can further increase confidence when identifying potential credential-reuse activity.

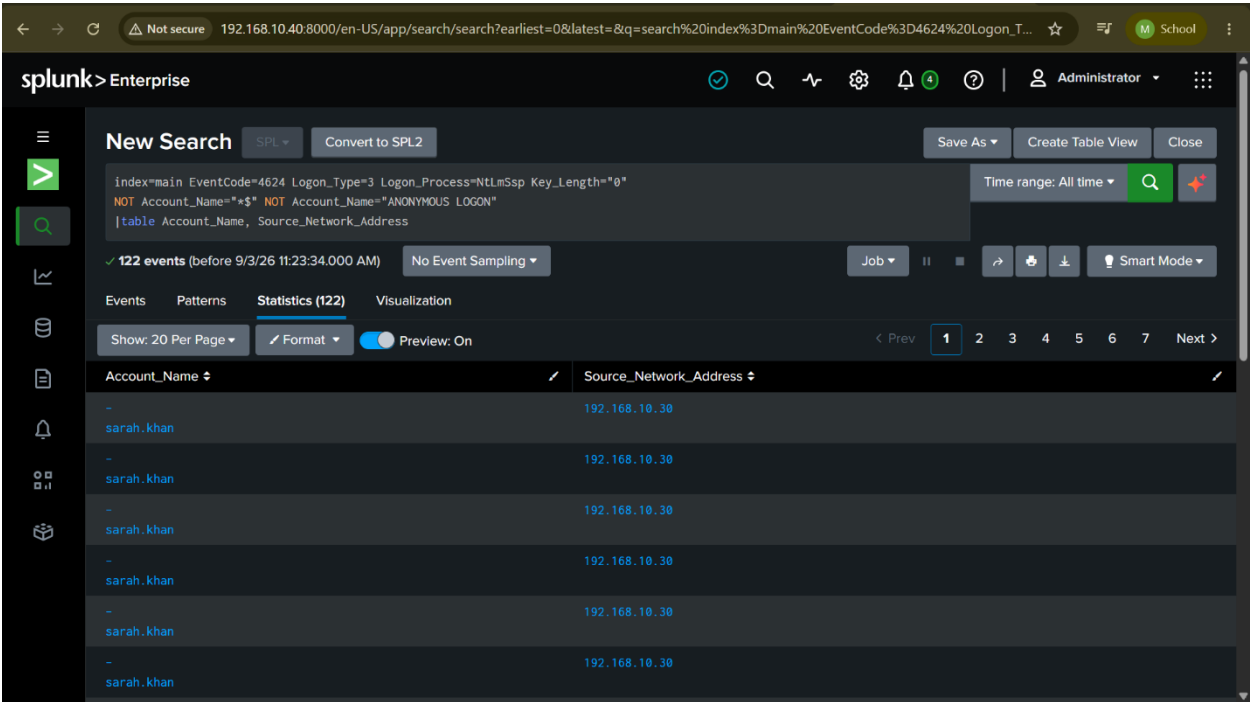


Figure 8: Splunk Identification of NTLM Network Authentication Activity from the Kali Attacker

3.4 Mitigation

Credential dumping and Pass-the-Hash attacks are best addressed by reducing both the availability and usefulness of privileged credential material.

Administrative accounts should follow the principle of **least privilege**, and privileged credentials should not be unnecessarily used on ordinary workstations or lower-trust systems. Separate accounts should be maintained for administrative and standard user activities so that compromise of a normal endpoint does not automatically expose highly privileged credentials.

Organizations should also restrict remote administrative access, protect credential-related Windows components, and reduce reliance on NTLM wherever operationally possible in favor of stronger authentication mechanisms such as Kerberos.

Strong password policies remain valuable, but they do not independently prevent Pass-the-Hash because the attacker may reuse the hash without recovering the original password. Consequently, protection of privileged credentials, account segmentation, restricted administrative access, and continuous monitoring are essential.

From a monitoring perspective, defenders should investigate unusual **Event ID 4624** activity involving privileged accounts, unexpected NTLM authentication, unusual source systems, and authentication patterns that differ from normal administrative behavior.

3.5 Result

The credential dumping and Pass-the-Hash simulation was successfully completed within the controlled Active Directory laboratory.

Credential material was successfully extracted from **DC01**, demonstrating the impact that privileged access to a Domain Controller can have on domain authentication security. The recovered Administrator NTLM hash was subsequently reused to authenticate remotely without requiring recovery of the Administrator's plaintext password.

From the defensive perspective, the resulting authentication activity was successfully recorded by DC01 and forwarded through the telemetry pipeline established during Phase 2. Splunk identified relevant successful network authentication activity originating from the Kali attacker system at **192.168.10.30**.

Overall, the simulation demonstrated the complete attack lifecycle of:

privileged access → credential dumping → NTLM hash acquisition → Pass-the-Hash → privileged remote access → Splunk investigation → mitigation

4. Conclusion

This project successfully designed and implemented a controlled Active Directory Blue Team laboratory that combines enterprise identity infrastructure, centralized security monitoring, attack simulation, and defensive investigation. The main objective was not only to build a functional Windows domain but also to understand how attacks against Active Directory generate security telemetry and how this telemetry can be used by defenders to identify suspicious activity.

The laboratory was initially established using VMware with Windows Server 2022, Windows 10, Kali Linux, and Ubuntu systems operating within an isolated network. Windows Server 2022 was configured as the Domain Controller for the corp.local domain, while the Windows 10 workstation was successfully joined to the domain. Organizational Units, domain users, security groups, and service accounts were configured to create a simplified enterprise-style Active Directory structure. DNS configuration was also implemented and validated to ensure reliable domain communication and authentication.

A major part of the project focused on developing security visibility across the environment. Advanced Windows auditing, PowerShell logging, process-creation auditing, and Sysmon were configured to generate detailed endpoint and authentication telemetry. Splunk Enterprise was deployed as the centralized SIEM, while Splunk Universal Forwarders were used to collect and forward Security, PowerShell, and Sysmon events from the Windows systems. This monitoring pipeline successfully provided centralized visibility into real Active Directory authentication and system activity.

After establishing the monitoring infrastructure, controlled Active Directory attacks were performed from the Kali Linux attacker system. These simulations demonstrated how weaknesses involving authentication, account configuration, and credential security can be abused by an attacker. Password spraying activity was successfully identified through patterns of authentication failures in Splunk, validating the ability of the monitoring environment to detect suspicious authentication behavior. AS-REP Roasting further demonstrated how an account with Kerberos pre-authentication disabled could expose authentication material for offline password cracking, while the initial Kerberos activity remained visible through Event ID 4768 within Splunk. Credential dumping and Pass-the-Hash testing also demonstrated the impact of privileged compromise, including the ability to reuse an NTLM hash for privileged remote authentication, with the resulting activity subsequently investigated through the centralized telemetry pipeline.

Overall, the project demonstrated the complete relationship between **Active Directory administration, adversarial activity, security telemetry, SIEM monitoring, investigation, and mitigation**. Rather than treating offensive and defensive security as separate activities, the laboratory showed how understanding attacker techniques can help defenders identify meaningful indicators within Windows and Active Directory logs. The completed environment therefore provides a practical foundation for further Blue Team experimentation, detection engineering, threat hunting, and Active Directory security research.